OECD Economic Surveys

Germany

OECD

Volume 2004/12 – September 2004

OECD ECONOMIC SURVEYS

2004

Germany

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ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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Publié également en français.

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BASIC STATISTICS OF GERMANY, 2003

LAND

Area, 2001 (1 000 km^2)	357
Agriculture (%)	54
Forest (%)	29

Major cities, 31.12.2001 (1 000 inhabitants):	
Berlin	3 384
Hamburg	1 726
Munich	1 228

PEOPLE

Population (1 000)	82 520
Inhabitants per km^2	231
Natural increase in population, 2001 (1 000)	–94
Net immigration, 2000 (1 000)	273

Labour force[1] (1 000)	42 093
Employment[1] (1 000)	38 246
Agriculture (%)	2
Industry (%)	27
Services (%)	70

PRODUCTION

GDP, current prices (billion euros)	2 129
GDP per capita (1 000 USD in current prices)	29
Gross fixed investment (% GDP)	18

Origin of value added (%)	
Agriculture	1
Industry	29
Services	70

GOVERNMENT

Public consumption (% of GDP)	19
General government total revenue (% of GDP)	45
Public debt, Maastricht definition (% of GDP)	64

Composition of Parliament	Seats
Social Democratic Party (SPD)	251
Christian Democratic Union (CDU)	190
Christian Social Union (CSU)	58
Alliance 90/The Greens	55
Free Democratic Party (FDP)	47
Party of Democratic Socialism (PDS)	2

Last general election: September 2002
Next general election: 2006

FOREIGN TRADE

Exports of goods and services (% of GDP)	36
Main exports (% of total merchandise exports):	
Machinery and transport equipment	51
Manufactured goods	13
Chemicals and related products	13

Imports of goods and services (% of GDP)	32
Main imports (% of total merchandise imports):	
Machinery and transport equipment	38
Manufactured goods	11
Chemicals and related products	13

CURRENCY

Irrevocable conversion rate (1 euro)	1.95583

Euros per USD:	
Year 2003	0.79
June 2004	0.82

1. Domestic concept.

Note: An international comparison of certain basic statistics is given in an Annex table.

Executive summary

With the effects of adverse external shocks diminishing, a strong and competitive export industry is helping the German economy out of a three-year period of near stagnation. Domestic demand has been declining over the last couple of years, as poor labour market performance has weighed on consumer sentiment and business confidence. The labour market still suffers from weak economic growth and distorted incentives, with both contributing to problems in taking up work and providing employment. Productivity growth is not high enough to compensate for the adverse effect of low labour utilization on economic growth. Fiscal targets have been missed on account of both cyclical and structural factors. The government has launched a major reform initiative to reinvigorate economic growth. These reforms are welcome, have to be continued and need to be broadened further to reduce government debt, remove fiscal distortions, and improve incentives to supply and demand labour. Furthermore, there remains considerable scope to foster the creation of new enterprises and widen product market competition, thereby also maintaining the strong innovative capacity of the economy. The major challenges are to link fiscal consolidation to public sector reform and to increase the capacity of the economy to create employment and increase productivity growth. To create confidence and to restore Germany's traditional economic strength it is necessary that reforms reflect a coherent vision about the reorientation of economic policy – combining a growth and stability oriented macroeconomic policy with structural reforms – and are implemented according to a transparent and predictable roadmap.

Linking fiscal consolidation to public sector reform

After phased income tax reductions that contributed to the rise in the structural deficit, priority should now be given to balancing the budget within a limited number of years. For the consolidation process to have maximum beneficial effects on economic activity, much will depend on casting it within a consistent framework that links budgetary improvement to public sector spending reform. The efficiency of the public sector should be increased by providing more scope for output performance budgeting. There remains room for a fully financed tax reform reducing the welfare costs of taxation by cutting high statutory tax rates while broadening the tax base by reducing tax expenditures. On current plans the

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structural balance is projected to improve by ¼ percentage point per year, which is clearly not enough to prepare for ageing related spending increases.

Important steps have been undertaken to reform health care and pensions, marking progress in re-establishing fiscal sustainability. Increasing co-payments, reducing the catalogue of fully funded health services and involving health funds in managed care should allow insurance funds to contain contributions. More could be done to reduce the power of special interest groups and increase the scope for insurance funds to play an active purchaser role. Progress in pension reform should continue by announcing soon a phased increase of the statutory retirement age. Fiscal subsidisation of the funded system should be reconsidered.

Increasing the capacity of the economy to create employment

Major reform steps to make the German labour market more conducive to employment creation have been legislated and are being phased in. Employment protection legislation (EPL) has been eased, including an increase in the employment threshold above which dismissal protection is binding for small firms. Benefit duration for older workers without job search requirements will be substantially reduced. Unemployment and social assistance will be merged. Active labour market programmes are to be evaluated more systematically. These reforms have to continue and should be broadened. In addition, fiscal obstacles to higher labour supply and demand should be removed. The efficiency of placement services must be further increased and job search requirements should be enforced regardless of age. Active labour market programmes which turn out to be unsuccessful should be terminated. The scope for individual agreements on wage and non-wage aspects of a labour contract should be increased. Opening tertiary education to more competition and widening the autonomy of educational institutions with respect to achieving targets should contribute to more effective human capital accumulation.

Improving the capacity to innovate and increase productivity growth

Competition in product markets should be further strengthened by removing entry barriers and making more progress in reducing administrative overheads. The part of the large German procurement market which falls below EU thresholds should be opened to greater competition, requiring *inter alia* making transparent the multitude of regulations on lower levels of government.

Key indicators show Germany belonging to the countries in the OECD with strong innovation activity even though some weakening in Germany's position relative to other OECD countries has occurred recently. Measures to improve the framework conditions for innovation should include removing disincentives to take risks and to provide capital to new firms and raising the efficiency of the higher education sector. Furthermore, making labour markets more flexible and increasing the scope for competition overall, as well as reducing administrative opacity, would improve the capacity of the German economy to innovate and contribute to higher potential growth.

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This Survey is published on the responsibility of the Economic and Development Review Committee of the OECD, which is charged with the examination of the economic situation of member countries.

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The economic situation and policies of Germany were reviewed by the Committee on 21 June 2004. The draft report was then revised in the light of the discussions and given final approval as the agreed report by the whole Committee on 13 July 2004.

•

The Secretariat's draft report was prepared for the Committee by Eckhard Wurzel and Andrés Fuentes under the supervision of Andreas Wörgötter.

•

The previous Survey of Germany was issued in January 2003.

Assessment and recommendations

The economy is picking up but the recovery is still held back by weak domestic demand

With the effects of adverse external shocks diminishing, the German economy is currently recovering, ending a couple of years in stagnation on the back of its traditionally strong, competitive and innovative export-oriented manufacturing sector. However, the economy is far from operating at full strength due to the weakness of final domestic demand. Poor labour market performance continues to weigh on consumer sentiment and business confidence remains volatile. The labour market suffers from weak growth and distorted incentives, with both contributing to problems in taking up work and providing employment. Although monetary conditions should remain supportive for GDP growth in the euro area, a persisting German inflation differential relative to the euro area average would mean that real short-term interest rates risk damping the recovery of demand in Germany to an extent, which may not be compensated by the corresponding gain in competitiveness. Cyclical weakness and the structural problems of the economy impact strongly on public budgets, while uncertainty about how public finances will be put on a durably sustainable path is a further factor undermining confidence. Re-establishing Germany's traditional economic strength requires a comprehensive policy response within a coherent framework.

Comprehensive reform is necessary and should continue

The German government has responded to this challenge by taking significant steps to reform labour and product markets, and substantial reforms to the social security system have recently been legislated. While not all measures are yet in force, the thrust of these reforms is appropriate and can be expected to improve labour market performance and business dynamism in the medium term. To create confidence it is necessary that reforms reflect a

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coherent vision about the reorientation of economic policy and are implemented according to a transparent and predictable roadmap. For economic performance to be raised in a durable way, it is indispensable that the reforms be continued and deepened. The main challenges in this respect are:

– *Linking sustainable consolidation of general government finances with public sector reform. This would include, inter alia, reforming budgetary procedures, including those governing federal relations, reducing distortions of the tax system, reforming long-term care and continuing with health and pension reform, as well as harmonising public and private sector labour and pension regulations.*

– *Enhancing employment creation through better incentives to supply and demand labour, a more efficient management of labour market policies and greater responsiveness of wage formation to labour market imbalances.*

– *Strengthening productivity growth by fostering product market competition and the economy's capacity to innovate.*

Moving in parallel to address these challenges simultaneously is essential for reaping the synergies across reforms in the different fields.

With the deficit at almost 4 per cent of GDP, fiscal consolidation has not been successful enough…

Fiscal consolidation resumed in 2003, with the cyclically adjusted balance improving by ½ per cent of GDP. This improvement did not translate into lower public deficits, however, mainly due to unforeseen cyclical and structural shortfalls in direct and indirect taxes. With economic activity stagnating, the general government deficit deteriorated to almost 4 per cent of GDP, nearly ½ percentage point higher than in 2002. Another fiscal consolidation package is being phased in this year and next, comprising *inter alia* cuts in subsidies and tax expenditures as well as revenue raising measures, including a tax amnesty with preferential taxation of repatriated assets. At the same time substantial income tax reductions will become effective, partly brought forward into 2004 from 2005. With further reductions scheduled for next year, income tax cuts amount to some one per cent of GDP. The overall deficit is likely to decline to close to 3 per cent of GDP in 2005, helped by strengthening economic activity. On balance the OECD projects that the structural

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deficit will improve by ¼ per cent of GDP both this year and in 2005, less than officially projected due to expected revenue shortfalls. Repeatedly under-achieving fiscal targets undermines confidence. *Public finances need to be put back on a sustainable path, which requires a balanced budget within a few years and surpluses further ahead.*

… requiring fundamental reforms

While most consolidation measures that are being phased in have a lasting effect, there is a need for more fundamental spending reform in order to address a number of counterproductive features of Germany's public sector. In their present setting, Germany's federal relations make it difficult to find the necessary political consensus for fiscal reforms, often involving complicated mediation processes between the Bundestag and the *Länder* chamber of Parliament (*Bundesrat*), whose outcomes are hard to predict. The working of market forces is hampered by a high tax-wedge on labour while persisting subsidy programmes reduce economic efficiency. Parameters of important ageing-related spending programmes are not sufficiently adjusted to predictable demographic changes and make future contribution changes likely. Special tenure and pension schemes for civil servants reduce labour mobility and make it more costly to reduce public sector employment. Measures which address these issues are explained in more detail below.

Budgetary procedures should better allow medium-term prioritisation and increase the efficiency of the public sector

Against the background of the various fiscal pressures that are impinging on the general government budget, strict prioritisation of public sector spending projects is required for the sustainability of public sector finances. This can be fostered by a simplified output-oriented budgeting framework that presents to parliament budgetary appropriations for programmes, together with analysis of associated costs and benefits. *Such a reform should be introduced and embedded in a medium term budgeting framework.* Moreover, untangling the responsibilities of the federal government and the states and communities is one of the most pressing tasks *in order to increase the speed and transparency of federal decision making. Reducing the degree of co-financing between the different levels of government and introducing a higher degree of tax autonomy for the Länder and communities*, as has been proposed in earlier *Economic Surveys* for Germany, would go some way towards making decision

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processes at the federal level simpler and more transparent. *In this context the setting up of a high level commission to look for a consensus on modifying those constitutional provisions governing federal relations is a welcome development.*

Further tax reform should focus on reducing distortions

Despite substantial income and corporate tax reductions, phased in since 1999, average and marginal tax rates on incomes in Germany are relatively high in comparison to major competitors and the new EU accession countries. At the same time, many special rules of taxation and tax concessions imply that the deadweight costs of taxation are high and the tax code is complex. In particular, the tax system disadvantages the financing of start-ups relative to investment in established firms. Also, certain tax rules hamper investment in equity capital and the broadening of equity capital markets. Abolishing such distortions is likely to improve not only the efficiency of the economy but also its capacity to innovate. *Further tax reform should therefore focus on reducing the deadweight costs of taxation by cutting both tax expenditures and statutory tax rates and should be fully financed.*

Fiscal and structural policies need to rely more on market forces to bring about higher growth and employment in the new Länder

Self-sustained growth in the new *Länder* is still lacking, unemployment remains persistently high, and productivity is still some 30 per cent lower than in the west. There is evidence that the structural weaknesses of the eastern German economy are unlikely to be remedied by special subsidy programmes. Indeed, the OECD has argued in the past that persistently high subsidisation and inefficient labour market policies have biased the structure of the economy and reduced its adaptability to market forces. Hence,

– *The objective of applying the same set of rules for subsidisation to the whole of Germany should be consistently pursued and enforced effectively.*

– *Regulatory reform should also apply to all of Germany. However, as a second-best solution, some discretion might be given to economically weak areas in Germany to establish pilot projects of regulatory reform.*

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Health care reform should be continued so as to reap the benefits from more competition

Health care spending in terms of GDP remains among the highest within the OECD while significant room for efficiency improvement remains. Recent reforms have reduced the share of health care services that are fully financed by the public system and have increased the scope for selective contracting between health care providers and insurance funds. Future reform should:

– *Further widen the scope for selective contracting.*

– *Consider reducing the collective role of the doctors' and insurers' associations in bargaining supply conditions.*

– *Strengthen quality control by securing the independence of the supervising bodies.*

Long term care is an important ageing related programme and measures are needed to contain mounting spending pressures

Since the end of the 1990s the long-term care insurance has been in deficit, and spending pressures are set to increase substantially, mostly as a result of population ageing. Measures that should be taken to increase the efficiency of the system include:

– *Rebalancing insurance payments so as to create incentives to substitute home and ambulatory care for more expensive stationary care.*

– *Re-channelling of funds in favour of better performing service providers.*

Progress in pension reform should be secured by further reducing incentives for early retirement…

From 2005 onwards pension adjustment will be linked to the ratio of dependent employees contributing to the scheme relative to the number of pensioners, and the minimum entry age for early retirement on account of unemployment will be increased from 60 to 63. On the other hand, the government decided to defer to 2010 the decision on whether or not to increase the statutory retirement age above 65. Linking pension adjustments to the ratio of contributors to pensioners, in particular, is an important step to put the public pension system on a sustainable base. Even so, pension contribution rates are set to rise significantly over the next decades due to ageing, driving up non-wage labour costs. Participation in the voluntary funded scheme is hampered by cumbersome regulations, while it is relatively expensive in terms of evolving tax expenditures.

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Hence, pension reform should be complemented by further measures in order to increase its effectiveness:

– *A phased increase in the statutory retirement age should be announced soon.*

– *The system should be made actuarially neutral around the statutory retirement age.*

– *The government should evaluate the extent to which tax-subsidies in the funded scheme actually induce households to save more rather than substitute between different savings instruments.*

… while pension schemes for civil servants are in urgent need of reform

Public sector reform will require a more flexible labour force. Pension rules and the terms of health care insurance are not sufficiently harmonised with the private sector scheme and within the public sector, respectively, which hampers the mobility of labour. Obstacles in this respect should be removed by a number of measures:

– *Rules for public sector pensions should be fully harmonised with the general scheme, and a transparent link between contributions and benefits according to individual contribution histories should be established.*

– *Special tenure for civil servants (Beamte) with a distinct pension plan should be reduced to the minimum or abolished entirely.*

– *Health care insurance for civil servants should be subjected to the rules prevailing for private sector employees.*

– *Raising contributions so as to pre-fund retirement outlays should be considered.*

Increasing the capacity of the economy to create employment is a second, related challenge

The weak capacity of the German economy to create more employment from economic growth stems from a complex interaction of various factors affecting the demand for and supply of labour with the wage setting system. While overall wage restraint has helped to improve competitiveness and contributed to price stability, wage formation is not flexible enough to prevent the various supply and demand constraints from affecting employment. Although work incentives have been improved, benefit programmes still induce older workers to withdraw from the labour market while high social security charges and cumbersome regulations add to the labour costs employers are facing. The education system is not sufficiently meeting skill require-

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ments of job market entrants, increasing structural unemployment. Some significant steps of labour market reform have been made, and first initiatives have been taken to increase the efficiency of the education system. While this marks progress, reform needs to continue and further measures – explained in more detail below – are necessary.

Various disincentives to higher labour supply should be abolished

Significant barriers to effective labour supply exist for older workers and second wage earners, in particular women. At present extended eligibility periods for unemployment benefits without obligation to search for a job for unemployed aged 58 and older serve as a channel into effective early retirement. From 2006 onwards the extended duration of unemployment insurance benefits for older unemployed will be cut significantly and older new unemployed will also be subject to job search requirement. While this marks substantial progress, significant age-related extensions of benefit duration and exemptions from job search requirements for the stock of older unemployed will remain. Second earners face larger average effective tax rates than single earners, mostly affecting women with full-time working partners. Households with dependent children find it difficult to reconcile work and family as long as proper child care facilities are lacking. The availability of free healthcare co-insurance for non-working spouses distorts incentives against working. Disincentives to higher labour supply should be removed by the following measures:

– *The possibility of stricter application of job search requirements regardless of age should be considered, and the impact of ending the preferential extended duration of entitlement for the older unemployed should be assessed.*

– *Personal income taxation and child benefits should be reformed so as to minimise distortions of labour-leisure choices across different household situations and average effective tax rates on labour income of secondary earners should be reduced.*

– *Resources for child support should be reallocated to some extent from cash to in-kind benefits like child care vouchers.*

– *Introducing charges for healthcare co-insurance for non-working spouses should be considered.*

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The management of labour market policies needs to become more effective

Important steps have been taken to increase incentives for taking up employment. In 2005 unemployment assistance will be combined with social assistance into one single means-tested benefit, and job acceptance criteria will be sharpened. This organisational reform of the Labour Office needs to be properly implemented; and reform widened, in particular:

– *The present legislation, which gives the local authorities the option of administering the new benefits and placement services instead of the Federal Labour Agency, should be rethought, as it would compromise the potentially large efficiency gains that could be achieved from having only one agency administering a unified set of placement services and means tested benefits. Instead only one agency should be involved.*

– *Initiatives to strengthen activation strategies providing counselling advice and support services associated with appropriate sanctioning should be pursued further.*

– *Active labour market programmes should be rigorously evaluated as foreseen and measures that are not effective should be immediately discontinued.*

– *First steps to provide labour offices with adequate resources and qualified personnel are welcome and should be pursued further while performance incentives for their management should be extended.*

While the labour market is becoming more flexible reform should continue

Progress has also been made in increasing the flexibility of labour markets. A number of measures reduce the restrictiveness of dismissal protection, in particular the range of social criteria that employers have to take into consideration in dismissal decisions has been reduced and the threshold number of employees above which dismissal protection legislation becomes binding has been increased from 5 to 10 employees. It remains to be seen whether these changes will reduce the legal uncertainty associated with the present legislation. Measures to increase labour market flexibility should include:

– *Evaluating the impact of recent reforms of* EPL *and considering further easing accordingly. Specifically this might include allowing the option of not adopting dismissal protection provisions as part of the work contract after a probationary period in exchange for higher severance pay and raising further the threshold of applicability of dismissal protection.*

– *Widening the scope for wage determination at the company level so as to better align wage contracts with labour market conditions.*

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– *Further confining the administrative extension of collectively bargained contracts to minimum wages and a minimum standard of working conditions.*

– *Discontinuing the practice of conditioning public procurement by Länder on firms paying wages according to local collectively bargained wage rates.*

– *Evaluating the impact of support for low paid jobs on labour supply and training and considering tighter targeting on benefit recipients.*

The vocational training system needs flexibility in both curricula and pay

Germany has low youth unemployment in international comparison, due to a relatively efficient school-to-work transition. However, excess demand for traineeships in recent years has led to serious concerns about rising youth unemployment. A contribution and subsidy scheme has been considered as a possible measure to increase the supply of apprenticeships at times of excess demand. As such schemes would increase non-wage labour costs, which are already high, alternative measures were sought to address the market imbalance between supply and demand directly, resulting in a voluntary agreement of the industry to supply 30 000 additional apprenticeship places. *This measure should be supported by allowing for a higher degree of flexibility of pay for trainees and reducing training costs to enterprises. Furthermore, efforts should continue to update training requirements and improve the attractiveness of apprenticeships by an increased flexibility in training periods.*

Broad reforms to raise the performance of tertiary education are necessary

The supply of well trained and highly qualified labour is a key contribution to innovation performance, in particular the generation and diffusion of knowledge. The efficiency of the German tertiary education system is relatively low, as witnessed by long study durations and high drop-out rates. While the government plans to promote a competition based selection of a small number of additionally funded high performance universities, a broader approach is required to raise the performance of tertiary education overall. Within the university sector itself, a greater degree of competition among institutions should be established. Major measures should include:

– *Institutional reform in the secondary education system toward a clearer focus on outcomes, as outlined in the* 2003 *Economic Survey.*

– *Better preparation in the non-academic track to direct university entry.*

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– *Bureaucratic allocation of students to universities should be given up and universities obtain more discretion to choose students.*

– *The allocation of public sector funding should be linked to performance indicators.*

– *Consideration should be given to allowing universities to levy tuition fees from their students, enabling students to act as purchasers of the universities' services at the institution of their choice. The distributional issue could be addressed by obliging students to pay such fees in instalments only after graduation and provided that their income exceeds a certain threshold.*

– *Universities should be given more autonomy with respect to the use of funds and their personnel management.*

Obstacles to the immigration of highly qualified workers should be removed

Net migration to Germany appears significantly biased toward low qualifications. On average, immigrants have significantly lower qualifications than the domestic population. The relatively high tax wedge is a disadvantage for Germany to be chosen as a location by internationally mobile high-skilled workers. New immigration legislation that aims at introducing qualification standards for immigrants seeking dependent and self-employment in Germany has been decided. *Legislation along these lines that raises the qualification profile of immigrants should be implemented.*

Innovation is strong in Germany but would benefit from broad based reforms

Key indicators show Germany belonging to the countries in the OECD with strong innovation activity, as reflected in one of the upper ranks in the number of patents filed per population and above average levels of R&D spending in terms of GDP. Some weakening in Germany's position relative to other OECD countries has occurred over the last one or so decade. While the redirection of resources towards unification-related spending as well as low economic growth has contributed to this development, more fundamental structural issues have also played a role. Germany has benefited less than other high-performing countries from the surge in new technologies, such as ICT and biotechnology, as innovation activities continue to focus on sectors such as machinery and automobiles in which Germany has a long record of strong export performance. Some features of the regulation of capital, product and labour markets are hampering the supply of risk capital, the

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creation of new firms and the reallocation of labour. In addition, firms are finding it increasingly difficult to recruit highly qualified labour. The government envisages a policy initiative to foster innovation. Several reform proposals addressed above are equally appropriate to improve the framework conditions for innovation. To summarise, the following points are of particular importance:

– *Reform the taxation of capital in order to remove disincentives to take risks and to provide capital to new rather than to established firms.*

– *Raise the efficiency of the higher education sector, inter alia by improving access and giving universities more autonomy and incentives to attract students.*

– *Evaluate government-funded innovation programmes more consistently in order to increase efficiency and eliminate ineffective subsidies.*

– *Remove hurdles to firm creation by reducing the complexity of the tax system, taking further steps to reduce administrative opacity and further reducing the costs imposed by employment protection.*

Efforts need to be continued to increase product market competition

Pro-competitive regulation of product markets contributes to improving productivity performance, stimulating investment and innovation. The government has embarked on a number of reforms designed to move the economy towards a more pro-competitive stance. The telecoms regulator's remit has been extended to the gas and electricity industries, improving the regulatory conditions for non-discriminatory network access for market entrants. Entry barriers to practicing a handicraft have also been reduced by removing the master certificate requirement in many crafts and exempting new craft businesses from contributions to the crafts association for four years. However, *deregulation needs to continue* as about 90 per cent of establishments in the craft sector remain subject to some entry requirements. Similarly, *compulsory contributions to enterprise associations should not act as a barrier to market entry*. Fee schedules of professional associations limit price competition among members and are legally binding in some cases. The government intends to allow private contracts to deviate from binding fee schedules for architects and engineers. This should be implemented and further deregulation should be considered. The government has also launched an initiative to

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reduce bureaucracy in order to reduce administrative opacity, including the planned simplification of complex and non-transparent rules on public procurement, which accounts for 17 per cent of German GDP. *Competition for public procurement should be strengthened further by removing the participation of business organisations in the procurement process.To prevent administrative opacity in future regulatory initiatives, regulatory impact analysis should be strengthened*. This would help, *inter alia*, to foster the establishment of new enterprises and company growth.

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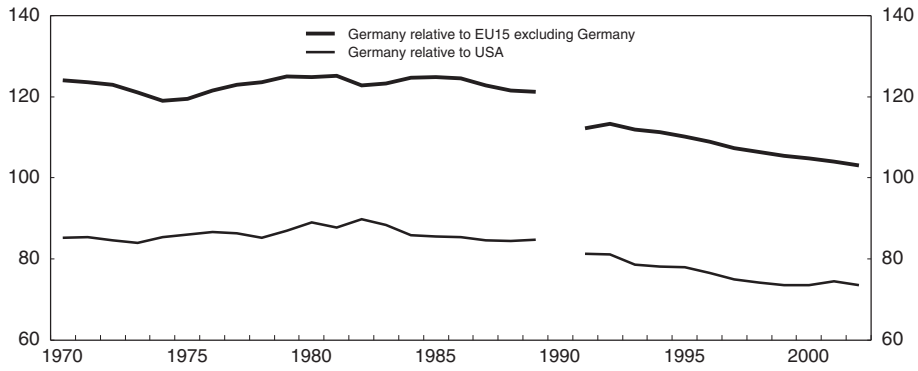
1. Main issues and policy challenges

Germany is still a strong economy in various respects, as reflected in its vigorous export performance and high innovative capacity. However, the economy has fallen back. For a period of more than ten years real GDP growth has fallen short of that in other European countries and North America. This is reflected in potential GDP having decelerated to a growth rate of just 1.5 per cent annually while structural unemployment has trended upwards. In addition actual GDP has tended to substantially undershoot potential. While convergence of GDP per capita with the US peaked already at the beginning of the 1980s, due to both lower production in the new Länder and weak German growth since the first half of the 1990s the gap widened rapidly, with per capita income in Germany now falling short of that in the US by some 30 per cent (Figure 1.1).

Figure 1.1. **GDP per capita**[1]

Per cent



1. Western Germany to 1989, Germany from 1991 onwards; 1990 omitted because of reunification break. Estimates based on GDP in 1995 prices and purchasing power parities.

Source: OECD, Annual National Accounts.

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Important structural reform steps have recently been legislated. Not all of these are in force yet, and further efforts are necessary. Indeed, structural reform to strengthen Germany's economic performance needs to improve conditions for labour utilisation and productivity growth. At the same time, fiscal consolidation needs to be given high priority and linked to public sector reform. The first part of this chapter highlights the main issues policy makers are confronted with, and the second part summarises important policy implications.

The growth contribution of labour is negative and not offset by higher productivity growth

As has been outlined in the 2003 *Economic Survey*, up to now only a negligible part of the subdued per capita growth performance relative to other European countries is attributable to demographic factors, and the negative demographic impact has been only little larger in comparison to the US. Most of the discrepancies in per capita growth are attributable to a negative growth contribution of employment.[1] Over the last decade total hours worked per inhabitant declined, accounting *ceteris paribus* for a weakening in real GDP growth per capita by 0.6 percentage points annually. By contrast, employment contributed positively to growth in other European countries and in the US. Moreover, while German labour productivity growth has been higher than in some other European countries, this differential was not high enough to offset the adverse effect on GDP growth of weak labour input (Figure 1.2).

Weak confidence in the economic recovery adds to the lack of economic resilience

The cyclical downswing following the growth peak in 2000 confirmed the weakness and low resilience of activity (Figure 1.3). For almost three years output growth remained close to stagnation, and over the same period Germany experienced the largest negative gap between GDP trend growth and final domestic demand among the larger countries within the OECD (Table 1.1).

For 2003 as a whole private consumption continued to contract as consumer confidence remained subdued and falling employment reduced disposable income growth. Equipment investment receded for the third year in a row, reflecting low levels of capacity utilisation, while construction investment continued its long-term decline. With the effects of adverse external shocks diminishing, the German economy is currently recovering on the back of its traditionally strong and competitive export sector. However, the economy is far from operating at full strength due to the weakness of final domestic demand (see Box 1.1 for the OECD short-term projections). Despite substantial income tax reductions in 2004 consumer confidence is recovering only slowly, reflecting a poor labour market performance, doubts about the recovery and perhaps uncertainty about how future incomes will

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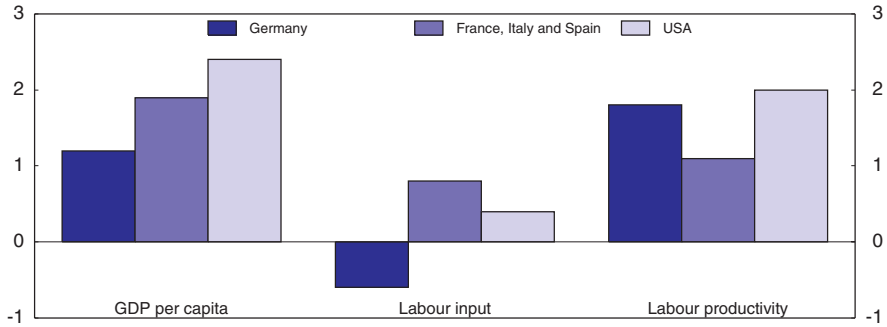
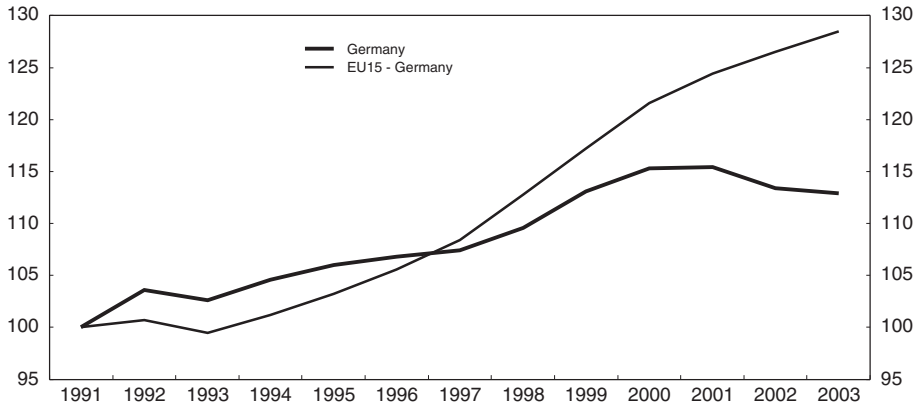
Figure 1.2. **Accounting for growth, 1993-2003**[1]
Average of annual changes, per cent



1. Growth in real GDP per capita is decomposed into growth in labour input and growth in labour productivity using the identity [gdpv/pop] = [(et X H)/pop]X[gdpv/(et X H)] in logarithmic differences, where gdpv is real GDP, pop is total population, et is employment and H is average annual hours actually worked by persons in employment.

Source: OECD.

Figure 1.3. **Domestic demand in Germany and the EU15**
1991 = 100



Source: OECD.

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Table 1.1. **Resilience of domestic demand, 2000-2003**
Per cent

	Potential growth of GDP	Largest negative gap between annual growth in domestic demand and trend growth in GDP
Germany	**1.5**	**–3.3**
Japan	1.2	–1.9
Italy	1.7	–0.9
United States	3.3	–1.7
Australia	3.6	–2.1
France	2.1	–0.6
Canada	3.1	–0.3
Spain	2.8	–0.2
United Kingdom	2.7	–0.2

Source: OECD.

be affected by the adjustments needed to put fiscal policy on a sustainable path. Monetary policy has remained supportive of economic activity, with the minimum bid rate in the Eurosystem's main refinancing operations maintained at 2 per cent since June 2003. Real interest rates still stand at historically low levels in Germany. However, as a result of a lower inflation rate, the short term real interest rate has been higher than in the other countries of the euro area. More recently the inflation differential *vis-à-vis* the other euro area countries has narrowed, mainly on account of one-off price effects associated with the health care reform and higher indirect taxes. A persisting German inflation differential relative to the euro area average would mean that real short-term interest rates risk damping the recovery of demand in Germany to an extent which may not be compensated by the corresponding gain in competitiveness.

For several years, exports have been the driving force in generating economic growth, although more recently concerns have been raised that the impact of strong exports growth on economic activity might be overstated due to a rising import content of exported goods (see Box 1.2). In 2003 exports decelerated, reflecting both slow growth in world trade and the adverse impact of the euro appreciation on German competitiveness *vis-à-vis* its main non-European trading partners (Figure 1.4). As domestic demand failed to cushion weakening export growth real GDP growth shifted into negative territory. Exports to North America fell sharply in 2003. Sales to South East Asia were also hit by the appreciation of the euro though exports to China remained strong, benefiting from China's vigorous growth. Within the euro area German competitiveness increased, however, owing to lower inflation und a more modest increase of labour costs than among major European competitors. Consequently, German exports to other euro area countries accelerated (Table 1.4). Both exports and imports firmed in the second half of 2003 and into 2004 as the economic recovery gained pace.

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Box 1.1. The OECD's short-term projections for Germany

Economic recovery will be largely driven by exports, which are projected to accelerate in the course of 2004 and into next year as world trade growth expands more rapidly. Private consumption should firm as disposable incomes rise, supported by the phased income tax reductions, even if some share of the tax relief is expected to be saved. However, reductions in health care contribution rates, associated with the 2004 health care reform will not fully compensate households for larger co-payments for health care services. Scheduled increases in indirect

Table 1.2. **Demand and output**

Percentage change from previous year, 1995 prices[1]

	1999	2000	2001	2002	2003	2004	2005
						Projections	
Private consumption	3.7	2.0	1.4	–1.0	–0.1	0.4	2.1
Government consumption	0.8	1.0	1.0	1.7	0.9	0.1	–0.2
Gross fixed investment	4.1	2.7	–4.2	–6.7	–2.9	1.3	2.5
Construction	1.4	–2.6	–4.8	–5.8	–3.4	–0.3	–0.7
Machinery and equipment	8.1	9.9	–3.6	–7.6	–2.3	3.3	6.2
Changes in stocks	–0.4	–0.1	–0.8	0.1	0.7	0.3	0.0
Total domestic demand	2.8	1.8	–0.8	–1.6	0.3	0.9	1.7
Foreign balance	–0.7	1.1	1.6	1.7	–0.4	0.3	0.5
Exports of goods and services	5.5	13.7	5.6	3.4	1.2	5.2	7.3
Imports of goods and services	8.4	10.5	0.9	–1.7	2.6	5.2	6.9
Gross domestic product	2.0	2.9	0.8	0.2	–0.1	1.1	2.1
Memorandum items:							
Private consumption deflator	0.3	1.5	1.6	1.3	1.0	1.1	0.6
GDP deflator	0.5	–0.3	1.3	1.6	1.0	0.8	0.8
Total employment[2]	1.2	1.8	0.4	–0.6	–1.1	–0.3	0.6
Unemployment (million persons)[2]	–7.3	–8.0	1.5	9.2	7.8	0.1	–3.1
Unemployment rate (% of labour force)[2]	8.0	7.3	7.4	8.1	8.7	8.8	8.5
Household saving ratio	9.8	9.8	10.3	10.6	10.8	11.1	11.0
Current account balance, % GDP	–1.0	–1.2	0.2	2.2	2.2	2.8	3.5

Note: Economic activity in 2004 and 2005 is subject to unusually large changes in the number of working days. The OECD projections are adjusted for this effect so that, other things equal, GDP is lower for 2004 and higher for 2005 than in the unadjusted projections presented by the German authorities. However, in the view of the OECD the positive impact of a larger number of working days on unadjusted GDP growth in 2004 is much smaller than the official German estimate of almost 0.6 percentage points.

1. Growth for all variables except change in stocks and foreign balance which are contributions to change in GDP (calculated as a percentage of real GDP in previous period) and memorandum items beginning with unemployment rate.
2. Employment and unemployment are according to national accounts concepts. Employment includes self-employment.

Source: OECD *Economic Outlook*, No. 75.

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Box 1.1. **The OECD's short-term projections for Germany** (*cont.*)

taxes will also adversely affect the propensity to consume. Rising foreign and domestic demand and gradually improving profits should lead to strengthening investment in machinery and equipment. Construction investment is projected to remain in recession over the projection period. All in all, in the OECD's spring projections, as published in the *Economic Outlook*, No. 75, GDP is foreseen to grow by 1.1 per cent, working day adjusted, in 2004 and as the upswing broadens at around 2 per cent in 2005, significantly above potential (see Table 1.2, including the note on working day adjustment). No rapid turn-around is expected on the labour market, with significant gains in employment only occurring in 2005. With the negative output gap remaining large, underlying inflation will remain low while headline inflation will continue to reflect increases in indirect taxes and administrative prices. Figures for the first quarter of 2004 indicate that exports accelerated somewhat more strongly than built into these projections, while domestic demand came in weaker. Export growth is therefore likely to turn out higher than in these projections, and it is likely that this will translate into somewhat higher GDP growth for the year as a whole. Recent oil price rises are likely to push inflation above the level shown below for 2004.

Risks to these projections largely arise from the uncertainty surrounding the recovery of world trade, the euro exchange rate and raw material prices in a context where consumer and investor confidence remain vulnerable. Exposure to external risks will remain significant but will be reduced once reform measures improve the resilience of the economy.

... while a turn-around in job creation has not yet occurred

The fall in overall employment since the beginning of the 1990s was predominantly attributable to employment reductions in manufacturing and construction. Over the same period, employment in the services sector increased significantly – reflecting the rise in the share of services in total value added – although this did not suffice to prevent the steep decline in the total volume of hours worked (Figure 1.5). Overall employment declined further in 2003 and into 2004, reflecting the stagnation in output. Employment in the services sector continued to grow, although at much lower rates than those seen in the years before. About 180 000 jobs have been shed in the course of 2003. Employment losses slowed, however, in comparison to the previous year, when employment declined by more than 500 000. The unemployment rate rose from 9.8 per cent in 2002 to 10.5 per cent in 2003,[2] but has levelled off since March 2003, while vacancies continued to fall (Figure 1.6). Some recently enacted measures, notably the stricter application of benefit eligibility rules for the unemployed, are likely to have increased the outflow from unemployment, although some of the exits from unemployment were into inactivity, which were further boosted by an increase of

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Box 1.2. The import content of exports is increasing

The main German exporting manufacturing industries appear to make increasing use of imported manufactured inputs of intermediate goods of the same industry. For example, between 1995 and 2000 the value of imports of chemical products used as inputs in the production of chemicals and chemical products in Germany grew by about 80 per cent, while the value added of the German chemical industry declined in nominal terms (at basic prices), according to national input-output tables. Imports of intermediate machinery and automobile products also rose more strongly than value added in each of these industries, respectively. Intermediary imports also grew faster than exports, with the exception of automobiles, and re-exports accounted for an increasing share of exports, although the levels remain modest (Table 1.3). Overall, exports net of all intermediate imports and re-exports grew less strongly than gross exports but more strongly than nominal GDP.*

Table 1.3. **Imports of intermediate goods, value added and exports**[1]

Per cent

	Chemicals and chemical products		Machinery and equipment		Motor vehicles, trailers and semi-trailers	
	1995	2000	1995	2000	1995	2000
Imports of intermediate goods as % of value added	29.0	54.2	11.2	18.5	24.3	36.8
Imports of intermediate goods as % of exports	19.6	23.7	9.8	13.3	15.6	14.7
Re-exports as % of exports	7.0	10.0	6.6	10.0	4.5	7.4

1. Imports of intermediate goods refer to the same goods as the column heading. For example in 1995, the ratio of imports of intermediate chemicals and chemical products was 29 per cent of the value added in the production of chemicals and chemical products.

Source: OECD using input-output tables (commodity groups) of the *Statistisches Bundesamt*.

For several years other industrialised countries, such as the US, have also been experiencing growing outsourcing activities of their enterprises and import penetration of their exports. This reflects the increasing degree of international economic integration. For example, large enterprises have outsourced business services, such as IT-related services, reflecting technological advances in communication technology as well as increasing international competition and openness to cross-border FDI flows. This process has helped enterprises deploy resources to those activities in which they are most productive and to gain access to foreign markets and develop new products. To reap the benefits of changing patterns of comparative advantage and minimise associated adjustment costs, policies should support competitive economic framework conditions and make room for structural change, for example by reducing barriers to company foundation and removing distortions in the tax system (see in particular Chapters 4 and 5).

* Nominal goods exports net of all imported intermediate products (including raw materials) and re-exports grew by 5.8 per cent annually on average between 1995 and 2000 while nominal exports grew by 9.4 and nominal GDP by 2.4 per cent.

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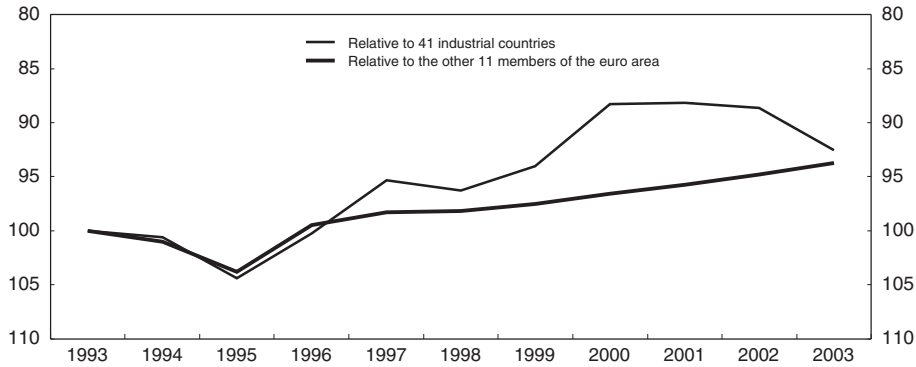
Figure 1.4. **GermanyÕs international competitiveness**
Real effective exchange rate, CPI based, 1993 = 100[1]



1. Calculated using weights which take into account the structure of trade. A rise (decrease in the index) indicates an improvement of the competitive position.

Source: OECD.

unemployed moving into early retirement. However, take-up of employment also rose, with the proportion of unemployed individuals moving to a job increasing by 9 per cent year-on-year. Easing of restrictions on employment of workers from temporary work agencies may have contributed to this development, improving matching efficiency. New schemes providing temporary subsidies for the unemployed moving into employment, notably for the unemployed to set up a business (*Ich* AG), account for part of the transitions from unemployment into employment, but may not bring about a durable reduction in unemployment, as most of the newly created firms are likely not to survive for more than a few years, and subsidy recipients may rejoin the unemployment pool once the subsidies lapse. The new scheme may therefore have shifted a part of the impact of weak economic activity on the labour market into the future.

... and convergence of the eastern Länder *to the west is progressing slowly*

While further progress has been made in convergence of per capita GDP and productivity of eastern Germany to the levels of the west, there is still no self-sustained upswing in the new states (see the 2001 *Economic Survey* for detailed analysis on the economic integration of the new states). In 2003 the new states recorded a positive growth differential after having undershot western German GDP growth in previous years.[3] On a per capita basis GDP has been expanding

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Table 1.4. **Trade by region**

	1996	1997	1998	1999	2000	2001	2002	2003	1996	2003
	Growth, per cent								Share, per cent	
Exports to[1]										
EU15	3.8	8.8	9.4	6.3	15.0	4.2	0.9	4.2	57.5	55.7
Of which: euro area	4.1	7.1	10.0	7.0	15.2	4.0	0.3	4.4	45.2	43.4
Central and eastern Europe[2]	17.3	28.9	12.4	–5.5	22.2	16.7	7.8	6.1	8.9	12.0
USA	10.1	27.5	17.1	12.1	20.1	9.8	0.6	–9.7	7.6	9.3
Canada	2.6	59.2	4.6	–5.7	27.0	22.2	7.4	–13.6	0.5	0.7
Japan	12.5	–3.4	–10.6	10.7	27.3	–0.7	–4.0	–5.9	2.7	1.8
Southeast Asia	5.8	9.0	–24.3	0.2	28.0	2.9	2.2	–2.9	5.6	3.7
China	1.0	–2.4	12.0	14.2	36.1	28.1	20.2	24.9	1.4	2.7
Other	1.7	14.1	5.9	–0.1	14.0	7.2	1.2	–1.6	15.8	14.1
Total	5.3	12.6	7.5	4.4	17.1	6.8	2.0	2.0	100.0	100.0
Imports from[1]										
EU15	3.6	9.2	6.5	3.7	14.3	1.1	–5.1	2.1	56.3	50.4
Of which: euro area	2.8	8.7	7.2	3.5	12.8	1.7	–4.5	2.9	45.4	41.0
Central and eastern Europe[2]	6.8	20.1	13.4	10.7	31.6	9.4	2.9	8.0	9.0	14.3
United States	9.3	19.3	15.7	5.3	28.1	–2.4	–12.2	–3.3	7.2	7.3
Canada	–7.0	3.1	7.7	–7.8	33.9	–4.3	–14.0	–16.2	0.7	0.5
Japan	–2.7	8.8	9.5	3.8	23.3	–14.7	–13.2	–3.8	5.0	3.6
Southeast Asia	0.7	12.2	5.5	4.4	35.0	–7.0	–6.0	–0.3	5.2	5.0
China	12.6	19.6	7.6	16.4	34.5	7.5	7.0	17.3	2.6	4.7
Other	3.9	13.1	1.2	5.0	25.4	1.9	–3.5	1.8	14.1	14.1
Total	3.9	11.8	7.3	5.0	21.0	0.8	–4.5	2.6	100.0	100.0

1. Special trade, according to the official foreign trade statistics (source: Federal Statistical Office).
2. Includes Czech Republic, Hungary, Poland, Russian Federation and Ukraine.
Source: Deutsche Bundesbank Zahlungsbilanzstatistik.

more rapidly in the new states for a couple of years, due to east-to-west migration combined with higher productivity growth. Downsizing of the construction sector continues to weigh on growth, although it decelerated somewhat in 2003. Services value added was held back by weak local demand. The lack of growth impulses from the west aggravate the situation. Unemployment continued to rise in 2002 and 2003, with the registered unemployment rate reaching 18.5 per cent last year, more than twice the western level (8.4 per cent). While the level of unemployment is higher, and despite larger reductions in the number of active labour market measures, unemployment has been rising by less than in the west. Migration explains some part of the differential development, and the number of residents in the east taking up jobs in the west appears to be increasing (Figure 1.7, panels A and B).

Since the beginning of the decade the productivity gap to the west, measured per employee, narrowedby 3.2 percentage points. Eastern German productivity stood at 72 per cent of the western level in 2003, although the differential is

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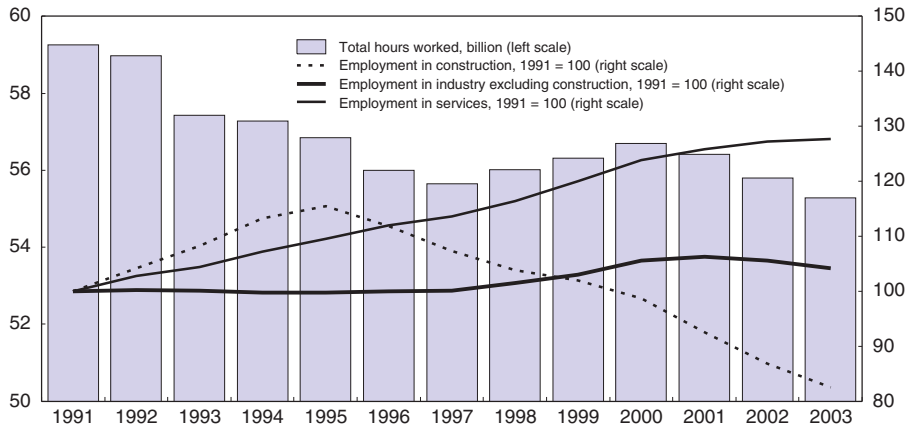
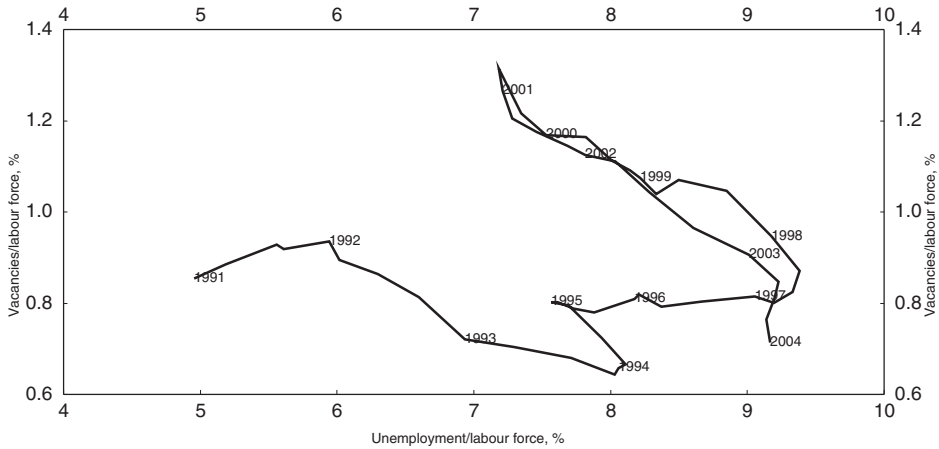
Figure 1.5. **Employment**



Source: OECD, *National Accounts*.

Figure 1.6. **Beveridge curve**[1]



1. Dates indicate the position of the first quarter of the year.
Source: OECD, *Quarterly National Accounts* and MEI Databases.

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Figure 1.7. **Economic activity in the old and new states**[1]

1. Old and new states exclude Berlin except for unemployment rates where total Berlin is included in the new states.
2. Variable for new states divided by variable for old states, expressed as a percentage. Labour productivity is GDP per person employed.

Source: Working Group of the Statistics Offices of the States; OECD.

slightly larger in terms of the number of hours worked (Figure 1.7, panel B). Since the second part of the 1990s the productivity catch up to the west has been slow, and on current trends, full convergence would take decades to be achieved. While high rates of firm creation made a major contribution to rapid productivity catch up in the first years after reunification, firm creation slowed thereafter and has approached lower western German levels. Both the survival and growth of newly founded enterprises even appear to be weaker than in the west. Convergence in output and productivity levels is proceeding at a brisker pace in manufacturing. In some export oriented sectors production is expanding at high rates. Competitiveness is being boosted by stagnant labour costs, as wage increases over the last couple of years remained substantially below pay rises set in collective agreements, in part reflecting the fact that the proportion of east German workers covered by collective agreements has declined substantially over the last ten years or so. By now unit labour costs in manufacturing might have fallen below the western German level. The productivity gap between the west and the east appears to be particularly large in business related services, as knowledge intensive services are less widespread in the new states. In part this relates to the fact that the relatively small size of eastern German manufacturing companies gives business services less room to exploit economies of scale. Also, regulation of some professional services has kept prices below western German levels.

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The EU accession of Central and East European countries poses additional challenges for the new states. The remuneration of labour in accession countries is much lower than in Germany *e.g.* in Poland, Hungary, and the Czech Republic labour costs per hours worked have been estimated to total between 13 and 22 per cent of the German level. While labour productivity is significantly lower than in Germany, unit labour costs substantially undercut those in Germany.[4] In addition, corporate taxes in accession countries are lower than in Germany, and the CEEs will benefit from regional funds of the EU. All of this reinforces the need to continue with structural reform in Germany overall, aimed at improving labour utilisation and enhance productivity growth.

Reunification and the structural problems of the economy have impacted strongly on public budgets

Between 1991 and 2003 general government debt in terms of GDP increased by more than 20 percentage points, to just over 64 per cent (Figure 1.8). Unification played a major role in this development, both in terms of spending increases and debt take-overs by the government sector. Annual transfers from western Germany to the new states – supplemented by funding from the EU – are estimated to total about 4 per cent of overall German GDP, financing roughly one third of total eastern German absorption (consumption and investment). While the size of the flows reflects the determination to create roughly equal living conditions across Germany, the marked increase in government debt as well as social security charges indicate deficiencies in the capacity to respond to new fiscal pressures through stronger prioritization of government spending. Total social security contributions increased from 36 per cent of the wage base at the end of the 1980s to 42 per cent in 1997, with non-unification related expansions in social benefits and the introduction of the long-term care insurance in 1995 adding to the increase. Social charges dropped slightly at the end of the decade, largely due to increases in indirect taxes that were earmarked to finance pensions, but charges rose again in 2002 and 2003. By 2003 57 per cent of general government outlays consisted of social spending, up from 48 per cent at the end of the 1980s, while investment accounts for 3 per cent of the spending share (5½ per cent at the end of the 1980s).[5]

The general government deficit in 2003 came in at 3.9 per cent of GDP, 0.4 percentage points higher than in 2002. The outcome exceeds the projected deficit on which budget plans were based at the end of 2002 by almost 1¼ per cent. According to the OECD cyclical adjustment method about half of the overrun is attributable to the fact that GDP growth was significantly lower than expected. The rest is due to the fact that the structural deficit of the general government fell by less than originally planned. While the government had aimed at reducing the structural deficit by 1 per cent of GDP, the actual drop is estimated to total only

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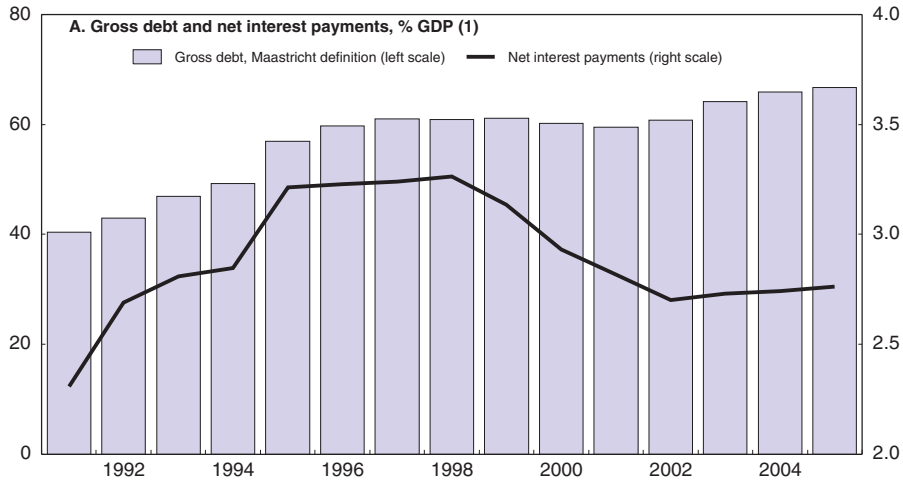
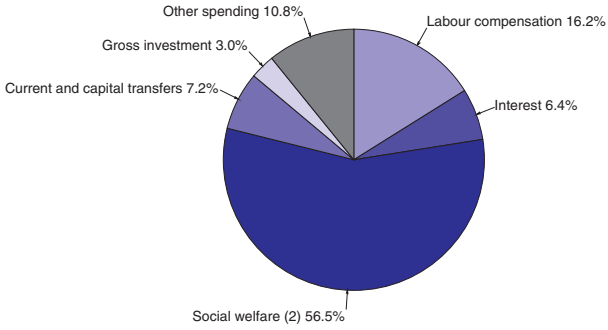
Figure 1.8. **General government debt and spending**



B. Composition of general government spending, 2003, % total expenditure



1. OECD estimates for 2004 and 2005.
2. Both benefits and benefits in kind.

Source: OECD, National Accounts and Economic Outlook Databases.

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0.5 per cent of GDP. The impact of the consolidation measures might have been over-estimated to some extent *a priori*. However, the shortfall is also attributable to other structural factors rather than a lack of implementation of announced consolidation measures by the German authorities. Tax revenues, both direct and indirect, turned out to be weaker than expected on the basis of lower economic growth. An unfavourable composition effect of GDP contributed to this outcome, with tax intensive private consumption falling by 0.1 per cent and largely untaxed stock building contributing 0.8 percentage points of growth. Moreover, the 2002 deficit was revised downwards, reducing the measured improvement in the structural balance relative to the budget plans.[6]

... while tax cuts have not yet paid off in form of higher consumption and investment spending

The reform of personal and corporate income tax that commenced in 1999 has led to substantial reductions in statutory tax rates in comparison to the mid 1990s and to some broadening of the tax base.[7] Personal income tax rates will fall further in 2005, when the last step of the present reform programme will have been phased in (Table 1.5). As has been noted in previous *Economic Surveys*, these steps mark considerable progress in reforming the tax system in favour of raising Germany's international competitiveness.[8] Nevertheless, effective average and marginal taxation of incomes and profits in Germany remains relatively high in comparison to major competitors and the new EU accession countries.[9] At the same time, many special rules of taxation and tax concessions imply that the deadweight costs of taxation are high and the tax code is complex. Thus, improving the attractiveness of Germany as a business location calls for reducing the distortions inherent in income and profit taxation by cutting tax concessions and subsidies while making further reductions in statutory tax rates. However, given Germany's high government debt, future tax reform needs to be fully financed so

Table 1.5. **Parameters of the personal income tax**

	Tax free minimum income	Threshold income for top-rate	Entrance tax rate	Top tax rate
	Annual, in €			
1999	6 681	61 377	23.9	53.0
2000	6 902	58 643	22.9	51.0
2001	7 206	54 999	19.9	48.5
2002	7 235	55 008	19.9	48.5
2003	7 235	55 008	19.9	48.5
2004	7 664	52 152	16.0	45.0
2005	7 664	52 152	15.0	42.0

Source: Federal government.

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as to conform with fiscal consolidation. Otherwise the credibility of fiscal policies would be likely to suffer and a reduction in today's tax burden might be perceived to imply new tax increases in the future. Indeed, empirical work by the OECD indicates that fiscal imbalances can be associated with offsetting movements in private saving that appear to be related to uncertainties about the sustainability of fiscal policies. The fact that in the recent past private consumption in Germany remained very modest despite substantial income tax reductions might partly reflect such aresponse.[10]

Significant steps in structural reform have been made…

Several measures that have recently been legislated within the government's "Agenda 2010" mark significant progress in structural reform. From 2005 onwards annual pension adjustment will be linked to a demographic factor. In the health care system co-financing of the insured has been increased and the scope for selective contracting between health insurers and health care providers been widened (Chapter 2). Disincentives for job search associated with certain features of unemployment related benefits are being reduced. This entails, in particular, a significant shortening of the eligibility period for unemployment insurance benefits for older employees, the combination of unemployment assistance and social assistance benefits into one single means-tested income replacement scheme for the long-term unemployed and a tightening of job search requirements. Restrictions for self employment in the crafts sector have been reduced to some degree so as to foster the birth of new establishments (Chapter 3). The thrust of reform is appropriate.

… but more coherence is necessary to create confidence

However, policy initiatives in 2003 seem to have had ambiguous effects on consumer and business sentiment. On the one hand, initiatives tabled in 2003 appear to have contributed to the upswing in business sentiment last year. Consumer sentiment remained weak, however, despite substantial income tax reductions in 2004. To a certain extent this is likely to relate to subdued labour market prospects and the fact that some measures weighed on disposable incomes, such as indirect tax hikes and cuts in tax concessions. Increased co-payments for health care services were also considered a burden.[11] However, to some extent a lack of clarity in the reform process itself, as reflected, for example, in short-term "emergency measures" in the pension system to avoid further hikes in pension contribution rates, might also have weighed on confidence. For economic performance to be raised in a durable way continuation of reform is indispensable. To fully reap positive confidence effects from structural reform it appears important that the rationale of future policy action is clearly signalled and much weight is put on the predictability and consistency of policy measures.

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… while policy challenges remain large

Fiscal consolidation needs to be given high priority and linked to public sector reform

Germany faces a considerable fiscal challenge, which is not only limited to balancing the budget within a few years, but, even more importantly, comprises dealing with considerable ageing related expenditure increases further ahead. This will require more restraint on the expenditure side. Related challenges include revisiting federal relations so as to improve the speed and transparency of federal decision making, better prioritisation of tasks within budgetary envelopes that are consistent with fiscal targets, reducing the distortions of the tax system, increasing the efficiency of public service, and better targeting of social security spending programmes. Furthermore fiscal policies need to rely more on market forces to bring about self-sustained growth in the eastern *Länder*. The respective recommendations are discussed in more detail in Chapter 2.

Obstacles to higher employment creation need to be removed

The economy's capacity to create employment needs to be raised. This includes reducing obstacles to higher labour force participation of older workers and women, a more efficient management of labour market policies, and allowing a higher degree of wage differentiation, at the firm level. An important aspect influencing demand for labour is how education, training and skills provided by education institutions contribute to the human capital of workers. The challenge in this respect is to improve regulations so as to increase the efficiency of the education system overall and to improve its capacity to respond swiftly to skill requirements of the economy. Major reform steps to make the German labour market more conducive to employment creation have been legislated and are being phased in. Notably, EPL has been eased. Benefit duration for older workers without job search requirements will be substantially reduced and unemployment and social assistance will be merged. These reforms need to continue and should be broadened. These challenges are investigated in more detail in Chapter 3.

Productivity growth needs to be strengthened

While German labour productivity growth is higher than in some other OECD countries, it is not high enough to compensate for the low growth contribution from labour. Entry barriers and administrative overhead hamper the creation of new businesses and mandatory membership in professional organisations together with legally binding fee schedules limit price competition. The challenge to foster competition on product markets is dealt with in Chapter 4. In order to maintain and further strengthen the innovative capacity of the economy it will be necessary to remove distortions in the tax system disfavouring innovative activities. Raising the efficiency of tertiary education will not only help improving the

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functioning of the labour market by reducing skill bottlenecks, but also the diffusion of new knowledge in the economy. Securing an efficient allocation of funds among various innovation programmes by more complete evaluation is a further challenge. Moreover, migration to Germany appears significantly biased toward low qualifications, and measures would have to be taken to raise the qualification profile of immigrants. The challenge to maintain and further strengthen the capacity of the economy to innovate is addressed in Chapter 5.

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Notes

1. See also OECD (2003a) and Deutsche Bundesbank (2002).
2. National unemployment definition. According to the internationally comparable concept, unemployment stood at 8.1 per cent in 2002 and rose to 8.7 per cent in 2003.
3. Berlin is excluded from the comparison.
4. Deutsche Bundesbank (2004).
5. To some extent the decline in the investment share is influenced by outsourcing of activities to the private sector.
6. For a similar assessment of the EU Commission see: European Commission (2004).
7. Already in 2001 Germany dropped its imputation system for corporate profits to reduce inequalities between the taxation of profits of tax residents and foreigners (see OECD, 2001).
8. See: OECD (1999) and OECD (2001).
9. However, effective taxation of personal incomes compares more favourable for families with single earners. See Chapter 2.
10. The savings ratio rose by one percentage point between 2001 and 2003, interrupting the earlier trend decline, and a further significant increase occurred at the beginning of 2004. Overall, this response absorbed a significant part of the phased income tax reductions that occurred over the period.
11. Increases in health care co-payments are likely not to be fully reflected in reductions of health care contributions, owing to health funds' efforts to reduce their debt.

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II. Linking fiscal consolidation to public sector reform

Over the last three years Germany's general government deficit has drifted upwards to almost 4 per cent of GDP. While part of the deterioration was caused by the cyclical downswing after 2000, it also reflected the fact that the underlying rate of growth is low. Following the phased income tax reductions, which contributed to the rise in the structural deficit, preference should now be given to balancing the budget over a limited number of years. For the consolidation process to have maximum beneficial effects on economic activity, much will depend on casting it within a consistent framework that links budgetary improvement to public sector spending reform.

The fiscal stance over the short and medium term

In 2004 and 2005 general government finances are subject to both income tax reductions and various consolidation measures on both the revenue and the spending side of the budget. The second step of the income tax reform foresaw phased tax reductions of some 0.3 per cent of GDP for 2004. Against the background of stagnating economic growth the government has envisaged a larger fiscal stimulus for this year, combined with budgetary consolidation measures. To this end, it tabled legislation that should both bring forward into 2004 the income tax reductions that were originally scheduled for 2005 and introduce new revenue-raising measures and cuts in tax expenditures. First inroads were made into reducing tax concessions for households. However, accelerated tax reductions as well as deeper cuts in tax concessions for private households were opposed to by the second chamber of Parliament. Finally, as a result of a parliamentary mediation process between the two chambers, only part of the tax reductions were brought forward into 2004. Simultaneously the size of tax expenditures to be cut was reduced. The compromise reached in the parliamentary mediation process reduces the volume of consolidation that would have been possible under the government's original proposal. This is so because the additional revenues foregone that would have been associated with a full shift of all tax reductions into 2004 would have been one-off while additional reductions in subsidies would have generated lasting savings.

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The size of cumulated income tax reductions this year and next has been estimated to total about 1 per cent of GDP. Overall, major revenue raising measures in 2004 and 2005 – some of which were legislated already in earlier years – comprise reductions in tax concessions for private households, tightening of depreciation rules for business, increases in the tobacco tax, and a tax amnesty to repatriate financial assets that had been transferred abroad for the purpose of tax evasion. Moreover, revenues benefit from a substantial expansion in corporation tax receipts, resulting from swings in the utilisation by corporations of tax refunds associated with earlier corporate tax reform (see the 2001 *Economic Survey*). At the same time, several measures have been introduced on the spending side of the general government budget. In the OECD's assessment government consumption will be almost flat in 2004 and increase little in 2005. Outlays for personnel will hardly increase in 2004 and 2005 as reductions in government employment are set to continue and supplementary pay for civil servants has been reduced. A profound effect on government consumption can also be expected from recent health care reform measures. Spending by health insurers is being curbed by increased co-payments of private households, reductions in health care services covered by the public insurance system and structural measures designed to improve the efficiency of the system (see further below). The larger part of these measures becomes effective in 2004, with some provisions implemented in 2005. Part of these savings is needed to repay the debt, which the statutory health funds have accumulated, and replenish the required minimum reserve. The remaining part is neutral with respect to the general government deficit – as in principle health care contributions are set so as to balance expected outlays – implying *ceteris paribus* corresponding reductions in social charges. Reductions in subsidies continue, notably for coal mining, and capital transfers are being reduced. The growth in monetary social benefits is also being curbed. Pension adjustment has been deferred by one year, and will be linked from 2005 onwards to the evolution of the number of contribution payers to pension recipients (see further below). Moreover, types of active labour market measures that have proven to be less effective in the past, are being reduced (see Chapter 3 below). Government investment is likely to decline further, although limited by additional funding by the federal government of investment in full day schooling.

All in all, the structural deficit is set to decline for the next two years. The consolidation plans of the federal government, as set out in the Stability Programme for the EU, are unlikely to be met, however. Estimates in spring 2004 of Germany's official tax estimation body indicate that tax revenues in 2004 and 2005 will come in significantly under the levels projected at the time the 2004 budget was legislated. In particular, the returns from the tax amnesty legislation appear to fall substantially short of the volume of five billion euro originally projected for 2004. Revenues will be further muted by the fact that economic growth in 2004 is likely to fall short of the initial official projections. In particular, relatively modest

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expansion of private consumption will weigh on VAT receipts. Preliminary information also indicates that reductions in contribution rates implied by the health care reform are likely to fall short of expectations. However, the health funds will record a surplus in 2004 reflecting the utilisation of revenues for debt redemption. On balance, the OECD projects that the structural deficit will improve by 0.3 per cent of GDP both this year and in 2005, less than officially projected due to expected revenue shortfalls. The overall deficit is likely to fall only little in 2004, to 3.7 per cent of GDP. In 2005 the deficit will drop further to around 3 per cent of GDP (Table 2.1).

The government's medium term consolidation plan, as laid out in Germany's Stability Programme to the EU, targets general government structural deficit reductions of 0.5 per cent of GDP annually. Based on the assumption that real GDP

Table 2.1. **Appropriation account for general government including social security**
Per cent GDP

	2000	2001	2002	2003	2004	2005
					Projections	
Current receipts	46.7	45.1	44.6	44.7	44.1	43.7
of which:						
Direct taxes	12.5	11.1	10.8	10.7	10.4	10.3
Social security contributions	18.6	18.5	18.4	18.6	18.4	18.2
Indirect taxes	12.0	11.9	11.9	12.0	12.1	12.1
Property and entrepreneurial income	0.6	0.7	0.7	0.5	0.4	0.4
Other current transfers receivable	2.9	2.8	2.9	2.9	2.8	2.7
Current disbursements	46.7	46.5	46.9	47.4	46.9	45.9
Of which:						
Government consumption	19.0	19.0	19.2	19.3	19.0	18.5
Property income payable	3.4	3.3	3.1	3.1	3.1	3.1
Subsidies	1.7	1.6	1.5	1.4	1.3	1.3
Social security outlays to households	18.7	18.8	19.4	19.7	19.6	19.1
Other current transfers	3.9	3.8	3.8	3.9	3.9	3.9
Saving, net	0.0	–1.5	–2.3	–2.7	–2.8	–2.2
Net capital outlays	–1.3	1.4	1.2	1.1	0.9	0.8
Gross investment	1.8	1.7	1.6	1.5	1.4	1.3
Net capital transfers received	1.5	–1.3	–1.2	–1.3	–1.1	–1.1
Consumption of fixed capital	1.6	1.6	1.6	1.6	1.6	1.6
Net lending	1.3	–2.8	–3.5	–3.9	–3.7	–3.1
Memorandum items:						
Net lending, Maastricht basis	1.3	–2.8	–3.5	–3.9	–3.7	–3.1
Cyclically-adjusted net lending (% potential GDP)	–1.6	–2.8	–2.8	–2.3	–2.0	–1.7
Debt, Maastricht definition	60.2	59.4	60.8	64.2	65.9	66.7

Source: OECD *Economic Outlook*, No. 75.

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grows by 2¼ per cent over the next three years, the government foresees reducing the deficit to 1½ per cent of GDP by 2007 (Table 2.2). However, several factors suggest that reaching the targets will require substantial further actions, and these need to be placed as part of a coherent consolidation strategy.

- First, not all measures are yet in place that would generate the envisaged reduction in the structural balance. Some of the reductions in subsidies still need to be implemented. Also, efficiency gains associated with social assistance reform may fall short of expectations. There appears to be a considerable risk that organisational reform of the Federal Employment Agency might not yet have gone far enough to ensure the administration of the new means-tested unemployment benefit that replaces social assistance (see Chapter 3).
- Second, over the last few years, restraining outlays for active personnel was a major source of budgetary savings. However, capping wage increases in the government sector is unlikely to be sustainable in the medium-term. Also, while government employment continues to be reduced since the beginning of the 1990s the rate of reduction eased over the last couple of years, indicating that the reorganisation of employment is increasingly difficult to achieve.
- Third, the deficit in 2005, as projected by the OECD, exceeds the government's target as set out in the Stability Programme by some 0.5 per cent of GDP.
- Fourth, there is a risk that weak economic activity may further depress future revenues. According to the estimate of the OECD potential growth stands at 1.5 per cent. While it is likely that actual growth will exceed potential in the economic upswing, average real growth of 2¼ per cent as

Table 2.2. **Medium-term fiscal objectives for the general government**
Per cent of GDP[1]

	2003		2004	2005	2006	2007
Total expenditure	48.9	(49)	48	46½	45½	44½
Total revenue	45	(45)	44½	44	43½	43
Net lending/borrowing	–3.9	(–4.0)	–3¼	–2½	–2	–1½
Government debt[2]	64.2	(64)	65	65½	65½	65
Memorandum item						
real GDP growth	–0.1	(0)	1½-2	2¼	2¼	2¼

1. Targets according to Germany's Stability Programme of January 2004. Figures in parentheses: official projections as of January 2004.
2. Definition of the Maastricht Treaty.

Source: European Commission.

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projected in the Stability Programme appears to be on the optimistic side. Moreover, the fact that growth is dominated by exports, with domestic demand remaining weak, depresses tax revenues. Indeed, the official tax projections of May 2004 indicate growing revenue losses relative to the assessment one year earlier until 2007.

– Fifth, aging-related spending pressure in the pension systems, including the system for retirement income for civil servants is building up, and provisions to cope with this development are only partially in place (see below).

Overall, for significant reductions in government debt to be realised in the medium-term more profound public sector spending reform will be required, based on strict evaluation and prioritising of government activity.

The framework for fiscal policy needs to be improved

As was outlined in the special chapter on public sector spending reform in the 2003 *Economic Survey*, linking fiscal consolidation to public sector reform requires a more rigorous approach to evaluating and prioritizing public sector spending projects. Improving the budgeting process itself and rebalancing intergovernmental fiscal relations are instrumental for progress on these counts. The latter also play an important role in determining the speed of response to policy challenges.

Towards better prioritization in the budgetary process

Against the background of the various fiscal pressures that are impinging on the general government budget strict prioritization of public sector spending projects is required for the sustainability of public sector finances to be secured. In the past, fiscal adjustment on all levels of government was largely made *via* cutting back public investment. While this is much easier to accomplish than reforming entitlement spending, the savings associated with curbing investment are relatively limited in the medium term and might be counter-productive (see Figure 1.8 in Chapter 1). Hence, such action needs to be evaluated against alternative spending strategies. Cuts in government employment were another major source of savings over the last decade or so. For these to be efficient choices on government tasks need to be made.

The present German budgeting system places little emphasis on policy outcomes. Legislation focuses on parliamentary control of inputs as opposed to budgetary appropriations on a programme or activity basis.[1] In contrast, since the 1980s, many OECD countries, such as Britain, Northern European countries, New Zealand and the US, have relaxed input controls and reoriented budget systems to focus on results.[2] In exchange for greater flexibility in budget management, in

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this approach decentralized managers are now more responsible for achieving results from a relatively small number of government programs. Accordingly, budget laws have been changed to simplify the structures of budget appropriations approved by Parliament. For example, in France full introduction of a programme-oriented budget is scheduled for 2006, with only about 150 programmes ("budget lines") to be approved by Parliament. Accrual accounting is supplementary to an outcome-based budgeting approach in that it makes the full cost of government activities more transparent, thereby improving decision making by using this enhanced information.[3] German legislation requires that budgeting is based on the cash rather than the accruals principle. While some moves were made by states and communities to introduce elements of accruals accounting, against the standard required by legislation these efforts are supplementary accounting practices and therefore unlikely to be adopted on a general scale within the present legal system.

Hence, facilitating the setting of spending priorities by policy makers suggests significant revisions of the input-oriented budget codes at the levels of the federal government and the states so as to adopt output oriented budgeting. This should entail basing budgetary appropriations around programmes, within a simplified budgetary framework, associated with the requirement to supply to parliament analysis of the costs and the benefits of such programmes. Within this setting, ministries would be held responsible for the programme management. This move might as well imply a re-orientation of the qualification structure within ministries with more emphasis on economic rather than legal issues as is presently the case.

Federal fiscal relations need to be untangled to reinvigorate decision making

With the receipts of almost all major taxes being shared between the different layers of government, modifications in the tax regime, including cuts in tax expenditures, require the approval of the second chamber of Parliament. Moreover, both the high degree of tax sharing and the significant degree of co-financing across government levels allows regional governments in principal to condition their approval to federal fiscal legislation on concessions in federal legislation that does not require the approval of the B*undesrat*, as experienced in the recent legislative process. Key elements of the government's recent legislative proposals that were scheduled to become effective in January 2004 were only decided in mid December 2003. This concerns legislation to bring forward income tax reductions into 2004 and to cut subsidies and tax expenditures. In each case, legislation was delayed by a complicated mediation process between the Bundestag and the L*änder* chamber of Parliament (B*undesrat*), which involved other draft legislation as well, such as deregulation in the handicraft sector and the combination of social assistance benefits and unemployment assistance benefits into one single means-

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tested income support scheme. In some instances the outcome of the mediation process proved hard to predict. For example, while some states questioned the appropriateness of bringing forward income tax reductions into 2004, given the need for fiscal consolidation, the volume of tax expenditures to be cut according to the legislative compromise is smaller than originally proposed by the federal government.

The organisational reform of the Federal Employment Agency (*Bundesagentur für Arbeit*), which commenced in spring 2003, stands as another example. The design of reorganisation is closely linked to the decision whether the responsibility for the provision of social assistance benefits would continue to rest with the communities or shift to the FEA as originally envisaged by the federal government. However, as with other legislation, the pertinent legislation passed the second chamber of Parliament only in December 2003, without a workable clarification of the issue. Instead, the new legislation introduced a new element of split responsibilities between the FEA and regional governments, which is likely to constitute a serious hurdle for efficient income replacement for long-term unemployed (see Chapter 3 below), and whose implementation arguably raised difficult legal questions. By spring 2004 follow-up legislation on the issue was still outstanding.

Hence, untangling the responsibilities of the federal government and the states and communities is one of the most pressing tasks in order to make meaningful progress on budget and structural reform. Reducing the degree of co-financing between the different levels of government and introducing a higher degree of tax autonomy for the *Länder* and communities, as has been proposed in earlier *Economic Surveys*, would go some way in this direction.[4] In October 2003 both chambers of Parliament commissioned an expert commission to make reform proposals, within a year, designed to improve the allocation of responsibilities within the federal system and to foster the efficiency of public sector actions. Federal fiscal relations, in particular issues of co-funding and taxation should be set on the commission's agenda. Moreover, constitutional reform, limiting the power of the *Bundesrat* to block federal legislation should be considered.

Important fiscal policy areas

This part of the chapter highlights a number of important policy areas where spending reform can be expected to contribute substantially to fiscal consolidation, allowing the tax burden on labour, in particular, to be reduced. In some areas – pensions and the health care system as well as income taxation – significant measures were recently legislated, but reform should continue. Other fields should be addressed soon. This is true for readjusting state aid for the new states, and responding to the looming fiscal pressure from pensions for civil servants and the long-term care system.

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Further tax reform should focus on reducing distortions

Income and corporate tax reductions, phased in since 1999, have been substantial. Moreover, German personal and corporate income tax revenues relative to GDP belong to the lowest within the OECD (national accounts basis). However, with social charges added in – which are high by international comparison – the tax-to-GDP ratio is somewhat above the OECD average (Table 2.3). At the same time, there is evidence that the effective taxation of labour and profits are relatively high by international comparison. The co-existence of subdued income and corporate tax receipts in terms of GDP and high effective taxation points to a substantial perforation of Germany's tax base. Thus, there remains much scope to foster economic activity by reducing the distortions inherent in the German tax system by broadening the tax base and lowering statutory rates.

Within the OECD Germany belongs to the group of countries with high tax wedges between gross wages paid by employers and the workers' take home pay net of personal income taxes and social charges. This is true for a wide wage band around the pay of an average production worker, although effective taxation is

Table 2.3. **General government tax revenue, 2002**[1]
Per cent of GDP

	Tax without social contributions			Social contributions	Total tax including social contributions
	Total	Direct[2]	Indirect[3]		
Denmark	46.9	29.4	17.5	2.7	49.6
Sweden	35.0	18.1	16.9	15.4	50.5
Finland	33.1	19.4	13.8	12.4	45.5
Australia	31.4	17.8	13.6	0.0	31.4
Belgium	30.6	17.7	13.0	16.7	47.4
Luxembourg	30.1	16.3	13.9	12.3	42.5
Canada	29.5	16.1	13.3	5.1	34.5
United Kingdom	29.2	15.5	13.6	7.4	36.6
Italy	29.0	14.3	14.7	12.8	41.8
Austria	28.9	14.0	15.0	16.5	45.4
France	26.6	11.5	15.1	18.2	44.8
Portugal	24.7	9.7	15.0	12.2	36.9
Netherlands	24.6	12.0	12.6	14.9	39.5
Greece	24.1	9.5	14.6	14.1	38.2
Ireland	23.7	11.6	12.1	5.7	29.4
Germany	**22.6**	**10.8**	**11.9**	**18.4**	**41.1**
Spain	22.6	10.9	11.6	13.6	36.1
United States	19.2	11.9	7.3	7.2	26.4
Japan	16.2	7.9	8.3	10.4	26.7
Average (unweighted)	27.8	14.4	13.3	11.4	39.2

1. Ordered by total tax without social contributions.
2. Current taxes on income, wealth, etc.
3. Taxes on production and imports.

Source: OECD, *Annual National Accounts*.

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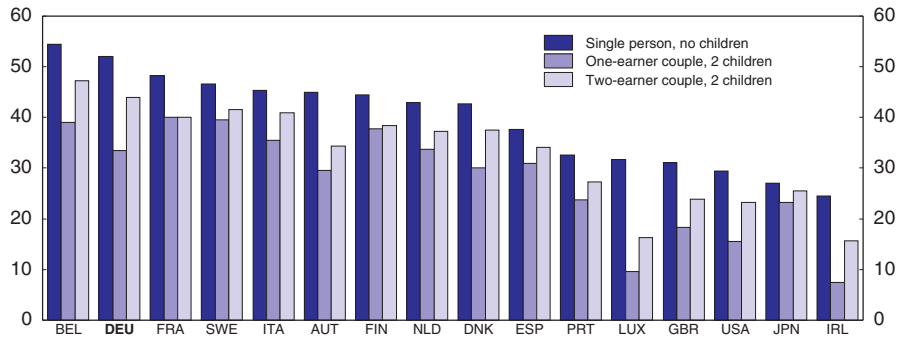
more favourable for families with single earners (Figure 2.1).[5] High social charges are mainly responsible for this outcome. Also, according to a study comparing six EU countries, Switzerland and the US up to the recent income tax reductions Germany belonged to the countries with the highest average effective taxation of wages for high-qualified personnel.[6] This position is set to improve, however, after the income tax reductions will have been fully phased in 2005. Moreover, various studies indicate that effective taxation of profits at the level of corporations is high by international comparison, which is important for location decisions. A study by the German Council of Economic Experts, which is based on 2003 tax codes covering six EU countries, concludes that Germany has the highest average effective taxation of the returns of investment both if the investment project were carried out domestically or across the border by a non-EU parent company. Germany also ranks in the group of countries with the highest marginal effective taxation of profits.[7] To some extent this outcome is influenced by a temporary increase in the German corporation tax rate by 1.5 percentage points in 2003 for the purpose to finance additional public sector spending associated with the flooding in autumn 2002. This measure has been reversed at the beginning of 2004. However, tax reductions in other EU countries have deteriorated Germany's relative position since the implementation of the business tax reductions 2001. In addition, corporate taxation in the countries that joined the EU in May 2004 is considerably lower than in Germany.[8]

Figure 2.1. **Effective taxation of an average production worker, 2003**[1]

Per cent of labour costs



1. Average production worker's gross wage for first earner, 67 per cent of this wage for second earner. The ratio is income tax plus employee and employer social security contributions less cash benefits as a percentage of labour costs and is also known as the tax wedge. Refer to source for details of changes in concepts and methodology.

Source: OECD, *Taxing Wages 2002-2003*.

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Cumulated tax expenditures (TE's) that are reported as subsidies – net of all tax concessions in favour of retirement savings – have been estimated to total some 1.6 per cent of GDP (€ 34 billion) in the personal and corporate income tax system for 2004, and another 0.9 per cent of GDP in other (indirect) tax codes.[9] Implied distortions of economic activity are likely to be substantial across various dimensions such as economic sectors, modes of financing investment projects and the labour-leisure decision. A large share of TE's accrues, directly or indirectly, to the construction sector. For example, the single largest tax concession for households, whose value is estimated to total € 7.5 billion in 2004, supports the construction of owner-occupied housing (*Eigenheimzulage*). Tax preferences in favour of the construction sector are likely to have contributed to the share of German construction investment in investment overall being relatively high by international comparison. While TE's were a major force behind the over-dimensioning of the eastern German construction sector, they presently interact adversely with substantial over-capacity in both office space and residential construction in Germany overall. Other sectors that are major beneficiaries of TE's comprise agriculture and fishery, mining, and transport. Examples for other major TE's linked to personal income taxation are the Saver's Tax Break (*Sparerfreibetrag*), which allows to deduct a fixed sum of capital income from the tax base, and a tax exemption of wage *premia* for work at Sundays, holidays and at night time. While the former affects the savings-consumption choice, the latter biases the labour-leisure decision of employees. The various special provisions make the German tax system overly complex and encourage rent seeking. According to a recent study on the personal distribution of income tax liabilities about half of the tax payers with positive annual income exceeding € 250 000 claim negative income from rents and leases (*Vermietung*, *Verpachtung*). Moreover, the size of negative income from these sources was found to increase with the level of positive income overall. This finding suggests that the losses are motivated by rent seeking, based on generous depreciation rules.[10] The complexity of the tax code also increases start-up costs of small business and makes tax auditing more difficult.

Over the last two or three years proposals for more radical income tax reform – aiming at profound reductions in the tax burden and substantial simplification of the tax system – attracted increasing attention in the public policy debate (Box 2.1). Revenues foregone – relative to the 2004 tax legislation – implied by a hypothetical introduction in 2005 of one of the major tax reform proposals, without a broadening of the tax base, have been estimated to total between 2.0 and 2.4 per cent of GDP (€ 43.7 and € 52.2 billion).[11] Given Germany's high level of debt and a deficit-to-GDP ratio that was close to 4 per cent in 2003, further unfinanced tax reductions might increase government debt further beyond sustainable levels and would be at variance with the government's consolidation targets. This risk is reinforced by the fact that the revenues foregone by changes in the tax system are hard to predict and were often substantially under-estimated in the past.

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Box 2.1. Recent tax reform proposals

The point of reference for the existing system of German income taxation is a comprehensive income tax with – ideally – taxation in the country of residence and no distinction being made between different types of income in either the tax rate or the tax base. The actual system deviates from this benchmark in several important respects. In particular, there are separate regimes for personal and corporate income taxation, which imply distortions with respect to both the firms' legal status (corporation versus partnership) and their financing decisions (credit versus equity financing and internal versus external financing). The replacement in 2001 of the imputation system for dividends by the half-income tax relief facility (*Halbeinkünfteverfahren*), which was motivated by reducing the differences in effective taxation of profits generated domestically or abroad, implied that distributed profits and personal income from other sources are not taxed at the same rates for all income levels.[1]

Major reform proposals for the income tax system include proposals by Parliamentary opposition parties, the German Council of Economic Experts and a university research institute.[2] They have in common that they envisage large cuts in statutory tax rates and a broadening of the tax base *via* substantial reductions in tax expenditures. According to some proposals Germany's continuous, formula-based, statutory income tax rate would be replaced by a step function with three tax rates. Depending on the proposal, the entry rate, above a tax-free income level, and the top rate vary between 12 and 15 per cent and 25 and 36 per cent, respectively. The proposals are less explicit to what extent tax expenditures would be cut. Major discrepancies between the systems suggested arise in the treatment of different types of income. While the Council of Economic Experts suggests introducing a dual income tax system with a uniform flat rate for all types of capital income and progressive taxation of earnings, the other proposals continue to be oriented at the benchmark of comprehensive income taxation. It is a major benefit of the proposal by the Council that it would terminate various distortions in the taxation of capital income. The system would also allow maintaining progressive taxation of earnings while setting tax rates for capital income at internationally competitive levels that take into consideration the high mobility of capital. A number of OECD countries has adopted dual income tax systems of this type, notably Denmark, Finland, Norway and Sweden. One of the trade-offs would be, however, that dual income taxation creates incentives to transform labour income into capital income (notably for self-employed and small partnerships),particularly when the gap between the rate of capital taxation and the top rate or earned income is large.[3]

1. Under the half-income tax relief facility (*Halbeinkünfteverfahren*) only half the dividends are taxable so as to reduce double taxation of profits at the levels of the corporation and individual share holders. See OECD (2001).
2. For a summary of the tax reform proposals see Boss (2003).
3. See Heady and van den Noord (2001).

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Hence, future tax reform should focus on reducing the distortions in the tax system by slashing tax expenditures and direct subsidies, associated with cuts in statutory tax rates. Certain state aid could be cut immediately without an imminent need to allow for extended adjustment periods. An example would be tax exemptions for work at night-time, Sundays and holidays, where a possible compensation for losses in disposable income would be left to wage bargaining. A list of such items should be identified and rapidly implemented. Other subsidies to be cut should be phased out over a few years, allowing the economy to adjust. Reductions in statutory tax rates should be phased in a way that secures compatibility with fiscal consolidation. Moreover, strengthening Germany as a business location would have to take into consideration that high-qualified labour is increasingly becoming mobile as well. This is particularly relevant in view of the fact that the federal government aims at strengthening the innovative capacity of the German economy (see Chapter 5 below on innovation). Hence, taxation of labour needs to be set at competitive levels as well, reinforcing the need to slash a large volume of tax expenditures and subsidies or other spending to reduce the tax burden for both labour and capital. While making the income tax schedule a step function would increase the transparency of the system, this would be of secondary importance compared to reducing tax expenditures.

Government spending in the new states needs to be adjusted

The volume of transfers devoted to the new states, both to the private and the public sectors, has been estimated to total € 125 billion since unification although attaching a precise figure is difficult. At the same time, government debt increased much faster in the new than in the old states, having surpassed the per-capita level of the western states. Government investment, by contrast, has been steadily declined since 1994 (Figure 2.2). Given that growth in the new states is not yet self-sustained and unemployment remains persistently high (see Chapter 1), the debate in Germany on how to adapt the support strategy to attain higher levels of activity in the east has recently intensified. In this context it has been suggested to give the new states the status of a "special economic zone", which would be exempted from certain regulations, such as stringent employment protection legislation. However, regulatory reform is necessary for total Germany so as to raise the growth potential for the country as a whole. Not least, the new states would strongly benefit from a strengthening of economic activity in the west. As a second-best solution, some discretion might be given to economically weak areas in Germany to establish pilot projects of regulatory reform that could be generalized at a later stage to the country as a whole. The government plans action along this line.

In a special chapter on the economic integration of the new states in the 2001 *Economic Survey* the OECD has argued that market forces needed to be

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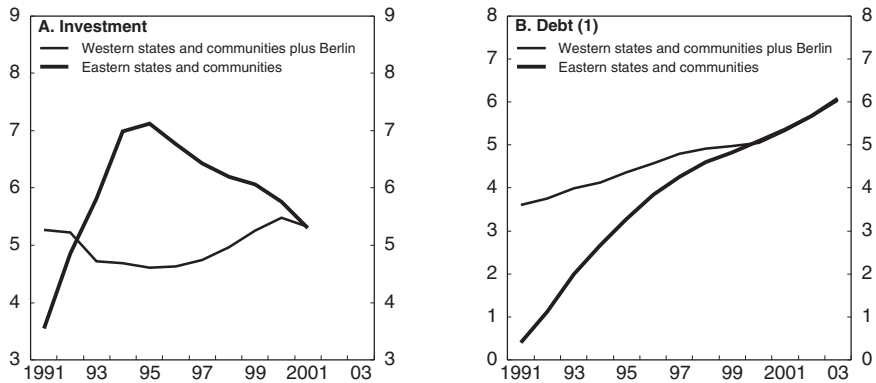
Figure 2.2. **Government debt and investment of the new states**
Thousand euros per inhabitant, current prices



1. Population for 2003 is estimated as equal to the 2002 level.
Source: Federal Statistics Office; OECD.

given much more scope than in the past to bring about adjustments in the eastern German economy that would generate higher growth and employment. Regarding government spending, important issues include:

– First, special investment subsidies to enterprises in the new states (*Investitionszulagen*) hamper fiscal consolidation while they do not appear to support the recovery of eastern Germany. Total factor productivity (TFP) has been estimated to stand at some 68 per cent of the level in western Germany.[12] Since, on the other hand, production technologies in industry were found to come close to those in the west, the gap in TFP is likely to reflect structural weaknesses of the eastern German economy that are unlikely to be remedied by subsidizing capital accumulation. The fact that despite massive government support investment per capita has fallen below the level in the west is further evidence for this view. Extended over a long period, subsidies are becoming more and more ineffective in reaching targets, and subsidy dependency increases. Indeed, the OECD has argued in the past that persistently high subsidisation to industry and the construction sector has significantly biased the structure of the economy and reduced its adaptability to market forces.[13] Some progress has been made in reducing subsidies. The trend toward applying the same set of rules for subsidization to the whole of Germany should be continued and reinforced. Phasing out special rules

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for subsidies in the east should be announced soon to trigger the necessary adjustment on the side of the enterprises. Credibility of announced policies is important for effective adjustment. Earlier plans to terminate special subsidisation have been repeatedly over-written, the prolongation of investment subsidies until 2006 being the most recent example (termination was originally scheduled for 2003). Unified rules for total Germany would still imply that the new *Länder* would obtain more support per capita, given the structural weakness of the economy.

– Second, notwithstanding the importance of infrastructure endowment for economic development, a general link between infrastructure investment and economic growth is hard to establish. Rather, links between infrastructure investment and productivity depend on the specific projects chosen and their degree of complementarity to private sector investment. Given the major progress in the infrastructure endowment in the new states since reunification it is therefore no longer appropriate to base infrastructure investment on east-west comparisons. For at least a decade infrastructure spending by the new states has been biased in favour of culture, social services, housing, health and central administration– areas which cannot be considered bottlenecks for economic growth.[14] Hence, infrastructure investment projects need to be prioritized on an individual basis, based on cost benefit analysis.

– Third, transfers to the new states that are extended for the purpose of infrastructure investment ("Solidarity Pact II") are in fact more than half used for consumption rather than investment.[15] To some extent this reflects the fact that government budgets in the new states are suffering from low tax revenues, which is being aggravated by the long phase of economic stagnation in Germany overall. Putting finances on a sounder footing requires, however, that government consumption be curtailed. In particular, the size of the government sector needs to be reduced further. Government employment per inhabitant in the new states exceeds the level in financially weak western states by one fourth. Also, the Solidarity Pact should include as an additional element a binding obligation for the new states to produce ex ante and ex post evaluations of spending projects that make transparent for what purpose certain projects are pursued and others not.

Reform of the general pension system needs to continue…

The 2001 pension reform aimed at confining spending in the public pay-as-you go system for the purpose of breaking the trend increase in pension contribution rates for the next several years and mitigating the impact on non-wage labour costs in the medium and long term when the old-age dependency ratio will

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increase steeply. The two major elements of the reform were a phased reduction in the system's net replacement ratio *via* changes in the indexation of pension adjustment and the promotion of private pensions to offset the impact of lower public pensions on income in retirement (see the 2001 and 2003 *Economic Surveys*). While the contribution rate was targeted to remain below 20 per cent until 2020 and 22 per cent until 2030, the legislation contained a conditional clause obliging the federal government to prepare more wide-reaching measures if it became evident that these ceilings would not be kept. In 2003 the system came under unforeseen pressure from both the spending and the revenue side, largely on account of weak economic growth, and it became clear that without further action contribution rates would have to rise above target already in 2004. Moreover, a new study, commissioned by the government from an independent expert commission, suggested that without further reforms in the medium and long term pension contribution rates would increase to 24.2 per cent in 2030 and 25.6 per cent in 2040, overshooting the government's target by a wide margin.[16] In the same vein, the Social Advisory Body on the government's pension policies recommended further structural reform to secure the viability of the pension system, suggesting *inter alia* that the statutory retirement age should be raised from the next decade onwards.[17]

The government responded by tabling two sets of legislation, both of which have been adopted by Parliament in the meantime, following a parliamentary mediation procedure between both chambers of Parliament:

- First, "emergency measures" were implemented for the purpose of stabilising contribution rates in the short-term. In particular, pension adjustment is being deferred from 2004 to 2005, the pensioners' contributions for long-term care insurance are fully charged on pensions (replacing the equal splitting of funding between the pensioner and the pension insurance), and the reserve requirement of the pension system has been reduced to 20 per cent of monthly pension outlays (from 50 per cent in 2003).
- Second, parliament passed measures linking pension adjustment from 2005 onwards to the ratio of dependent employees relative to the number of pensioners, and increasing the minimum age for early retirement on account of unemployment from 60 to 63, to be phased in between 2006 and 2008. On the other hand, the government decided to defer to 2010 the decision on whether or not to increase the statutory retirement age.

Moreover, legislation is being prepared to simplify eligibility criteria for support of funded pension savings (*Riester-Rente*). Finally, from 2005 onwards, public pensions will be subject to a new tax regime, in which contributions are tax exempt in the investment and accumulation phase while pensions will be taxed (EET). At present, the employees' pension contributions to the public pension

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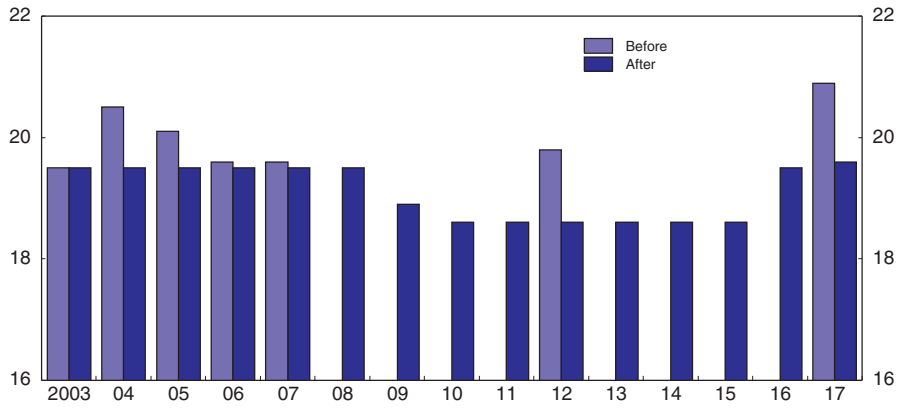
insurance are tax free only up to a certain threshold, and pensions are largely untaxed. The new regime, which was instigated by a ruling of the Federal Constitutional Court, is designed to harmonise the taxation of public pensions with pensions for civil servants, since the latter are being taxed in full.

Linking pension adjustment to contributions is an important step to put the public pension system on a sustainable base. According to simulations by the federal government, the new legislation would stabilize pension contribution rates at 19.5 per cent of the wage base until 2015, falling temporarily to 18.5 per cent between 2010 and 2015, and would rise to 19.6 per cent in 2017, while reserves would be increased over the next years to a level of 0.9 of monthly spending (Figure 2.3).

Projections of contribution rates based on the government-commissioned study confirm that the legislated measures are likely to cap contribution rates at the present level over the next 10 years and at 22 per cent until 2030. The targeted pension contribution rate of 22 per cent in 2030 would however still entail a substantial increase in the tax burden on labour which is likely to coincide with increases in other social charges, notably for health care and long-term care. Moreover, there is a risk that future increases in charges will be larger than projected today. The fact that emergency measures were legislated to stabilise the contribution rate in 2004, two years after the previous pension reform, stands as a recent example. Indeed, the targeted cap of contribution rates at 22 per cent would, on

Figure 2.3. **Projected pension contribution rates before and after the reform**

Per cent of gross wage



Source: Federal Government.

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the basis of current projections, be passed by 2030 if the effect of the deferred adjustment in 2004 were partly offset by new adverse developments on the spending or revenue side. Further emergency measures would increase perceived uncertainty about future pensions and fiscal policies more generally. Hence, additional measures to curb the future rise of pension contribution rates should be taken soon, ensuring that reaching the government's target of 22 per cent will not require further unforeseen action. In fact, the need for adjustment that policy makers are currently facing would have been less if more determined action had been taken over the last 15 years.[18]

This suggests that a phased increase in the statutory retirement age – implying that a given reference replacement rate at retirement is reached only at a higher age – should be announced soon so as to rebalance the portions of the lifetime spent employed and as retiree. The average duration of retirement increased between 1960 and 2002 by 66 per cent, to 14.3 years for men and 18.3 years for women. Moreover, life expectancy is estimated to rise by close to 3 years between now and 2030. Early and gradual introduction of the measure would help to ensure that the larger number of older workers is absorbed into employment. In addition, decisions to curb old-age-related spending might become politically more difficult to make in the future than today. A rise of the statutory retirement age from 65 to 67 years, phased in between 2011 and 2025 as proposed by the government-appointed commission, has been estimated to lower pension contribution rates by 0.6 percentage points in 2030. Statutory retirement above the age of 67 is already in place in other OECD countries such as in Norway and the US.

In addition, discounts and bonuses in case of earlier or later retirement should be made more actuarially neutral. At present discounts are not large enough to remove financial incentives to retire before the statutory retirement age.[19] OECD estimates suggest that achieving actuarial neutrality with respect to the individual's pension wealth could raise the overall participation rate in Germany by between 0.5 and 1.2 per cent in 2025.[20] Within an actuarially fair system the employees' choice for earlier or later retirement around the statutory retirement age would be facilitated by establishing individual retirement accounts allowing the insured to identify their claims given their elapsed employment and contribution histories. Benefits that are unrelated to individual contributions by the recipients are best financed out of general tax revenues rather than wage-based contributions. Moreover, effective early retirement on account of unemployment-related benefits needs to be curbed further. Measures to remove disincentives to work among women would also help offset the adverse effects of ageing on labour supply (see Chapter 3 below).

The government has taken further steps to strengthen private household savings for retirement, but policies could become more consistent in pursuing this objective. Steps to simplify application procedures for a subsidized private pen-

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sion provision scheme that became effective in 2002 (R*iester*-R*ente*) will help boost hitherto modest participation in the scheme. For example, participants will only have to go through application procedures when they enrol in the scheme, rather than every year. However, a planned regulation requiring equalisation of *premia* for men and women might discourage the R*iester* saving plans.

More generally, owing to the large fiscal cost of the subsidies to private retirement provision in future years, as take-up of the scheme will increase, the government should evaluate to what extent the subsidies induce households to save more rather than substitute between different savings instruments. Indeed, relatively generous pension provision in Germany suggests that substitution effects might be considerable.

... while reform of retirement income systems for civil servants needs to begin now

A major building block of the government's consolidation policies over the last years has been the containment of outlays for active personnel. For many years in a row employment at the general government level declined at an annual rate of 1.5 per cent or more, validating the federal government's target for the federal administration. Over the last couple of years, however, employment reductions eased. Wage restraint added to containing government consumption. As has been highlighted in the 2003 E*conomic* S*urvey* steps have also been made to improve human resource management, although more is required. Public sector reform should also embrace dealing with the mounting pressure stemming from personnel outlays for inactive civil servants.

Retirement incomes for tenured civil servants (B*eamte*) are paid out of the governments' budgets rather than the general public pension scheme. Also, non-tenured civil servants (A*ngestellte*) obtain a supplementary pension in addition to the pension obtained by the general public scheme. Combined outlays total some 1.5 per cent of GDP in 2004, net of pension payments for former employees of the Federal Railways and the Post Office (0.4 per cent of GDP), for which the federal government also pays a share. Outlays accelerated recently, growing by more than 40 per cent since 1995. Pension outlays and dynamics are uneven across the government sector. Spending of the states and communities totals more than four times that of the federal government and rose faster in the past. To a considerable extent the past increase in overall outlays is attributable to pay increases and changes in the qualification structure of employees towards higher qualifications, increasing pension outlays per person. However, changes in the age structure of the employees and earlier retirement are also at work. The average retirement age of tenured civil servants is about 60 years, one year above that in the general public scheme. Early retirement programmes associated with the incorporation of the Post Office and the Railways have contributed to higher outflow into inactivity, although they terminated at the end of the 1990s. At present,

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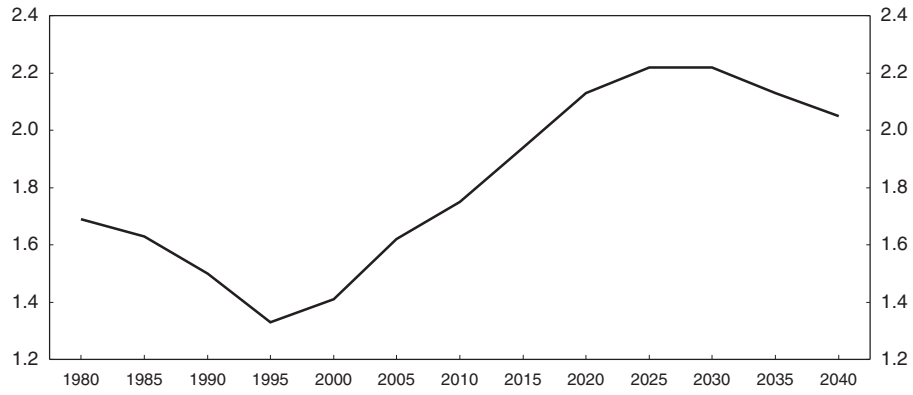
only some 18 per cent of civil servants retire at the statutory age of 65. However, the average age of active civil servants in the public sector overall increased continuously over the last years – despite early retirements – standing at 42 years in 1999.

Over the next years and decades to come the ageing of the public sector workforce will imply a steep acceleration in retirement spending. Based on projections by the federal government on transitions into retirement and adjustment of pensions, and assuming nominal GDP growth of about 3.5 per cent until 2009, in line with the OECD's medium-term baseline scenario, and 3 per cent thereafter, pension spending for civil servants would increase from some 1.5 per cent of GDP presently to 2.2 per cent by 2025 (Figure 2.4).[21] Supplementary pension outlays for non-tenured civil servants are projected to accelerate more vigorously than those for tenured employees. Also, outlays will increase much faster on the level of the states and communities than on the level of the federal government. Within the next ten years the former is projected to rise to seven times the level of the latter. In the longer run the impact is particularly pronounced for the new states. At present outlays for inactive civil servants, net of obligations stemming from the time of the GDR, are relatively modest in eastern Germany (€ 0.3 billion). Spending is projected to increase gradually to about one billion euro by the end of the decade and to accelerate thereafter, with the level in 2040 exceeding eleven times the one projected for 2010.

Figure 2.4. **Retirement outlays for civil servants**[1]
Per cent of GDP



1. The scenario contains OECD projections for 2005 and assumes an annual growth of nominal GDP of 3.0 per cent from 2009 to 2015 and of 2.7 per cent from 2016 onwards.

Source: Federal Government; OECD.

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Health related outlays for public employees aggravate the fiscal pressure. The federal and the *Länder* governments cover part of the health care costs of active and retired civil servants (*Beihilfe*), as opposed to the health insurance scheme. This is a substitute for the private sector regulation where employers cover half of the premium to the health insurers. Since per capita health care spending is positively correlated with both age and the length of the retirement spell, the adverse age dynamics will profoundly impact on the government's health care spending.

The projected developments call for policy action. Public sector reform should facilitate labour re-allocation within and outside the public sector so as to increase public sector efficiency and avoid early retirement programmes as a means to bring about reductions in personnel (see the special chapter on public sector reform in the 2003 *Economic Survey*). Certain features of the pension system and health care financing for civil servants impact adversely on labour supply and hamper the reallocation of labour both across the public sector and between public sector and private sector employment. Major features are:

- Pension claims of tenured civil servants are linked to seniority wages rather than the wage base over the entire employment spell as is the case in the public pay-as-you-go system. *Ceteris paribus*, this favours employees with minimal employment contracts over the first part of their working life relative to those with more steady (full-time) employment as both groups receive the same level of pensions. This discourages labour supply over the life cycle, the more so the more flexible employment contracts become. In addition, there is an incentive to maximise pension claims by pushing up pay with seniority with adverse budgetary implications.

- The rules for granting support to tenured civil servants for health care insurance (*Beihilfe*) are dispersed across the federal and the *Länder* governments.

- The supplementary pension insurance for non-tenured civil servants (*Angestellte*) can generally not be continued if the employee moves to private sector employment, except for special cases where the new and the former employer are linked *via* a public sector participation.

Hence, to foster labour re-allocation and contain pension outlays, the following measures should be implemented:

- Special tenure for civil servants (*Beamte*) should be reduced to the minimum or abolished entirely (see Chapter 5 below on the university system). In Switzerland the long-standing status of tenured civil servant was recently abolished.[22]

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- Rules for public sector pensions should be fully harmonised, with the general scheme, and a transparent link between contributions and benefits according to individual contribution histories should be established. This would be instrumental in improving the mobility of employees across sectors.
- The system of *Beihilfe* payments should be given up and health care insurance for civil servants should be subjected to the same rules as prevail for private sector employees.

Progress on all these counts cannot be achieved by policy initiatives of the federal government alone, but requires concerted action involving the states. The retirement income scheme for members of Parliament should also be reconsidered. At present pensions for parliamentarians are subject to special rules that generate high claims already after a few years. As a result activity income and retirement income are highly unbalanced in favour of the latter. To reduce disincentives for labour supply the system should be brought into line with the rules governing the general pension scheme. To the extent total remuneration for deputies is considered too low given this adjustment, compensation could be increased accordingly. Moreover, pre-funding pensions of civil servants would contribute to cushion the projected increase in spending. It has been proposed to fund a pension reserve *via* lengthening the civil servants' weekly working time without associated adjustment in salaries.[23] However, applied in isolation the revenue elasticity of this measure might be relatively small. Hence, to be fiscally effective, extending the working time should be associated with additional cuts in personnel, based on revised priorities of public sector tasks. More generally, the fiscal pressure arising from the ageing of the public sector work force needs to be seen as another serious challenge reinforcing the need for genuine savings in government spending so as to achieve pre-funding *via* rapid reductions in general government debt.

As others, Germany faces the challenge of improving efficiency of the health-care system

Germany's health system provides universal health-care insurance coverage of the population and access to an excellent health care infrastructure. At the same time, Germany is among the countries with the largest health care spending to GDP ratios in the OECD. Health care activity levels are relatively high, as reflected in hospital case numbers, duration of hospitalizations, the frequency of expensive treatment – such as cardiovascular surgery – and the number of patients' contacts with physicians.[24] Although previous cost-containment measures helped to keep health care expenditure growth below the OECD average since 1992, evidence on unnecessary diagnostic procedures and surgical interventions suggests that substantial room for efficiency improvements remains.[25] Estimates suggest that spending on unnecessary or poor quality services might

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account for one quarter of outlays in the health care system.[26] Against this background, the government introduced a reform of the publicly financed health care system, which came into effect in January 2004, aiming to reduce costs as well as improve efficiency of publicly financed health care provision.

Recent reform contributes to containing costs and improving the efficiency of the health care system …

The reform generates savings as a result of extending the reference price system (*Festbetragsregelung*[27]) to some patented medicines. Before the reform took effect, all patented medicines were exempt from the reference price system and therefore benefited from higher prices, which encouraged patenting of new medicines even when they did not add therapeutic benefits to existing medicines. As a result of the reform, patented medicines which do not provide additional therapeutic benefits are included in the *Festbetragsregelung*, reducing the cost of pharmaceuticals to health funds without discouraging innovation in drugs improving therapeutic performance. In addition, the discount the public sector health insurance system receives on purchases of pharmaceuticals, which do not fall under the *Festbetragsregelung*, was increased from 6 to 16 per cent. Steps taken to deregulate pharmacies, as recommended in the 2003 *Economic Survey*, include the liberalization of the prices for non-prescription medicines and permission to operate small chains of up to 4 pharmacies and mail-order pharmacies. These measures are likely to lead to a fall in prices of non-prescribed medicines.

The health care reform measures unify co-payment rules across services, widening the range of services to which co-payments apply. Specifically, adult patients will pay 10 per cent of the cost of medical services and prescribed medicines, subject to a € 5 floor and a € 10 ceiling. Co-payments have been extended, in particular, to physicians' and dentists' consultations, for which patients have to pay a € 10 fee in each quarter in which consultations are made. Preventive care consultations are exempt from co-payments and lower co-payments apply to recipients of social assistance. Total co-payments are limited to 2 per cent of yearly income and 1 per cent for the chronically ill.[28] Data for 2002 suggest that the proportion of health-spending paid out-of-pocket in Germany was among the lowest in the OECD prior to the reform. Given the relatively large degree of discretion patients exercise over doctors' visits, the extension of limited co-payments to these services is likely to reduce inefficiency resulting from excessive demand for services, although the ceilings on co-payments – introduced to safeguard universal coverage of access – limit the effect. In addition, the new co-payment regime for medicines is likely to strengthen incentives for the use of inexpensive medicines, as previously co-payments did not depend on price. Health fund payments to pharmacies for the dispensing of medicines will cease to depend on price, removing pharmacies' incentives to sell expensive medicines. Thus the widening

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of moderate co-payments to ambulatory care is a welcome contribution to efficiency, while ensuring continued wide access of the population to health services. However, while recent data suggest a decline in doctors' consultations – probably reflecting to some extent a normalisation after high service utilisation at the end of last year – the impact on demand for health services is likely to be moderate.

Health funds have also ceased to cover some services. Specifically, they have stopped paying death and birth cash benefits as well as for glasses and prescription-free medicines. Coverage of artificial insemination, sterilisation, and transport to health service providers in outpatient care has been reduced. As is the case with the co-payments, private payment is likely to lead to more efficient demand levels, and the cuts are unlikely to worsen health-related financial risks of insured individuals to a significant extent.

Diagnosis related paymentsfor hospitals (DRGs) – which were legislated in a previous health care reform law (see the 2003 *Economic Survey*) – have become compulsory for hospitals this year and will begin to affect hospital budgets in 2005. As noted in the *Economic Survey*, experience in other countries suggests that DRG schemes may generate cost savings as a result of shorter durations of hospitalizations but require monitoring of care to counter incentives to lower costs through a reduction in quality of care and cream-skimming of patients. Quality control exercises have been strengthened with the introduction of the DRGs in Germany and reports on the quality of hospital care, compiled by health funds and doctors' associations, will be available from 2005 onwards.

The health care reform has increased the scope for selective contracting between insurers and health care providers, a recommendation that was made in the 1997 and 2003 *Economic Surveys*. While this allows insurers to develop managed care programmes, the admissible range for selective contractors is limited. Managed care has been virtually absent in Germany so far.[29] Some evidence available from other OECD countries suggests that managed care programmes provided by insurers can be a useful tool to reduce inefficient health care provision without compromising health outcomes. Examples of managed care include programmes in which patients pre-commit to limiting their choice of provider, such as in a gatekeeping in scheme, in which insured individuals consult one general practitioner who takes decisions about referrals to other health care providers. Managed care programmes have also achieved efficiency improvements through benchmarking of provider performance, disease management programmes or comprehensive electronic patient records.[30] Through allowing the use of selective contracts, incentives are given for health funds to compete not only on administrative cost but also on the cost-effectiveness of services purchased on behalf of their patients and provides individuals with some scope to choose healthcare coverage according to their preferences.

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Specifically, the recent reform package allows health funds to negotiate contracts with hospitals for a few ambulatory services, which are otherwise scarcely provided, as well as for services for which collective contracts allow selective contracting. Health funds will also be able to negotiate selectively with integrated service networks (*integrierte Versorgung*) which are yet to be created. The selective contracts for the *integrierte Versorgung* are meant to give scope to improve efficiency through encouraging coordination among health care providers, given that the high decentralisation of the German health system coupled with unlimited patient choice on ambulatory care contribute to unnecessary diagnostic and therapeutic procedures. To further promote the *integrierte Versorgung*, the health care reform includes financial incentives for setting up integrated networks and allows doctors to be employed by practices. Health funds can also negotiate selective contracts for "gatekeeper" schemes. Participation in the new schemes, *integrierte Versorgung* and gatekeeper schemes, is voluntary for patients. Health funds can offer discounts for the participation in such schemes, as well as in return for participation in health check-up programmes and preventive care programmes and chronic disease programmes. However, most bargaining for ambulatory services will continue to be contracted in a bilateral monopoly setting – the association of health funds on the one hand and the doctors' associations on the other hand. It remains to be seen whether competition between health funds is effective enough to push them to take measures that generate efficiency gains.

The reform also aims at strengthening peer review and evidence-based medicine, but the independence of the institutions involved is not always assured. A new institute for quality assurance (*Institut für Qualität und Wirtschaftlichkeit im Gesundheitswesen*), to be established by health funds, doctors' associations and hospitals, provides guidance on the benefits of therapies to patients and providers. However, its lack of independence from stakeholders in healthcare may compromise its credibility. The control commissions reviewing the doctors' prescribing process have been strengthenedby heading them with independent experts. However, the new quality review exercises for the doctors' practices will be supervised by the doctors' associations, which might cast doubts on their credibility as well. Similarly, in its ongoing programme for health promotion and disease prevention the federal government established a council, comprising 65 institutions from the different layers of government and civil society, to elaborate measures to meet health targets. While policy initiatives in the area of health promotion and disease prevention can yield substantial benefits in terms of health outcomes and reductions of health care spending, the inclusion of groups representing disparate interests risks hindering the implementation of health targets.

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… but needs to be followed up

Widening the scope for selective contracts with health care providers would turn the health funds into more active purchasers of health services on behalf of the patients. This would increase the scope for competition among health funds to generate additional efficiency improvements and involve health funds more strongly in managed health care. Indeed, all ambulatory care contracts could eventually be moved from the still dominant bilateral monopoly bargaining between health funds and doctors associations to selective bargaining.

Competition for individuals with low risk characteristics among health funds ("cream skimming") can undermine incentives to cut administrative costs and improve cost effectiveness in health care provision. The present risk adjustment mechanism (*Risikostrukturausgleich*) aims at preventing such adverse selection, compensating health funds for risks associated with gender, age, dependants and income, as well as the number of insured patients participating in programs for the treatment of chronic conditions. Moreover if unusually large expenditures arise for any one insured individual, 60 per cent of costs are born jointly by all health funds. However, as in other countries with similar schemes, the German risk adjustment mechanism captures a fraction of risk differences across individuals and scope for improvement remains. A review of the risk compensation mechanism is scheduled this year and the German government plans to implement a morbidity-oriented compensation scheme in 2007.

Some countries in which providers of hospital care are involved in disease prevention have experienced adverse incentives of DRGs on health prevention efforts, as the number of hospital admissions typically increases hospital revenue under the DRGs. The introduction of DRGs has also led to increases in hospital admissions in some countries with spare hospital capacity. Hospital capacity in Germany, as measured in the number of acute-care beds per head, is among the highest in OECD countries for which comparable data are available. While the German DRG scheme takes readmissions into account, reducing incentives to increase hospital admissions, the hospital caseload may require careful monitoring. To prevent costs from rising through an increase in the hospital caseload, some countries have complemented DRGs with budgetary controls or have reduced prices set within the DRG scheme.[31] It will also need to be ensured that health promotion and disease prevention are not negatively affected.

Two reform options to replace the current system of payroll-based financing of the public sector health funds have been discussed in Germany: first, levying health fund contributions on all personal income (*Bürgerversicherung*) and, second, a system of lump sum per capita health fund *premia* (*Kopfprämien*), the level of which would be set by each health fund.[32] To make the per capita *premia* affordable for individuals on low incomes and offset the adverse effect of the lump sum *premia* for low-paid employment, their introduction would need to be coupled with

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an income redistribution mechanism financed from tax revenue which would refund part of the per capita *premia* paid by low income households. Assuming the redistribution mechanism would be financed from personal income, a trade-off between marginal tax rates over the distribution of personal income would have to be considered. The per capita *premia* can only generate lower marginal tax rates on medium and higher incomes than those associated with the B*ürgerversicherung* if marginal tax rates on lower incomes are higher.[33]

Both schemes would have the advantage of allowing income redistribution to be conducted through income taxation rather than through payroll taxes. The per capita *premia* system would suggest additional advantages: First, redistribution of contribution revenues among health funds to compensate for differences in wage income among the insured would become unnecessary, simplifying the risk adjustment mechanism. This could make competition among health funds more effective, as the current risk adjustment mechanism imperfectly compensates the health funds for differences in wage income. Second, the finances of health funds would cease to depend on labour market developments. Third, the per capita *premia* would make the cost of public-sector financed health care more transparent to the insured. Fourth, the introduction of per capita *premia* could help reduce disincentives to work for second earners in households, such as married women (see also Chapter 3). However, there appears to be little empirical evidence yet how changing the financing system in this direction would affect efforts to achieve cost containment. More generally, care needs to be taken that the reform process does not focus on opening additional sources of finance while diminishing implementation of other policy measures designed to generate efficiency gains.

T*he looming spending pressures in long–term care need to be dealt with*

The mandatory public long-term care insurance system, introduced in 1995, has helped to mitigate individuals' financial risks related to long-term care needs, providing universal coverage and reducing dependency on means-tested benefits (social assistance) to pay for long-term care. While the system is in principle pay-as-you-go, financed out of wage-based contributions, surpluses in the first few years were used to build up a small asset base, which peaked at € 5 billion, equivalent to about 30 per cent of current yearly spending. However, with declining employment, the extension of long-term care benefits, and increasing utilisation of the scheme, the surplus melted down. As a result in 1999 the system slid into deficit, totalling € 0.7 billion in 2003.

Spending pressures on long-term care are set to increase substantially, mostly as a result of population ageing.[34] According to the projections of an expert commission appointed by the government, demographic aging will push the number of care recipients from 1.9 million in 2002 to 2.7 million in 2020 and 3.4 million

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in 2040. Current legislation fixes the payroll contribution rate to the current level of 1.7 per cent. However, with the cost of care likely to rise faster than the general price level as well, the deficit of the system is projected to rise to about € 10 billion by 2020, if the current level of services to care recipients is to be maintained and in the absence of policy measures. Government plans to improve service provision in some areas covered by the insurance scheme may further add to spending pressures.

Hence, policy action is required in order to prevent rising contribution rates to cover a mounting deficit. There appears to be scope to increase the efficiency of the system. First, substantial progress has been made over recent years in increasing the share of long-term care spending devoted to home care rather than to more costly stationary care. Further progress could be achieved along these lines by rebalancing insurance payments to create stronger incentives to substitute home and ambulatory care for stationary care. Second, while steps have been made to improve the information base on the quality and cost-effectiveness of the services provided, mechanisms need to be installed to re-channel funds to more efficient service providers on the basis of this information. To this end, consideration could be given to extend personal allowances to the purchase of professional care. The current system of caps on payments for services provided to each care recipient, coupled with the opportunity to choose among providers, generates some incentives for cost considerations to influence demand decisions. However, the extension of personal allowances for professional services would give care recipients more scope to better meet their individual care needs and generate more room for market-determined prices for care services to develop, as prices are currently set through collective bargaining between providers and long-term care funds.[35] In such a system, availability of guidance for the care recipients on the efficiency of providers as well as monitoring of the quality of purchased services would be required.

Moreover, the demographic spending pressures in the long-term care insurance underline the need to establish priorities in government spending programmes, including ageing-related spending, in order to identify scope for offsetting spending reductions which would allow payroll contribution rates to remain constant. The commission has estimated that the equivalent of, initially, 0.5 percentage points of the payroll contribution would need to be devoted to pre-funding in order to create sufficient room for ageing-related spending growth in order to keep payroll contribution rates at their present level of 1.7 per cent.[36] The build-up of assets would be phased over a period of twenty years, reaching an amount equivalent to 7.5 per cent of projected GDP in 2040, giving an indication of savings that would have to be cumulated within general government over this period to keep long-term care contribution rates constant. An option suggested by the commission to finance the build-up of assets is to levy an additional contribution on pensions, as pensioners are the main beneficiaries of long-term care services.

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Notes

1. Details on proposed changes in the number of civil servants and salaries are contained in the budget. The input orientation of the budgeting process is illustrated by the fact that the parliament's discussions on resources relative to staffing are often the most contentious part of the budget debate. See: *Deutscher Bundestag* (2003).
2. See Blöndal (2003); Kristensen *et al.* (2002). See also: Lienert and Jung (2004).
3. See: Lüder (2002); Blöndal (2002).
4. See OECD (1997 and 2003a).
5. For analysis for different wage brackets see OECD (2003b).
6. Elschner *et al.* (2003).
7. See: Sachverständigenrat zur Begutachtung der gesamtwirtschaftlichen Entwicklung (2003). For other studies on average and marginal taxation of profits see: European Commission (2001); Spengel (2003); and Elschner *et al.* (2003).
8. However, most German companies are partnerships, subject to personal income tax. As the study by the Council of Economic Experts shows, effective taxation of these companies, marginal and average, is considerably lower than that of corporations.
9. The tax expenditures under consideration are those noted in the federal government's biannual Subsidy Report and the list by Koch and Steinbrück that formed the base for the negotiations on the 2004 income tax reductions in the parliamentary mediation process between the Bundestag and the Bundesrat. For a compilation see: Boss and Elender (2004).
10. See Sachverständigenrat zur Begutachtung der gesamtwirtschaftlichen Entwicklung (2003).
11. Revenues foregone are about 0.2 per cent of GDP smaller if the rate reductions that are already scheduled for 2005 are taken into consideration. See: Boss and Elender (2004). See also Boss (2003).
12. See Beer and Ragnitz (2002) and Ragnitz *et al.* (2001).
13. See OECD (2001), and for more detail: Wurzel (2001).
14. See the discussion in OECD (2001). For a recent comparison in infrastructure endowment between the new and the old states see Sichelschmidt (2003).
15. On the Solidarity Pact see OECD (2003a).
16. BMGS (2003).
17. BMGS (2003).
18. The introduction of the demography factor stands as an example. A similar measure, linking annual pension adjustment to the development of life-expectancy, was already

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legislated to come into operation in January 1999 but was reversed on social grounds before it became effective. See OECD (1999).

19. See BMGS (2003).

20. Burniaux *et al.* (2003).

21. For nominal GDP at the starting point in 2005 the projection from the OECD E*conomic* O*utlook*, No. 75 is taken. In the OECD medium-term baseline scenario for Germany nominal GDP grows at a rate of around 3.5 per cent annually until the output gap is closed in 2009. Thereafter, nominal GDP growth slows to 3 per cent. This reflects a growth of the GDP deflator of 1.5 per cent annually, as assumed by the government as a base for projected pension adjustments. Real GDP growth is assumed to grow at a potential rate of 1.5 per cent. This might well over-predict true growth in the decade between 2020 and 2040, given the severe adverse impact of the shrinking of Germany's working-age population within these two decades. The spending item does not include the government's pension outlays for tenured civil servants employed with the Federal Railways and the Post Office. Upon incorporation of these facilities an agreement was made that the federal government would cover the pension liabilities of its former tenured employees to the extent these are not covered by revenues of the corporations. While pensions outlays for the Railways are projected to decline until 2040, those for the Post Office are expected to increase by about the same amount.

22. See OECD (2002).

23. Abstracting from the funding of pensions, flexible employment schemes that allow for a lengthening of the working time of civil servants is likely to contribute to making public sector services more customer friendly.

24. OECD (2003c).

25. For example, a third of all x-rays taken are superfluous according to the Ministry for Health and Social Security.

26. See: BMGS (2002).

27. The F*estbetragsregelung* imposes ceilings up to which the price of a medicine is paid by a health fund.

28. See also Docteur and Oxley (2003).

29. See, *e.g.* Felder (2003).

30. The Veteran's Health Adminstration (VHA), a public health care provider in the US, based a successful strategy of lowering cost and improving quality on such records.

31. Docteur and Oxley (2003). Austria and Sweden reintroduced budget caps.

32. A detailed proposal for each of the two options is provided in BMGS (2003).

33. In principle, the system of the per capita *premia* could replicate the marginal tax burden implied by the B*ürgerversicherung* through a corresponding specification of the income redistribution mechanism.

34. BMGS (2003).

35. BMGS (2003).

36. The commission assumed that efficiency savings from the rebalancing of insurance benefits would be used to finance improvements in selected services. BMGS (2003).

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III. Increasing the capacity of the economy to create employment

The government has embarked on a substantial reform programme, improving incentives for individuals out of work to take up jobs and for firms to hire workers. Indeed, the protracted weak performance of the labour market continues to underline the need for labour market reforms. Further progress is necessary to ensure that the reforms achieve a substantial and lasting increase in employment, as outlined below. The Annex summarises recommendations for structural reform and action taken since the 2003 *Economic Survey*.

Job search counselling with appropriate sanctioning improves incentives for employment…

The government has implemented measures proposed by the *Hartz* commission to improve the effectiveness of the public employment office's placement activities. Specifically, the German public employment service (PES, *Bundesagentur für Arbeit*[1]) has begun to enforce benefit eligibility criteria more strictly. The number of penalties imposed on individuals for not accepting a job offer or a placement in an active labour market policy measure (ALMP) more than doubled in 2003 to almost 200 000 (equivalent to 4½ per cent of unemployment registered at the employment office).[2] Job seekers have been obliged to accept jobs further away from where they live and to provide evidence themselves that refusing a job offer was justified. Moreover, recipients of the new second tier unemployment benefit (see below) will have to be prepared to accept any legal job regardless of pay.[3] Measures the government has taken to improve job placement, include steps towards the general introduction of profiling (see the 2003 *Economic Survey*), and shifting resources towards placement activities, including an increase in counselling staff, while reducing administrative costs through streamlining benefit payments and cutting spending on those ALMPs which have been ineffective in placing the unemployed in unsubsidized jobs. Indeed, spending on active labour market policies declined by 15 per cent in 2003, to about € 12 billion, and is projected to fall by a similar proportion this year. While progress has thus been made in raising the effectiveness of the public employment office in placing the unem-

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ployed in jobs, the process is as yet incomplete. More steps are necessary to improve its organization and to focus its activities towards further improving the flow of the unemployed into jobs. For example, placement performance incentives within the public employment service could improve further (see below).

Empirical evidence from OECD countries suggests that efforts to improve job search counselling and placement services can make a substantial contribution to reducing unemployment if accompanied by enforcement of rules making benefit eligibility conditional on search efforts, readiness to accept job offers or a placement in an ALMP, as well as a credible sanctions regime.[4] High quality search counselling, including the use of ICT and regular follow-up interviews, were found to be effective. Indeed, measures to enforce benefit entitlement rules are especially important in Germany, as relatively generous unemployment benefits diminish incentives to take up work (Table 3.1). In Denmark, for example, such policies have helped reduce structural unemployment in the presence of high unemployment benefit replacement rates. Survey evidence from the UK also suggests that the introduction of defined job seekers' obligations increased the exit rate from unemployment into work by 40 per cent. Empirical evidence also indicates that the increase in exit rates from unemployment from such measures reflected seekers accepting jobs at lower wates.[5]

... as do measures to improve the effectiveness of the public employment service

Experience from other OECD countries suggests that incorporating performance incentives in the management of the public employment service can have a marked impact on the success of moving the unemployed into employment.[6] The Federal Labour Office in Germany has begun to monitor placement performance of local employment agencies, improving scope for identifying agencies which need to improve placement activities. Local employment agencies have also been given more independence in implementing active labour market policies. Temporary contracts for management staff in the agencies give scope to reward performance, but no explicit incentive scheme is as yet in place. For example, such performance incentives could help to foster an effective use of integration agreements, (I*ntegrationsvereinbarungen*) which specify the obligations of the job searchers, as the design of the agreements is left to local labour offices.

First steps have been taken to limit the use of active labour market policies to those measures which are effective in placing workers in unsubsidised jobs, but participation of the social partners in the decision making process may hamper further progress in this direction. New legislation requires evaluation of programmes, and availability of data for the purpose of evaluation has improved. ALMP programmes have ceased to qualify participants for unemployment insurance benefit, preventing enrolment in programmes in order to re-establish a benefit entitlement. While unions and employers are no longer represented on the

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Table 3.1. **Effective tax rate for an unemployed person (2001)**[1]

previous work = 67 per cent of an average production worker returning to work at a wage equivalent to:

% of APW	Single				1 earner couple with two children				2 earner couple with two children[2]			
	50%	67%	100%	150%	50%	67%	100%	150%	50%	67%	100%	150%
Belgium	100	89	78	71	87	76	68	64	99	89	78	70
Denmark	107	91	78	73	76	87	82	75	107	91	78	73
Germany	**100**	**88**	**77**	**70**	**100**	**85**	**75**	**65**	**114**	**98**	**83**	**72**
Greece	101	79	60	50	107	84	61	51	107	84	61	51
Spain	100	81	65	53	100	82	61	50	103	83	65	54
France	103	87	70	59	78	84	76	59	104	87	68	55
Ireland	87	73	59	54	95	87	72	57	72	61	51	43
Italy	69	60	53	49	78	54	53	53	80	70	62	55
Luxembourg	107	88	71	63	101	104	83	64	115	89	68	58
Netherlands	93	85	72	60	94	90	78	63	89	78	67	57
Austria	88	75	64	57	100	97	78	67	86	74	63	56
Portugal	111	88	67	55	55	55	57	56	111	86	65	52
Finland	90	81	69	63	100	99	89	76	91	78	67	61
Sweden	105	87	70	62	100	100	84	71	105	87	70	62
United Kingdom	78	70	58	49	66	72	74	62	54	49	43	39
Czech Republic	80	67	54	46	100	96	77	63	77	65	55	46
Hungary	84	75	64	61	89	71	61	59	84	73	62	60
Poland	92	77	63	53	100	91	80	65	77	66	55	51
Slovak Republic	88	77	59	49	100	100	96	72	85	69	58	48
Norway	89	75	62	57	99	92	73	63	94	78	64	59
Switzerland	115	92	71	58	100	100	75	59	109	88	68	56
United States	85	71	57	51	59	62	60	50	83	69	56	47
Japan	94	74	56	45	84	86	71	56	91	71	56	44

1. The table indicates the rate at which taxes increase and benefits fall relative to gross pay in a new job. The unemployed person is assumed to have worked for 22 years before becoming unemployed and to be in the second month of his/her unemployment spell. He/she is assumed to be entitled to means-tested social assistance benefits in addition to unemployment insurance benefits. Income tax rate reductions in some countries since 2001, including in Germany in 2004 and 2005, reduce average effective tax rates below the figures indicated in the table. An average production worker in Germany earned € 32 384 in 2001.
2. The wage level of the first earner is fixed at 67 per cent of the APW, while the wage level of the second earner (who moves from unemployment to employment) is indicated in each column.

Source: Joint European Commission-OECD project, using OECD Tax-Benefit models.

governing body of the PES, which takes operational decisions, they continue to play a role in the supervision of decision making. The role of the social partners in the labour office appears to have contributed to spending on ineffective labour market policies in the past (see the 2003 *Economic Survey*).

A new ALMP, paying subsidies to the unemployed who decide to set up their own business (*Ich* AG), has been introduced on a large scale but has yet to be evaluated with regard to its success in moving the unemployed into unsubsidized jobs. The *Ich* AG scheme accounted for a substantial proportion of newly founded firms in 2003, with more than 90 000 individuals having participated. A few features are likely to boost the number of scheme participants. The viability of the business plan of the new enterprise is not tested for individuals to qualify for the sub-

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sidy and the unemployed may pick up *Ich* AG transfers to extend receipt of benefits, also participation in the scheme can be combined with participation in the *Mini-Job* scheme (see below). Given high exit rates of newly founded firms, even if they are unsubsidised, many participants are likely to return to unemployment, especially when the subsidies expire after at most three years.

Further progress is necessary in redeploying employees of the PES from benefit administration to counselling and placement as well as in limiting use of active labour market policies on programmes which are proven to be effective. Reducing the role of the social partners in the decision-making process may improve incentives for evaluation-based use of active labour market policies. Further development of performance incentives is necessary to ensure that operational autonomy of local labour offices generates gains in placement effectiveness. The *Ich* AG scheme should only be continued if evidence supports its effectiveness in moving the unemployed into unsubsidized employment.

Unemployment benefit duration has been reduced…

An important element of the "Agenda 2010", legislated by Parliament, is the reduction in the duration of unemployment benefit entitlements. From 2006 onwards, unemployed individuals who become unemployed when they are aged 55 and older will be entitled to a maximum of 18 instead of 32 months of unemployment insurance benefit (*Arbeitslosengeld* I). Job seekers up to 55 years of age will receive unemployment insurance benefit for at most 12 months. Under the current system the unemployed between 45 and 55 years of age can receive between 18 and 26 months of benefits, while younger workers' unemployment benefit is already limited to 12 months. Moreover, a new second tier unemployment benefit (*Arbeitslosengeld* II), to be introduced in 2005, replaces two previous benefit systems (*Sozialhilfe* for employable persons and *Arbeitslosenhilfe*) as recommended in the 2003 *Economic Survey*. The new flat-rate means-tested benefit will pay about as much as the previous *Sozialhilfe* and is generally less generous than the *Arbeitslosenhilfe*. Up to now the latter provides means-tested benefits after the first tier unemployment insurance benefit entitlements lapse at slightly lower replacement ratios. Since the benefits of the *Arbeitslosenhilfe* are open-ended, its abolition further reduces the overall duration of unemployment benefits provided at a fixed replacement rate for many workers. These are important steps towards improving labour market performance, as empirical evidence across OECD countries[7] shows that the duration of unemployment benefit eligibility plays an important role in the determination of unemployment. However, to ease the transition to the less generous second tier benefit, the unemployed receive a supplement for two years after the expiration of the unemployment insurance benefit.[8]

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... generating scope for more effective placement of the unemployed into jobs

Unification of the two main means-tested social assistance benefits into a new second tier means-tested unemployment benefit (*Arbeitslosengeld* II) marks a potentially substantial improvement of the efficiency of benefit administration and job placement. Empirical evidence shows that responsibility for benefit entitlement, active labour market policies and placement activities is best placed in one body, allowing these measures to be coordinated.[9] The German reform of the second tier unemployment benefit moves in this direction, as it unifies, in principle, the responsibility for the recipients of this benefit in the PES, which has so far been split between the federal employment office and local authorities, giving rise to fiscal externalities.

Present legislation, however, gives local authorities the option of providing tier II unemployment benefits and placement services instead of the Federal Labour Agency, potentially compromising the efficiency gains that could be achieved from the unification of means tested benefits. Local authority responsibility for second tier unemployment benefit results in split responsibilities over the duration of an unemployment spell, as individuals move from unemployment insurance benefit to the second tier unemployment benefit, given that the tier I unemployment benefit remains under the control of the federal employment office. Moreover, job search beyond the confines of a local authority may also require coordination across several agencies. The transfers to local governments to finance benefit payments and placement activities could also further complicate fiscal federal relations. In addition, the transfers would need to be designed in a way that ensures that local authorities have incentives to place workers into unsubsidized jobs.

The option for local governments to provide the second tier unemployment benefit and placement activities is thus likely to complicate the ambitious task of reorganizing the labour office ahead of the introduction of the new second tier unemployment benefit in 2005. This legislation should be rethought, assigning the responsibility for all unemployment benefit recipients to the federal employment agency.

Measures reducing marginal effective taxation of low paid workers need to be more effectively designed

Effective marginal tax rates of around 100 per cent for low-paid workers result from the withdrawal of benefits as earnings rise, especially for workers with dependants (see the 2003 *Economic Survey*). The rate of withdrawal of social assistance (second tier unemployment benefit from 2005 onwards) has been reduced to 70 per cent in a range of gross income between € 400 and 900 and to 85 per cent between € 900 and € 1 500,[10] so that full withdrawal of benefits will begin at a monthly income level of € 1 500, up from about € 700.[11] While this measure, pro-

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vides some incentives for benefit recipients to take up low-paid jobs, full withdrawal of benefits above earnings of € 1 500 continues to generate effective marginal tax rates of 100 per cent. Also, since social assistance benefits are now paid over a larger range of income, effective marginal tax rates have risen at higher income levels at which workers qualify for social assistance payments as a result of the changes, reducing their incentives to work. Indeed, reducing effective marginal tax rates at low income levels inevitably entails raising marginal tax rates at higher income levels if high fiscal costs are to be prevented, unless benefits are cut. Empirical evidence suggests that reductions in the withdrawal of social assistance would need to be combined with substantial reductions in benefit levels to generate positive employment effects.[12] This reinforces the need to complement these measures with a credible sanctions regime, lowering benefit payments to social assistance recipients who can work and do not accept job offers.

In order to further increase incentives to take up low-paid employment, reduced social security contribution rates, summing to 23 per cent paid by the employer, are now available for jobs paying up to € 400, instead of up to € 325 (M*ini*-J*obs*).[13] Moreover, the 15 hours working time limit for the reduced social security contribution rateshas been abolished and the scheme has been made available for individuals taking up a second job. Workers who do not benefit from the basic income tax allowance when working as a result of additional sources of household income, notably married women and workers taking up additional jobs, also benefit from a reduced income tax rate of 2 per cent. The contribution subsidies are now gradually phased out over a range of incomes up to € 800 (M*idi*-J*obs*).[14]

Unlike the partial withdrawal of social assistance, the benefits of this scheme are not linked to means-tested benefit entitlement, making the M*ini* and M*idi*-J*obs* a less effective instrument in reducing benefit dependency and the ensuing disincentives to take up jobs than the reduced withdrawal rates from social assistance. While the measures improve incentives for those out of work to pick up a job, they also generate incentives to reduce hours worked and reduce incentives for training, given that reducing gross earnings allows some employed workers to qualify for the benefits. This effect generally arises from low paid jobs subsidies. Since the scheme is not targeted to benefit recipients, these adverse incentives will affect more workers than the benefit withdrawal in social assistance. Moreover, individuals taking up M*ini*-J*obs* as a second job may crowd out low-skill workers who supply a sufficiently large number of hours for their jobs to remain unsubsidized.[15] Past experience suggests that this may be the case, as cuts in the tax advantages for small jobs in 1999 were accompanied by an unusual fall in the unemployment rate for the low-skilled, possibly because fewer small jobs were taken up as second jobs. Available evidence suggests that most participants were already employed when joining the scheme, reducing hours worked or taking up a second job.[16] While marginal employment can facilitate adjustment of labour sup-

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ply over the life cycle and might provide an avenue into part-time or full-time employment,[17] the *Mini-Job* scheme is hence likely to be considerably less effective in raising employment than expected on account of the substantial take-up of the scheme[18] and may lead to a further reduction in total hours worked as well as a decline in social security and tax revenues.

The impact of the *Mini-Job* scheme should therefore be evaluated with respect to its effects on labour supply and training with the purpose of removing distorting effects on labour supply and achieve better targeting of wage subsidies. Consideration should be given to abolishing public pension contributions for *Mini-Jobs* altogether, while removing corresponding benefit entitlements. The tax subsidies for second jobs should be revisited. The taxation of second earners should be more comprehensively reformed (see below), with a view to improving incentives for female labour supply, replacing the tax subsidies in the *Mini-Job* scheme. Tighter targeting of subsidies for low-paid jobs on individuals entitled to means-tested benefits should be considered, combined with the cuts in benefits for recipients who are able to work but reject job offers within a credible sanctions regime. Replacement ratios across demographic groups should be revisited, with a view to lowering benefits to recipients whose replacement ratios are particularly high. Measures to raise basic skill levels, improving the productivity of low-skill workers, also help reduce disincentives to work resulting from generous benefit systems. While graduation rates at the secondary (or higher) level in Germany are large in comparison to other OECD countries, basic skills among 15 year olds are among the weakest in the OECD.[19] Stepping up intensive education from an early age onwards for at-risk social and ability groups would improve employment outcomes later in life.[20] This includes language training for individuals with an immigration background.

Adverse demographic developments interact with low labour force participation among older people

Germany's old-age dependency ratio is among the highest in the OECD and will increase substantially over the next four decades (Figure 3.1). This calls for reducing disincentives to labour supply as one important element of structural reform. While there is still a perception in the society that economic growth is not high enough to prevent higher labour supply from translating into higher unemployment, abolishing disincentives for labour force participation and reducing impediments for higher labour demand impacts positively on employment and growth. In OECD countries where participation rates of older workers are high, so are their employment rates, suggesting few inherent barriers to employment at an old age (Figure 3.2).

Total labour force participation in Germany stands at 72 per cent of the working age population, 2 percentage points above the OECD average. Female

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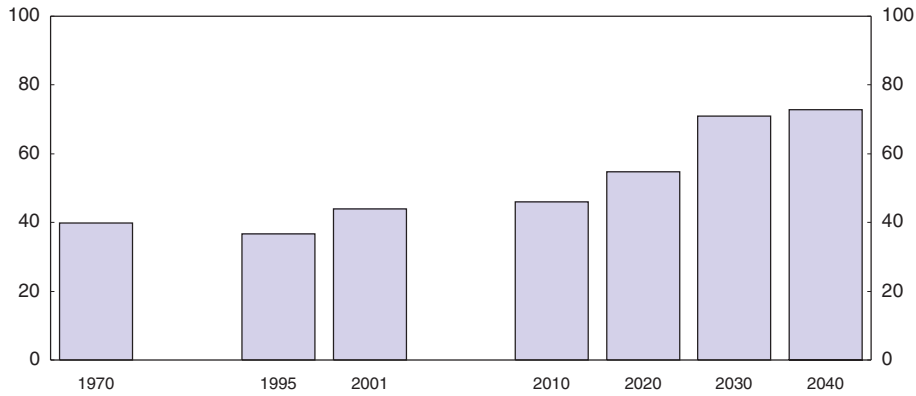
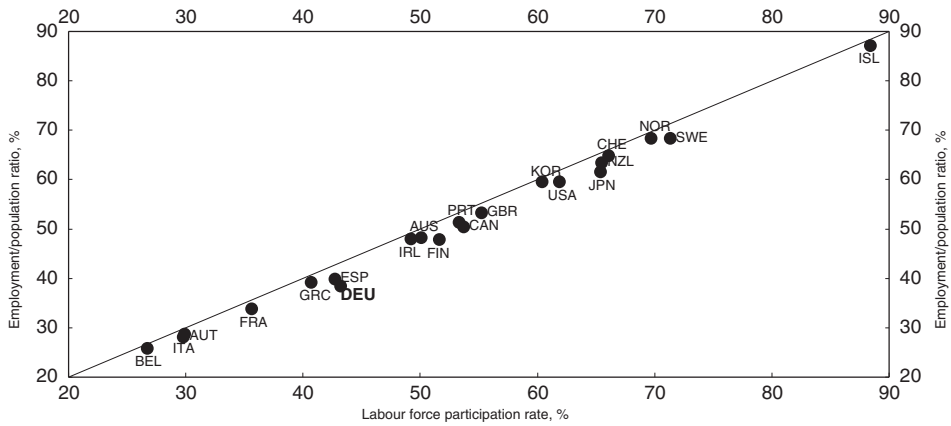
Figure 3.1. **The old age dependency ratio**[1]
Per cent



1. Persons aged 60 and over as a percentage of persons aged 20 to 59.
Source: Federal Statistical Office, In the spotlight: population of Germany today and tomorrow, 2003.

Figure 3.2. **Participation and employment rates for 55-64 year olds, 2002**[1]



1. Participation and employment rates are, respectively, labour force and employment of persons aged 55-64 years divided by population aged 55-64.
Source: OECD, *Labour Force Statistics;* for EU member countries data is from the European Labour Force Survey.

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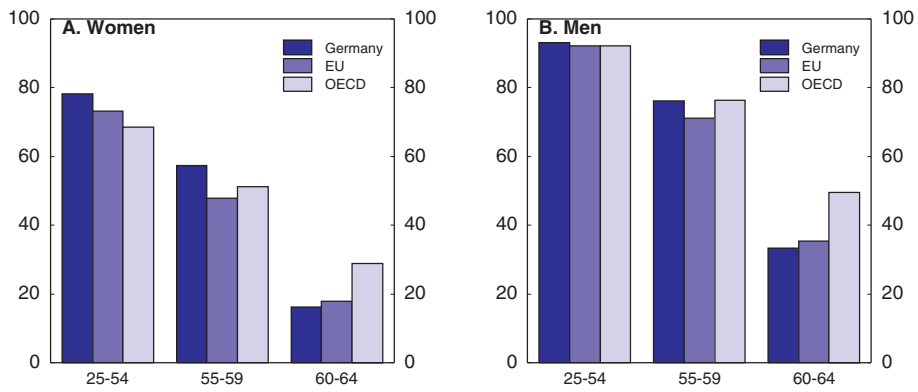
labour force participation increased from 61 per cent in 1991 to 64 per cent in 2002 (persons aged 15 to 64 years), despite the downwards adjustment in the new states from the very high rates that prevailed in Eastern Germany before reunification. For the age group of the 25 to 54 year-olds, 78 per cent of the women participate, which is in the upper half within the OECD. However, labour force participation of older people is weak by international comparison, amounting to 33 per cent for men and 16 per cent for women aged between 60 and 64 years (Figure 3.3). Hence, low participation rates in Germany relative to other OECD countries are associated with age rather than gender, although, as in other countries, scope also remains for female labour supply to increase further. Several measures were put in place over the last decades to reduce labour force participation of older people. This includes different paths to early retirement within the general pension system – such as early retirement on account of unemployment – as well as effective early retirement programmes outside the pension system – such as extended eligibility periods for older unemployed without the obligation to search for employment. Occasionally, early retirement of civil servants was also encouraged when public sector companies were shifted into the enterprise sector, as was the case for the German Railways and the Federal Post Office.

Given that labour force participation is much lower at higher age, the rising share of older people in the population exerts further downward pressure on

Figure 3.3. **Labour force participation, 2002**[1]

Per cent



1. Labour force aged as a percentage of population aged in the respective age group.
Source: OECD Employment Outlook.

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the aggregate labour force participation rate. OECD simulations indicate that this demographic effect alone – assuming all other things are unchanged – would instigate a drop in the proportion of the population above 15 years of age who participate in the labour market by some 6½ percentage points between 2000 and 2025. Entry of new age cohorts into the labour force mitigates this effect, as women in these cohorts are more likely to take up work than women from earlier cohorts, so that the participation rate is projected to decline by 3½ percentage points over this period. As a result, the labour force is expected to fall by around 4 per cent (Table 3.2).[21] Between 2025 and 2050 the participation rate would fall by another 3.3 per cent, with the decline in the labour force accelerating to 13 per cent. In addition long tertiary education has led to late labour force entry of qualified workers. The probability that a German male worker enters the labour force at an age of 30 is about 8 per cent, compared to less than 5 per cent in France and the United States.[22] Overall, OECD estimates suggest that measures to strengthen labour force participation of older workers and women and shortening the duration of higher education can reverse the projected decline in the participation rate. This would slow the decline in labour supply between 2000 and 2050 substantially.[23]

Table 3.2. **Projected changes of participation, population and the labour force (15 years and over)**

	Change in participation rate, % points[1]			Total growth, % in:	
	Total change	*of which*:		Population aged 15 and over	Labour force
		Demographic effect	Cohort effect		
2000-2025	–3.5	–6.4	2.7	2.5	–3.7
2025-2050	–3.3	–3.1	–0.3	–7.8	–13.4
2000-2050	–6.8	–9.5	2.4	–5.5	–16.6

1. The "demographic effect" denotes the impact of the change in the age composition of the population on the participation rate, given current participation patterns. The "cohort effect" denotes the effect of new cohorts entering the population of working age, in which women have a higher likelihood of participating and includes a small "fertility effect" resulting from projected changes in fertility on female labour supply decisions.

Source: Burniaux *et al.* (2003), Table 6, pages 72-73.

Measures to reduce impediments to labour supply among women and older workers are necessary to cope with economic pressures resulting from an aging population

The government has taken steps to reduce disincentives to supply labour among older workers. Recent legislation will reduce the duration of unemployment benefits (see above) and raise the minimum age of early retirement on account of unemployment (see Chapter 2). Moreover, to improve the employability of workers, all restrictions on the use of temporary contracts have been lifted for workers above 52 years of age. The public employment service pays a premium for workers above 55 years of age who reduce full-time employment to part-

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time employment to guarantee an earnings level above the part-time remuneration, provided the firm compensates the reduced working hours by hiring unemployed or young workers who have just completed their apprenticeship. The scheme (*Altersteilzeit*), is intended to ease the transition into retirement. Participants can choose how to distribute part-time employment over a period of up to 10 years and most decide to work full-time in the first half of the period and not to work in the second half. Effectively, this scheme allows early retirement to be subsidised by the Federal Labour Office provided older workers are replaced by unemployed or newly graduated apprentices. Tier I unemployment benefit recipients aged 58 years and above will continue to be exempt from job search and acceptance requirements until 2006, when the exemption will begin to be phased out. The government has increased the minimum early retirement age after participation in the scheme from 60 to 63 years. Consideration should be given further to cut back incentives to retire early stemming from the *Altersteilzeit* scheme. The possibility of stricter application of job search requirements regardless of age should be considered before the exemption of older workers from search and job acceptance requirements expires, and the impact of ending the preferential extended duration of benefit entitlement for the older unemployed should be assessed.

Second earners face higher effective tax rates than single earners in Germany. This mostly affects women with full-time working partners. Furthermore, female labour supply is more sensitive to variations in the net wage than male labour supply. Disincentives for women to work are especially strong for full-time jobs, as the *Mini* and *Midi-Jobs* provide incentives for small-scale part-time work. Moreover, spouses are entitled to full public-sector provided health insurance regardless of their labour market status, but are required to pay health insurance contributions if taking up a job.

In addition, the German childcare support system, while relatively generous, is geared towards child benefits rather than childcare support. Whereas the former have negative labour supply effects, especially if the benefits are income dependent, as in Germany, the latter increases female labour supply, in particular full-time labour supply,[24] as subsidies for childcare reduce the private monetary costs of childcare as well as commuting times. The federal government is co-financing investment in full-time schooling, easing disincentives for women to work full-time resulting from part-time schooling which predominates in Germany.

Steps to foster female labour supply should include a reduction of effective tax rates on labour income of secondary earners as well as reducing the extent to which child benefits are phased out with income and shifting a larger proportion of childcare support to subsidize childcare facilities. Reallocating resources along these lines would help reduce marginal effective tax rates associated with child benefit withdrawal as income rises. Giving financial support to the users rather than the providers of childcare facilities, for example through childcare vouchers, would help to improve the efficiency of childcare provision. Consider-

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ation should be given to introducing health-care insurance charges for non-working spouses. Steps to limit free co-insurance for spouses have been taken in Austria.

OECD estimates show that measures to reduce disincentives to work for secondary earners would increase full-time female labour supply in Germany substantially. According to OECD estimates, equalizing the average tax rate of primary and secondary earnings could increase the full-time female participation rate by 6.2 percentage points by 2025,[25] helping to offset the adverse effects of population ageing on participation. Increasing spending on childcare facilities would raise full-time female participation by 2.6 percentage points.[26]

Employment protection legislation was eased for new and small firms, but uncertainty remains

The cost of hiring and dismissing workers generated by EPL is likely to affect small and young firms more strongly than other firms. Small firms have less scope to reallocate labour internally. In addition, survival rates of young firms, which tend to be small, are particularly low, so that EPL-related potential costs of closure or downsizing are likely to weigh particularly on young firms.[27] Recent legislation has increased the threshold number of employees above which dismissal protection legislation becomes binding from 5 to 10 employees with regard to new hires and the time limit for temporary contracts without specific justification has risen from 2 to 4 years for newly founded firms. These measures are likely to increase firm entry with positive effects on the labour market across all firms, as firm entry increases product market competition, reducing rents that can be captured by workers in wage setting.

While temporary contracts have successively been liberalized in recent years and scope for small scale employment, such as in the *Mini Jobs*, has increased, EPL remains stringent for workers in established firms. This might German EPL has also given rise to large legal costs, as severance payments may be high and are associated with considerable uncertainty (see the 2003 *Economic Survey*). Unpredictable dismissal costs are, however, hard to pass on to wages, so are more likely to affect employment adversely.[28] Some measures have been taken recently to reduce legal costs and uncertainty in EPL. *Inter alia*, new legislation limits the range of social criteria that employers have to take into consideration in dismissal decisions to disability, job tenure, age, and the presence of dependents in the household, and also limits judicial review of firm-level agreements on the implementation of the social criteria. Complaints to the courts are subjected to stricter time limits.[29] Employers can offer a payment equivalent to half a month's salary per year of job tenure to dismissed workers, which workers can choose to accept instead of challenging the dismissal in court.

While temporary contracts have successively been liberalized in recent years and scope for small scale employment such, as in the *Mini Jobs*, has increased,

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EPL remains stringent for workers in established firms. This might foster labour market segmentation, suggesting benefits to broader reform of EPL for permanent employment contracts.[30] Moreover, stringent EPL also reduces voluntary job-to-job movements because the degree of employment protection depends on job tenure, potentially slowing down structural change.

It remains to be seen whether the simplification of EPL results in a substantial decline of legal uncertainty, as reflected in the number of cases dealt with by labour courts. If necessary, the rules will need to be simplified further. A further increase in the threshold of applicability of EPL could have substantial economic benefits. Measures to ease EPL for permanent contracts generally could reduce labour market segmentation, improve incentives to quit voluntarily and might remove barriers to firm growth. To this end, workers and firms could be allowed to opt out from employment protection provisions as part of the work contract after a probationary period. A more fundamental reform option would be to link firms' contributions to unemployment insurance to the number of their dismissals, as practised in the US, replacing or partly replacing EPL. Such a system would safeguard workers' incentives to quit voluntarily while facing firms with the social costs of dismissals.

Some regulatory restrictions for temporary work agencies have been lifted, but problems remain

As noted in the 2003 *Economic Survey*, temporary work intermediated by private agencies provides an avenue for reintegrating unemployed job seekers in the job market, as they recruit many workers from the pools of unemployed and inactive individuals. Empirical evidence suggests that TWAs can contribute to reducing the rate of unemployment consistent with stable inflation, as they improve matching, preventing wage inflation in response to skill bottlenecks, and increase competition in the labour market.[31]

Recent changes to the legal framework have eased regulation on TWAs, including the abolition of the *Synchronisationsverbot* which required that the duration of the contract between the agency and the worker had to differ from the duration of the contract between the TWA and the employing firm (see the 2003 *Economic Survey*). However, as a result of the application of temporary contract rules, TWAs cannot generally hire workers on a temporary contract for more than 2 years[32] and cannot generally reemploy workers on a temporary contract they have employed at any previous point in time on a temporary contract for two years.[33]

Recent legislation stipulates that work and pay conditions of TWA workers have to be the same as those of comparable regular workers employed at the same firm, unless a collective agreement for TWA workers makes different provisions.[34] Unemployed individuals are exempt from this rule. Nonetheless, employment of unskilled workers in TWAs might fall as a result. Following the introduction of the legislation, the trade union associations negotiated a collective contract

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with TWA employers which reduced the gap in gross pay between TWA workers and regular workers while allowing unemployed workers to be hired at temporarily lower wage rates. According to a survey of TWAs, prior to the new collective contract TWA wage rates of many unskilled workers were substantially below pay rates of similar workers with regular contracts,[35] suggesting that the collective contract may reduce chances of employment especially for young unskilled workers, who have been the key target for recruitment by TWAs and who record an unemployment rate far above average. The differences in pay between TWA workers and regular workers before the introduction of the new legislation may also point to a lack of adjustment of regular workers' pay to labour market conditions, suggesting scope to improve wage setting more generally (see below).

The effect of the collective contract on low-skill employment in TWAs should be evaluated and, if necessary, the legal requirement stipulating harmonization of pay of TWA workers with respect to regular workers in the absence of a collective bargain repealed, as it affects the bargaining outcomes in the TWA industry. In addition, further efforts should be made to facilitate the use of temporary contracts by TWAs. In particular restrictions on the reemployment of workers on temporary contracts previously engaged in the same TWA should be eased.

A new programme providing subsidies to TWAs taking on unemployed job searchers has so far had limited success, as participation is modest and displacement effects are likely. The TWAs participating in the scheme are given a monthly subsidy (determined through competitive tendering among agencies) for the duration of nine months for each seeker they take on as well as for each placement to a regular job. The subsidy decreases with the duration of the spell at the TWA to encourage early placement into a regular job. In June 2004, 26 000 participants were enrolled in the scheme. The monthly exit rate of participants into regular employment stood at about 5 per cent,[36] suggesting only limited success of the scheme so far, which however should be subject to evaluation. The skill profile of workers and incidence for long-term unemployment in the subsidized programme appears to be similar to the skill profile in conventional TWAs and the incidence of long-term unemployment does not appear to be substantially higher, suggesting that displacement effects are likely although contracts are meant to reduce such displacement by targeting the subsidies at specific skill groups.[37] On the other hand, participation may reveal availability to work, as TWAs dismiss workers on the grounds of uncooperative behaviour, but there is no evidence to indicate that local labour offices link these dismissals to the sanctions regime.[38]

Differentiation of wages to reflect conditions at the firm level should be supported

Despite progress in recent years, collective contracts do not yet give sufficient room to allow firm-specific and regional conditions to determine bargained wage outcomes. The social partners in east Germany, for example, adopted the recent collective contract in the metal and engineeringindustry, first negotiated in

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west Germany, without any changes, notwithstanding much higher unemployment in the eastern *Länder*, suggesting that local labour market conditions do not sufficiently influence bargaining outcomes. Indeed, in industrial sectors where coverage by sector-wide wage bargaining is high and which are little exposed to competition little pressure on wages has been found.[39] The lack of flexibility for firms to deviate from collectively-set pay and working time has prompted firms to leave the collective bargaining framework (see the 2001 *Economic Survey*). Ineffective ALMPs as well as subsidies to industry, notably construction, have also contributed to the lack of responsiveness of wages to labour market conditions in east Germany.[40] Moreover, wage differentiation by qualification has not increased over the last two decades, despite rising unemployment among the low-qualified.[41]

Legal restrictions prevent employers and workers to strike firm-level agreements on pay and working time without the agreement of the parties involved in the collective agreements. Firms which are members in the employers' associations can only deviate from collective contracts without the agreement of social partners if the firm's pay is higher or the working time lower than stipulated in the collective contract (*Günstigkeitsprinzip*). The scope for wage determination at the company level should be widened, so as to better align wage contracts with labour market conditions. This might require relaxing pertinent legal restrictions. This would preserve the advantages of the collective bargaining framework, such as lower transaction costs in wage bargaining.

Government action in support of collectively bargained contracts contributes to insulating collectively bargained wages from labour market conditions, as the option for firms to leave collective bargaining arrangements is likely to help moderate collective wage settlements. Specifically, powers of the federal government to impose the collective bargain outcomes on all firms[42] (*Allgemeinverbindlicherklärungen*) reduce the scope for firm-level agreements outside collective bargains, notably in the construction industry. In addition, some states have adopted legal provisions restricting public construction procurement to firms which pay collectively agreed wage rates prevailing locally. Further legal provisions on public procurement aim at eliminating contractors who offer wages which are substantially below competitors' wages or who do not meet quotas for the employment of women.[43] These legal provisions increase the cost of public procurement, transaction costs, and reduce competition. Legal provisions linking procurement to collective wage agreements should therefore be abolished. The administrative extension of collectively bargained contracts should be further reduced.

The apprenticeship system requires more rapid adjustment to changing skill requirements

The apprenticeship system has benefited labour market performance in Germany, contributing to relatively low youth unemployment and producing skills of high quality. Excess demand for traineeships in recent years has given rise to

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demands for a contribution and subsidy scheme aimed at increasing the supply of apprenticeships at times of excess demand, but would entail more costs than benefits. The proposed contribution would be levied on enterprises providing relatively few apprenticeships and its proceeds would subsidize enterprises providing many apprenticeships. While externalities can, in principle, justify reducing the marginal cost of apprenticeship supply,[44] the contribution scheme could introduce distortions within the apprenticeship system or encourage contribution evasion activities, adding to administrative opacity. Employers, for example, might respond by supplying inexpensive but less productive apprenticeships. Although the scheme under discussion would only be implemented in those years in which supply falls short of demand, it may entail substantial administrative costs, estimated at € 50 million per year by the government. Following a commitment of German employers to increase the number of traineeships, the government has decided that the scheme should not be introduced.

The supply of apprenticeships has tended to reflect labour market conditions, suggesting that measures to support weak job creation are required rather than specific measures to address the current apprenticeship shortage. Flexibility of pay for trainees would go some way in preventing imbalances of supply in demand. Indeed, in the 1990s the supply of apprenticeships outstripped demand by a large margin, pushing up pay rates. A recent initiative of the government, adopted in 2003, to reduce the costs of licensing of staff for teaching of trainees within the plant will increase the supply of apprenticeships as well as help firms become more flexible in adjusting to fluctuations of demand for apprenticeships.

The system also requires a continuous upgrading of formal training profiles. For example, for 2004 the curriculum for 30 professions is scheduled to be updated. While the federal government is responsible for the definition of new curricula, according to an agreement based on a consensus they are involving the social partners. In the recent past, the consensus process appears to have slowed the adoption of new curricula, with the introduction of new training paths of shorter duration having been opposed by the unions. This is at variance with the requirements stemming from rapid technological change and innovation. The government is planning, as part of its programme to reduce red tape, to accelerate the modernisation of training programmes, streamlining the decision-making process. Delays in adapting to new skill requirements should also be avoided by deviating from consensus-based decisions if necessary. As a step in this direction, the government has introduced training programmes with shorter durations in spite of the resistance of the trade unions, helping to further reduce costs of training and improve the matching of employers and searchers of traineeships.

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Notes

1. Previously named Federal Labour Office (*Bundesanstalt für Arbeit*).
2. Bundesagentur für Arbeit (2004). Penalties for rejecting a job amount to 30 per cent of benefits for up to three months. Young unemployed job seekers' benefits can be fully withdrawn for this period.
3. However, in the first 3 months of an unemployment spell the unemployed receiving tier I unemployment benefit do not have to accept job offers which pay less than 80 per cent of the previous salary. After six months, recipients of tier I unemployment benefit do not have to accept a job paying less than the unemployment benefit.
4. Martin and Grubb (2001).
5. OECD (2003c).
6. OECD (2002).
7. See, *e.g.*, Scarpetta (1996), Nickell (1999). Shortening the duration of benefits has a larger impact on employment than cutting the level of the benefit.
8. The supplement makes up two thirds of the difference between tier I and tier II benefit, up to a maximum of € 160 per month for a single worker, with additional supplements for dependants, and is halved in the second year.
9. This was recommended, for example, in OECD (1994).
10. The withdrawal rate indicates how many cents of social assistance a benefit recipient looses as her net income increases by one euro.
11. This measure replaces the wage subsidies of the *Mainz* model. See the 2003 *Economic Survey*.
12. See Boeters *et al.* (2003). According to the results of this study, a 0 per cent withdrawal rate for earnings up to € 300 (for a single worker without children) and a 50 per cent withdrawal rate for all higher earnings, combined with a 50 per cent cut of social assistance for all individuals who are able work would increase unskilled employment by 50 000, and total employment by about 270 000. Steiner and Jocobebbinghaus (2004) also estimate modest employment effects of a subsidy of low wage earners combined with a cut in social assistance.
13. 12 per cent for employees providing domestic services to private households. See Rudolph (2003b).
14. The tax advantage for second earners, such as married women with a working spouse, and individuals on second jobs are fully eliminated once earnings exceed the *Mini-Job* threshold.
15. See Reinberg and Hummel (2003).

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16. On the basis of a micro-simulation model Steiner and Wrohlich (2004a) estimate that the *Mini* and *Midi* jobs only result in a small increase of employment. According to data from the PES, 88 per cent of individuals taking up *Mini- Jobs* between June and September 2003 took up them up as second jobs.
17. Kolb and Warwatz (2001).
18. The number of individuals on *Mini-Jobs* was estimated at 6.4 million in March 2004.
19. According to the OECD PISA study, 10 per cent of 15 year olds in Germany did not reach basic literacy competencies, the third highest proportion in the OECD (OECD, 2003b).
20. Martin and Grubb (2001).
21. However, these projections do not include recent pension and unemployment benefit reforms which will have some positive impact on labour force participation of older workers.
22. This probability has been approximated as the probability of men aged between 26 and 29 entering the labour force 5 years later, when they are between 31 and 35 years old.
23. Burniaux *et al.* (2003).
24. Jaumotte (2003).
25. Increase in the participation rate of women aged 25-54, 2000-2025. See Burniaux *et al.* (2003). Steiner and Wrohlich (2004b) also show that measures to equalize tax rates between primary and secondary earners would increase female labour supply in Germany. While labour supply of men would fall, this would only offset a fraction of higher female labour supply.
26. See Burniaux *et al.* (2003). The study simulates an increase in childcare subsidies to the level in Denmark, as a percentage of GDP, the highest in the OECD.
27. Bielinski *et al.* (2003) analyze survey evidence from personnel managers in Germany indicating that large firms have lower worker turnover rates than small firms. Small firms are more likely to resort to temporary contracts and to personnel from temporary work agencies in order to avoid EPL-related costs OECD (2003a) provides cross-country evidence that more stringent EPL has adverse effects on the creation of small firms.
28. Elmeskov *et al.* (1998), Nicoletti *et al.* (2001).
29. In addition, the new legislation allows firms to override social criteria to keep workers who are essential for firm performance.
30. OECD (2002).
31. Katz and Krueger (1999) show that in the US the spread of TWAs shifted the Beveridge curve inwards, contributing to the reduction of the NAIRU over the 1990s.
32. The rules on temporary contracts without specific reason (*sachgrundlose Befristung*) are generally applicable to contracts between TWAs and workers.
33. Bothfeld and Kaiser (2003).
34. Rudolph (2003a). Up to 2002, equal conditions only applied to TWA workers employed for more than 12 months by the same firm.
35. ZEW (2003). About 50 per cent of the sampled TWAs paid unskilled TWA workers (*Hilfsarbeiter*) at least 10 per cent less than the wage of a similar worker on a regular contract. 35 per cent of TWAs paid skilled manual workers at least 10 per cent less than regular workers, with most TWAs paying at least 25 per cent less than a regular worker's wage. According to Kvasnicka and Werwatz (2002), the average wage differential between TWA and other workers with otherwise similar characteristics amounted to about 20 per

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cent on average before the reform, with the differentials being particularly large for unskilled workers.

36. Estimate based on data up to October 2003 in Jahn and Windsheimer (2004b).
37. 14 per cent of participants are long-term unemployed (compared to 8 per cent in unsubsidized TWAs) and 61 per cent have vocational qualifications at secondary level (54 per cent in unsubsidized TWAs). See Jahn and Windsheimer (2004a).
38. Jahn and Windsheimer (2004b).
39. See Franz and Steiner (2000).
40. Wurzel (2001).
41. Empirical evidence suggests that the wage distribution spread out only in eastern Germany immediately after reunification (see Christensen, 2003).
42. Including firms which are not organised in employers' associations to which the collective contracts do not apply in the absence of the *Allgemeinverbindlicherklärungen*.
43. Besenthal (2004).
44. Externalities result from reduced risk of benefit dependency of trainees, a reduction of social exclusion and, as with all human capital investments, productivity spill-overs. Disincentives for the supply apprenticeships also result from the transferability of apprenticeship-provided human capital, as apprentices are paid wages above their marginal product. Coordination within employers' associations has thus far prevented such disincentives.

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IV. Fostering product market competition

As outlined in the 2003 *Economic Survey*, pro-competitive regulation of product markets can contribute to improving productivity performance, in part through stimulating innovation (see Chapter 5). Recent empirical evidence also shows that a regulatory stance conducive to product market competition increases investment, as producers in competitive markets have stronger incentives to expand production capacity when product demand increases.[1] Competitive product markets, by reducing monopoly rents, also encourage wage moderation, especially in countries with unionised wage bargaining, such as Germany. The government has embarked on a number of reforms since the previous *Economic Survey*, notably in the handicrafts sector, which move the economy towards a more pro-competitive stance. However, scope for improvement remains. In the network industries, the key challenge continues to be securing competition through unrestricted access at non-discriminatory non-excessive charges.

Barriers to competition need to be lowered further

Reform of handicrafts regulation has lowered entry barriers, removing the master certificate requirement in many crafts. Prior to reform, owners of handicraft businesses were generally required to hold a master certificate,[2] in addition to completing basic vocation training in the apprenticeship system, which takes 2 to 3 years to complete. The certificates resulted in high entry barriers, as the training and examination costs amount up to € 8 000 and require more than a year of full-time training in some engineering-related crafts. Even in less technical handicrafts, such as hairdressing, the costs of the master certificate are considerable, with the monetary component estimated at € 5 000 and six months of full-time training. As a result of the reform, in force since the beginning of 2004, 53 out of 94 craft trades cease to be subject to entry requirements. The new legislation has retained restrictions in 41 trades which were selected on the basis of two criteria: first, the presence of health and safety risks that would arise if the work were not supervised by staff trained to the *master* level, and, second, the provision, by the handicraft in question, of a relatively large number of vocational apprenticeships.[3] In 35 of these crafts, however, work experience (a minimum of six years) can replace the "master" degree. In the six remaining crafts a "master" remains neces-

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sary, but the requirement ceases to apply to the owner and an employee may hold the degree instead. Entry requirements have also been lifted for individuals exercising related professions, such as engineers. However, about 90 per cent of craft businesses are active in the 41 crafts where entry requirements remain in place, employing 85 per cent of handicraft workers. The government originally envisaged to remove entry requirements for 12 additional crafts, suggesting further scope to ease business setup costs. In particular, the provision of vocational training should not be a criterion for limiting firm entry to product market segments.

Firms remain subject to compulsory contributions to business associations and binding price schedules for professional services from the respective associations hamper competition. New legislation exempts new craft businesses from contributions to the crafts association for a duration of four years. However, the compulsory contributions over the whole expected lifetime of a crafts business deter entry, so they should be reduced further. Professional associations' fee schedules, which are exempt from anti-collusion provisions in the competition law, limit price competition among members. In some professions, these fee schedules are legally binding. The government intends to both limit the coverage of a legally binding fee schedule for architects and engineers and to allow private contracts to deviate from the fee schedule. The legally-set price schedule, should be phased out as soon as possible, and should not be substituted by a recommended price schedule in order to ensure fully market-determined price setting. Regulation of some professional services in Germany is stricter than in most OECD countries, suggesting further room for reform (Figure 4.1). A recent study confirms that professional services are more heavily regulated in Germany than in most EU-15 countries.[4] Scope for extending market-determined prices for other professional services should be explored.

In a further initiative to foster competition, parliament has passed legislation removing restrictions on discounting in the retail sector. However, the prohibition of below-cost pricing remains in place. This restriction should be abolished in the absence of predatory pricing or monopoly power. Moreover, relatively low retailing outlet density may limit competition in retailing in Germany, on which regulatory action, such as zoning regulation, is likely to exert influence. At the same time, productivity in retailing falls short of best-performing European countries by 20 per cent (Table 4.1). The government also plans to strengthen competition law enforcement by giving consumer associations the right to litigate against collusion on behalf of consumers.

State aid to support product markets impairs economic performance, protecting supported producers from competition and preventing market exit of unproductive firms. While some targeted subsidies in transport may be justified *e.g.* as a result of increasing returns to scale in public transport, subsidization distorts relative prices and, to the extent that it is granted to selected producers, may

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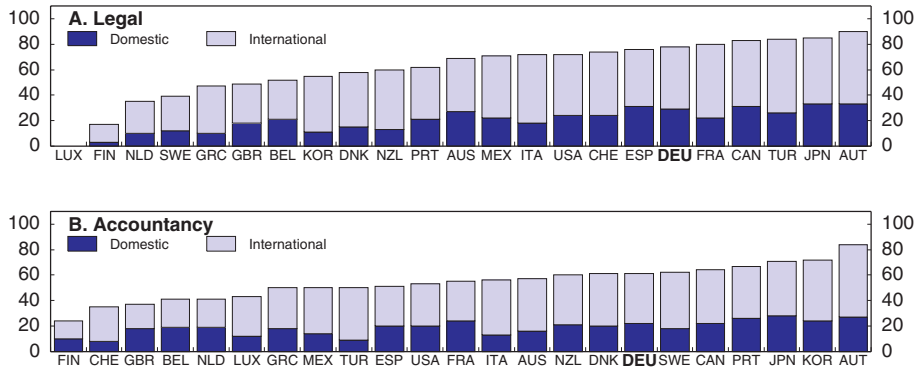
Figure 4.1. **Regulations of professions: restrictiveness indices for OECD countries**[1]



1. The restrictiveness index ranges from 0 (least restrictive) to 100 (most restrictive). International (domestic) indices indicate the ease or difficulty for a foreigner (domestic resident) to establish and operate in a country.

Source: Nguyen-Hong (2000).

Table 4.1. **Key structural features of the retail distribution sector, 2000**

	Outlet density[1]	Retail distribution, total value added per employed person[2]
Austria	43	96
Belgium	80	96
Denmark	47	91
Finland	46	116
France	64	118
Germany	**35**	**100**
Ireland	36	84
Italy	130	72
Netherlands	54	71
Portugal	150	38
Spain	133	65
Sweden	65	115
United Kingdom	36	88
EU15	71	88
excluding Italy, Portugal, Spain	51	98
Norway	68	99
Switzerland	56	88

1. Number of enterprises per 10 000 inhabitants.
2. Germany = 100.

Source: Eurostat, New Cronos.

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generate rents, thus raising wages and reducing employment, in addition to weighing on the government's finances (Chapter 2). State aid to specific sectors in Germany is high relative to other high-income European countries (Table 4.2) and should be phased out as soon as possible, except to the extent that subsidies clearly help correct market failures. In the same vein, efforts to privatize government owned firms should be stepped up.

Table 4.2. **Share of state aid by sector, 2001**[1]

	Per cent GDP				
	Manufacturing	Transport	Agriculture and fisheries	Coal	Total
Finland	0.24	0.26	1.00		1.53
Denmark	0.35	0.55	0.12		1.33
Luxembourg	0.14	0.92	0.16		1.24
Ireland	0.40	0.31	0.20		1.13
Germany	**0.37**	**0.45**	**0.10**	**0.20**	**1.12**
France	0.23	0.43	0.23	0.06	1.07
Italy	0.32	0.55	0.10		0.98
Austria	0.23	0.30	0.41		0.97
Netherlands	0.14	0.58	0.20		0.93
Sweden	0.15	0.39	0.16		0.76
Belgium	0.02	0.09	0.01		0.13
United Kingdom	0.01	0.04	0.01		0.07

1. Defined by declared primary objective of the aid. Ranked by total aid as per cent of GDP.
Source: European Commission.

The regulatory framework in the telecoms industry needs to be strengthened

Competition in the telecommunications market is especially relevant for productivity performance. ICT use contributes to capital deepening and can spur innovation, triggering complementary innovations which give rise to total factor productivity growth (see Box 5.5, Chapter 5).

A new law on telecoms regulation addresses some issues which have thus far hampered competition in the sector.[5] The law clarifies rules on abusive behaviour of market domination, defining below-cost pricing and bundling of services as abuse of market power, helping to enhance the effectiveness of regulatory action to prevent such practices. In addition, procedures to be followed by the regulator to prevent non-competitive pricing have been streamlined.

The law will give more discretion to the regulator in determining whether or not and how to regulate access to telecoms networks than the law currently does, potentially generating uncertainty among market participants, which might adversely affect investment. In particular, while the law lists obligations on the incumbent to ensure access for competitors to the incumbent's network, the list is

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not exhaustive and the regulator need not resort to all the measures listed. Uncertainty on the future regulatory stance is likely to have more adverse effects on market entrants than on incumbents, reducing competition.[6] The law will strengthen the powers of the chairperson of the regulatory authority, whose appointment is a political decision, increasing the scope for political influence of the federal government on the regulator, potentially exacerbating perceptions among market participants of a conflict of interests within the government, which both oversees the telecoms regulator and owns 42 per cent of DTAG. The government should privatise its remaining stake in DTAG in order to address concerns over the government's conflict of interest. Lengthening the chairperson's tenure of office, while removing the option of a reappointment, could strengthen the perceived independence of the regulator.

The regulatory framework should ensure that *ex ante* regulation is applied to services where firms exercise significant market power over a significant time period. In contrast to many other OECD countries, Germany does not regulate rates for fixed-to-mobile connections. By contrast, the UK, for example, has concluded that price cap control is the only remedy to prevent excessive pricing of mobile network call termination. Moreover, the new legislation might limit *ex ante* regulation further, even if the market is dominated by one player.[7] *Ex ante* regulation should continue to be prescribed whenever market domination is present, as recommended in the 2003 *Economic Survey*.

The regulatory framework in other network industries has improved but important challenges remain

The telecoms regulator's remit has been extended to the gas and electricity industries, providing scope for improving competition in both sectors. Germany previously relied on self-regulation under the oversight of the competition authority (*Bundeskartellamt*) which, however, is not equipped with *ex ante* regulatory powers widely used by network industry regulators. This system reduced market entry, contributing to high gas prices and high electricity prices for private households.[8] Indeed, network access fees have appeared to be high in both industries. The new regulator may be better placed to ensure that network access prices provide more scope for competition and to address discriminatory behaviour of network operators toward competitors, promoting further separation of network activities from other activities and improving accounting information, to help prevent cross-subsidization being used as an entry barrier against potential competitors.[9] In addition, the creation of a multi-sector regulator may reduce the scope for regulatory capture (see the 2001 *Economic Survey*).

Further liberalisation in the postal and water services sectors could secure more competition and efficiency. The government has decided to prolong the monopoly of the mostly state-owned incumbent, *Deutsche Post*, on domestic

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letter delivery below certain weight thresholds to 2007. *Deutsche Post* is also exempt from sales tax on revenues generated in competitive markets, distorting competition. Steps to create a more level playing field would increase the effectiveness of competition. Competition is expected to increase once all postal services are deregulated in 2007. While competition is more difficult to introduce in water services, greater use of benchmarking and competitive tendering for water contracts could improve efficiency. As steps in this direction, the government plans to require that all companies in the wastewater disposal sector are treated equally and to set up guidance for the development of a benchmarking system.

Burdensome regulations need to be reduced

Lowering administrative costs helps increase labour productivity, as firms can shift resources to productive use, and promote firm entry, as administrative costs tend to be independent of firm size, so are particularly burdensome for small enterprises. The Federal Ministry of Justice estimates that 17 per cent of federal regulations are unnecessary. Examples for unnecessary duplication of regulation include government collection of data from enterprises and individuals which are already available within public sector institutions as well as multiple sources of legislation, for example on public procurement. The government has taken legislative steps towards their elimination.

The government has launched a number of initiatives to reduce existing unnecessary regulatory burdens. Recent legislation has simplified accounting and tax procedures on enterprises, lifting size thresholds above which full accounting obligations as well as turnover tax liability apply and streamlining rules concerning the determination of tax liabilities of small enterprises. Draft legislation will become effective this year to remove duplication of legislation concerning product and food safety as well as environmental regulation of construction. The government is also planning to ease specific regulatory burdens on service industries and has simplified statistical reporting duties on enterprises, preventing duplication of existing administrative data. Revisions of income and payroll tax rules, to be introduced in 2006, aim at replacing paper-based by computer-based procedures, and the government is increasing the use of information technology to streamline the provision of administrative services. More use of evaluation and quantitative, evidence-based assessments should be integrated in decision-making to prevent the adoption of new burdensome rules. Guidelines to underpin regulatory impact analysis are in place. Indeed, in 2000 regulatory impact analysis became compulsory for all new legislation. However, formal sanctions to enforce the guidelines are lacking and resources devoted to their implementation appear to be insufficient. The adoption of a central regulatory quality unit has proved useful in a number of OECD countries to improve regulatory impact analysis, helping to disseminate guidance and advocate policy initiatives. Cost-benefit analysis as

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well as temporary introduction of regulation, to ensure its subsequent review, could be applied more frequently. A more transparent, systematic approach to public consultations could help widen stakeholder participation beyond well-organised interest groups.[10]

Improving transparency and facilitating access for foreign suppliers in public procurement rules would contribute significantly to competition and reduce the cost of public sector purchases, as the public procurement market accounts for about 17 per cent of German GDP.[11] The German procurement framework is complex, hindering market entry, as it is based on a wide range of different regulations. In addition, the authorities frequently split contracts. While the purpose of this practice is to facilitate participation of SMEs in procurement, small procurement contracts do not have to be published EU-wide, so that this practice has the effect of curtailing foreign firms' access to the contracts.[12] In addition, procurement rules below the EU threshold are set by the *Länder*, leading to a multitude of regional differences in regulations. These rules are set by committees, which represent business associations and government officials, favouring incumbent market participants to the detriment of supra-regional market entrants. The government is planning to simplify public procurement legislation, removing the role of business associations in most sectors, although on current plans business associations would continue to play a role in construction contracts below € 6 million. Provisions in some *Länder*, linking construction procurement to adherence to local pay conditions, further limit competition (see Chapter 3). Measures to increase the transparency of the procurement process, eliminating the role of business associations completely and ensuring wider publication of procurement contracts at the level of the EU would further benefit competition for public contracts.

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Notes

1. Alesina *et al.* (2003).
2. Following a constitutional court ruling, some exceptions had been introduced before the reform. See OECD (2003a).
3. BMF (2004).
4. Paterson *et al.* (2003).
5. See OECD (2001 and 2003).
6. M*onopolkommission* (2004).
7. The new telecoms law, if enacted, might result in ex-ante regulation of prices for network access being restricted to cases of pure monopoly (M*onopolkomission*, 2004).
8. Electricity prices for private households, both including and excluding the tax component are among the highest in a list of sixteen OECD countries. See OECD (2003b). Industrial customers have managed to exercise more pressure on electricity prices as a result of their market power. See OECD (2001).
9. See the 2003 E*conomic* S*urvey of* G*ermany* and OECD (2004) for related recommendations.
10. OECD (2004).
11. OECD (2004).
12. OECD (2004).

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V. Improving the capacity to innovate

Reviving the dynamism of the German economy also requires policy measures that strengthen productivity growth. Total factor productivity growth – the growth in output, which can be achieved given all inputs combined – decelerated in Germany in the 1990s, as it did in several other countries within the OECD. Innovation is a key element stimulating total factor productivity. This, in turn, can potentially spur higher employment of both capital and labour, which is a source of higher economic growth in its own. Indeed, empirical studies establish a strong positive relationship between research and development (R&D) and per capita GDP growth.[1] The recent policy debate in Germany has therefore focussed increasingly on innovation as a means to foster economic activity. This led the federal government to establish a commission whose task is to propose policy measures, in the second half of this year, on how to enhance innovation. Indeed, "innovation" spans a wide range of activities ranging from invention to its diffusion and related organisational changes (Box 5.1), with private firms being the main actors in Germany (Box 5.2). While several factors impact on innovation in a complex way – with knowledge about relevant interactions often still being quite limited – regulatory reform of product, capital and labour markets can have significantly beneficial effects on innovation. This chapter aims at highlighting a number of policy issues affecting innovation and proposes options to improve framework conditions so as to raise the capacity of the German economy to innovate. Setting framework conditions conducive to innovation is also a necessary ingredient for government spending for innovation to be effective. The German government plans to increase public spending from 0.8 per cent of GDP in 2002 to 1 per cent by 2010. Since a large part of R&D spending is on personnel this requires, *inter alia*, that framework conditions support the supply of high-qualified research personnel. By contrast, regulatory supply constraints – also with respect to new firms – would risk wasting resources.[2] The next section of the chapter provides an overview about stylised facts that position German innovation performance in an international context. The following sections examine important aspects of the German economy which are relevant for innovation. This concerns product, capital and labour markets as well as taxation and the tertiary education system. The final section highlights a few aspects that are relevant for government support to enterprises.

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Box 5.1. Innovation – concepts and data

What is "innovation"?

Innovation comprises activities which eventually lead to the introduction of new products – "product innovation" – or the introduction of new production techniques – "process innovation". Innovation thus spans a wide range of diverse activities, including, for example, basic research, the acquisition of new production equipment or the invention of a new medicine. Innovation covers, in principle, both the application of existing knowledge in a new setting as well as the production of new knowledge.[1] Empirical evidence shows that the application of existing technology – for example, ICT – requires additional innovations to generate productivity gains, such as changes in work procedures and the organisation of production processes.[2] Innovation is a main driving force for economic growth. It generates higher income levels directly by increasing the productivity of inputs in the production process (total factor productivity). In addition, improvements in total factor productivity motivate capital deepening which also contributes to economic growth.[3]

What are characteristic innovation processes?

The characteristics of the innovation process differ across different types of industrial activity. Empirical evidence[4] suggests that industrial innovation patterns might be characterized either by radical innovation or cumulative innovation.[5] Firms innovating in a cumulative way build on previous innovation efforts in an incremental fashion. Fundamental changes in the technology are less frequent. Cumulative innovation is therefore more likely in technologies with a well-established technological platform. Cumulative innovation often builds on firm- or individual-specific competencies. Specific competencies may be important in the innovation process, for example, when skills are costly to be codified (and thus costly to be transferred), or when they are embedded in complex organizational routines.[6] Cumulative innovation helps firms erect barriers to entry tending to generate a prevalence of established firms in the innovation process. Technological stability tends to be reflected in stability of the hierarchy of firms contributing to innovation and thus in high concentration in innovating firms.

If innovation is *radical*, fundamental changes in technology occur at high frequency, so they are more likely to predominate in new technologies (*e.g.* biotechnology). This innovation pattern is characterized by *creative destruction*, with a large contribution of entry of firms in the innovation process which replace firms whose technologies become obsolete. Technological instability is reflected in more frequent changes in the firms which contribute to the industry's innovative output.

While innovation in specific industries does not always clearly fall under either type of innovation, automobile manufacturing, the chemical industry and parts of machinery production – which make up a large proportion of German exports – are generally considered to be characterized by *cumulative* innovation.[7]

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Box 5.1. **Innovation – concepts and data** (*cont.*)

How is innovation measured?

The main indicator used to measure innovation inputs is spending on R&D, often expressed as a share of GDP. For Germany, R&D spending has been estimated to cover about 50 per cent of spending on innovation. R&D spending is often considered a useful indicator of overall innovation effort, assuming a positive correlation between the two.[8] Total R&D spending data are available in time series that are fairly comparable across countries. However, the widening in the coverage of R&D surveys in some countries affects the comparability of business R&D spending in the service sector.

Manufacturing industries are classified in *high-*, *medium-high*, *medium-low* and *low-tech* industries, depending on their R&D intensity. *High tech* industries have the highest R&D intensities. R&D intensity is measured as R&D spending relative to turnover or relative to value added.[9] New technologies, such as ICT and biotechnology, fall into the category of *high tech* industries. The traditional German export industries – automobile manufacturing, chemicals and machinery – are medium high-tech industries.

The main indicator of the output of innovation processes are patent statistics. However, individual patents differ considerably from one another in usefulness, and thus in value, and the propensity to use patents to protect inventions also differs across industries. For example, survey evidence from Germany indicates that more than 30 per cent of innovating firms engaged in transport manufacturing (including production of automobiles) and in the machinery manufacturing sectors use patents to protect their inventions, whereas this is true for less than 5 per cent of innovating firms in the ICT and service sector industries.[10] Also, patent coverage varies considerably across countries and over time, partly being the result of changing regimes of intellectual property right protection. This chapter draws on "triadic patent family" statistics, with approved patents broken down by the nationality of the applicant.[11] These are applications for patents at the three most important patent offices in the world,[12] so that they are likely to have substantial economic value. Patent families also help to prevent "home bias" in international comparisons of patent statistics, as innovators are more likely to file patents at their own country's patent office. It turns out that patents are closely correlated with R&D spending.

Surveys of enterprises provide additional information on both inputs and results of innovation in the enterprise sector based on self-assessment. However, time series of data generated by these surveys in Germany are short and can for the most part not be compared to data from other countries. Cross country comparable survey data have been analysed for the EU for 1998-2000 and changes in survey design limit their comparability over time. These surveys provide information on, for instance, spending on all innovation activities. Innovative output measures provided by these surveys include cost reductions achieved as a result of innovations and proportions of sales of new products in total turnover.

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Box 5.1. **Innovation – concepts and data** (*cont.*)

In addition, data on the spending on new technologies, notably ICT, provide information about the extent to which the economy is able to incorporate existing knowledge in production processes.

1. Products and production techniques are typically defined as innovative if they are new to the respective firm, rather than new to the world or to the market in which the firm operates. See the definition of innovation in the European innovation surveys (European Commission, 2001, 2003).
2. See *e.g.* Licht and Moch (1999) and Hempell (2002a, 2002b).
3. See, *e.g.* Jones (1998).
4. Using principal component analysis. See Breschi, Malerba and Orsenigo (2000); Malerba and Orsenigo (1997).
5. Following the terminology in Soskice (1997). Bassanini and Ernst (2002a, b) use the terms entrepreneurial instead of radical and routinized instead of cumulative.
6. See Bassanini and Ernst (2002b) and the references therein.
7. See, *e.g.* Breschi *et al.* (2000).
8. See, *e.g.* Legler (2003).
9. See OECD (2003b) and Legler (2003). Notwithstanding differences in the criteria to determine R&D intensity, the actual classification of industries is similar across methods used.
10. Rammer (2003).
11. If several applicants have different nationalities, the patent is attributed proportionally to the countries of origin of the applicants. Patents are attributed to the year in which the first approval is granted ("priority date").
12. The European, Japanese and US patent offices.

Innovation in Germany – stylised facts

Innovative activities in Germany are strong…

Key indicators show Germany belonging to the countries in the OECD with strong innovation activity, as reflected in one of the upper ranks in the number of patents filed per population and above average levels of R&D spending in terms of GDP (Figure 5.1). Similarly, survey information from the *European Community Innovation Survey* on the proportion of enterprises reporting product or process innovations places German manufacturing firms second, behind Ireland, and first in services, while it ranks third in the proportion of turnover spent on innovation in both manufacturing and services.[3] Germany is also one of the most attractive destinations for foreign-based R&D activities and R&D intensive industries contribute a larger share to exports in Germany than in OECD countries on average.[4]

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Box 5.2. Key players in German R&D

The enterprise sector accounts for about two thirds of total R&D spending in Germany. As of 1999, it provided about 7 per cent of the funding to the public research institutions, while the government financed about 5 per cent of private sector-conducted R&D for civilian purposes. Germany occupies a middle position in comparison to other high-income countries with respect to the size of these cross-sector flows. The bulk of private sector R&D is conducted by large enterprises.

Germany's federal structure gives rise to a complex institutional structure of public research Most of the R&D financed by the public sector is conducted by the public research institutions, about half of which by universities. The largest of the institutes is the *Helmholtz Association of German Research Centres* (HGF) which conducts long-term oriented basic research in key technologies and includes large scale research facilities. The *Max Planck Society* also conducts basic research in selected areas, comprising 80 institutes. The 79 institutes of the *Leibniz Association of Research Institutes* conduct research in a broad range of disciplines. The *Fraunhofer Society* consists of 57 institutes. Its main objective is to promote technology transfer to industry. 40 per cent if it's funding is through contract research form enterprises. All research institutes are co-financed by the federal government and the respective states at varying rates and are independent from the federal government. In addition, the federal ministries and the *Länder* run more than 200 research laboratories. These offer routine services, such as measuring, testing and standardization, and are also engaged in applied research.

According to evidence from the European Innovation survey few enterprises in Germany report a lack of technological knowledge as a factor limiting their innovation activities. Evaluations of the four research associations, conducted by *ad hoc* commissions between 1996 and 2001, noted that the research output in most research institutes is often excellent and internationally competitive. A few institutes whose performance was found lacking were subsequently closed or restructured. However, the evaluations also noted scope for allocating a larger proportion of research funds through competitive tendering among institutes. Also, the research laboratories of the federal and *Länder* ministries have not comprehensively been subjected to evaluation.

The volume of government funding for the *Fraunhofer* institutes from the government is linked to success in obtaining research contracts from the private and public sector, allowing the institutes to both engage in basic research and technology transfer to private sector enterprises. Patented innovations of *Fraunhofer* institutes are often developed within projects financed by industry. The financing mode appears to be successful, as reflected in a nine fold increase of turnover compared to the 1960s and a strong patenting record. To secure the efficiency of the financing system in the future, agents should continue to observe that the output of the institutes are offered at market conditions.[1]

The government has taken steps to increase the scope for competition for research funds among institutes and plans to subject all research laboratories of the federal ministries to evaluation. A timetable of regular reevaluations would

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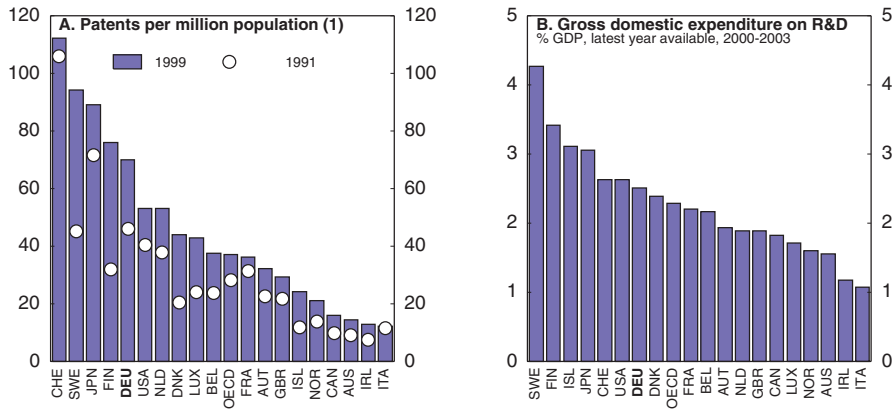
Box 5.2. **Key players in German R&D** (*cont.*)

further strengthen incentives for performance. The independence of evaluation commissions should also be ensured.[2] Steps should be taken to ensure market-determined prices for the use of patents of public research institutes.

1. See Rammer *et al.* (2004).
2. The evaluation of the Fraunhofer Institutes included representatives of German industry.

Figure 5.1. **Innovation activity**



1. Patent families filed at the European Patent Office, the US Patent and Trademark Office and the Japanese Patent Office. Data for 1999 are OECD estimates.

Source: OECD, Patents and Main Science and Technology Indicators Databases.

A relatively large proportion of patenting and R&D spending in Germany takes place in industrial sectors with a long-standing record of strong export performance. Patenting is largely concentrated in transport equipment (including automobiles), machinery, and in the chemical manufacturing industries, where concentration has increased further in recent years relative to other countries.[5] These industries are predominantly characterised by cumulative innovation patterns. Almost all industries with a high share in German patenting are *medium-high-tech*. A relatively small proportion of innovative activity, by contrast, is devoted to *high-tech* industries.[6] Innovation activities have also started to play an important role in eastern Germany (Box 5.3).

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Box 5.3. Innovation in the new *Länder*

Research-intensive sectors in the east contributed over-proportionately to the strengthening of eastern Germany's export market share. In manufacturing, since the middle of the 1990s export turnover of these sectors increased by some five percentage points more rapidly per year than the manufacturing average.[1] Between 1998 and 2002 production in research intensive sectors in manufacturing rose at an average annual rate of 8 per cent, as opposed to 5.9 per cent in eastern German manufacturing overall. In comparison to the west, manufacturing production in research-intensive sectors expanded more strongly, although the growth differential in favour of the east is larger in sectors that are not research-intensive.[2]

Empirical investigations also show that on the level of enterprises whose characteristics are comparable between the new and the old states research activity in the east is higher, largely owing to substantial government support. Eastern German research intensity is lower on the macro level, however: The share of the research-intensive sectors in industrial value added is almost 10 percentage points smaller in the east than in the west, and employment shares are significantly smaller than in the old states.

1. There is still a considerable gap between the export share in comparison to the west, however. While the export share in research intensive sectors stands at some 40 per cent in the new states it is 54 per cent in western Germany. See: Deutsches Institut für Wirtschaftsforschung (2003).
2. See Beer (2004).

... but appear to have weakened somewhat

However, there are signs that innovative activity has weakened somewhat. In terms of R&D spending as a share of GDP Germany has lost its leading position among OECD countries since the beginning of the 1990s. Spending dropped by some 0.3 per cent of GDP while it increased in the OECD as a whole (Figure 5.2, panel A). Business-financed R&D spending accounts for this relative decline, as it fell in the first half of the 1990s without fully recovering later to its former level. R&D spending of the government sector, net of public sector enterprises, also declined as a share of GDP over the last twenty years, but not more so than in the OECD at large (Figure 5.2, panel B).[7] Patenting statistics also suggest some weakening in Germany's relative position (Figure 5.1, panel A), although German patenting growth was well above the OECD average between 1991 and 2001, and exceeded patent growth in the US and Japan.[8] The country's position as a recipient of R&D spending of foreign-based US enterprises dropped from first to second (after the UK).[9]

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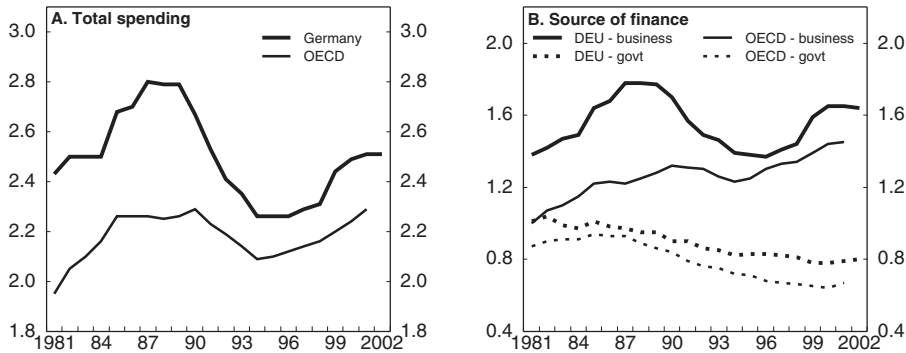
Figure 5.2. **Development of R&D spending**
Gross domestic expenditure on R&D as a percentage of GDP[1]



1. Data up to 1990 is for western Germany only. Business source means financed by private and public enterprises and institutes serving such enterprises. From 1992 onwards, government finance for Germany includes the private non-profit sector.

Source: OECD, Main Science and Technology Indicators Database.

To some extent the drop in innovative activity in the first part of the 1990s relates to the effects of reunification. R&D spending dropped as resources in both the public and the private sectors were shifted towards unification-related spending.[10] Weak economic growth commencing in the first half of the 1990s and temporarily squeezed profits are also likely to have adversely affected R&D spending. Structural factors contributed to the relatively slow recovery of German R&D activity as well. In the OECD as a whole, high-tech industries – such as ICT and biotechnology – experienced much faster growth in innovation, as reflected in patents, than other sectors. Therefore, countries devoting a relatively large proportion of their R&D spending to high-tech industries, such as Finland, Ireland, Canada and the US, also tended to experience larger increases in overall R&D spending as a proportion of GDP.[11] By contrast, Germany devoted a much smaller proportion of its innovation effort to high-tech industries than the OECD average.[12] The shares in patent applications of ICT and biotechnology are below the averages of both the OECD and the EU (Figure 5.3). Thus, Germany's share in the R&D surge in new high-tech technologies was moderate, contributing to the decline of its R&D spending relative to others.

At the same time, Germany's world export market share in pharmaceuticals declined substantially since the beginning of the 1990s. Medium-high-tech industries have a long record in Germany's strong export performance, and attract the largest share of German R&D spending and patenting. Nevertheless, Germany's export shares in machinery production also declined, with the notable exception of

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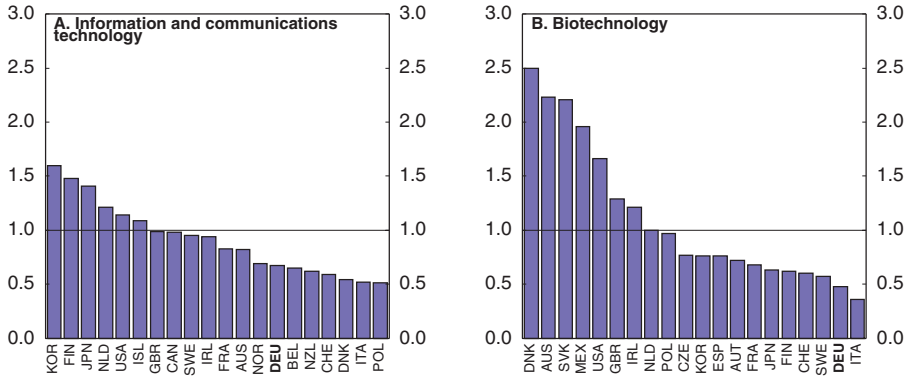
Figure 5.3. **Innovation in Òhigh-techÓ industries**
Specialisation index, average 1992-2000[1]



1. Share of country X (in OECD total) in ICT or biotechnology divided by share of country X (in OECD total) of all patent applications to the European Patent Office. Index greater than 1 indicates relatively specialised.
Source: OECD, Main Science and Technology Indicators Database.

automobiles. Mirroring this development, middle-income countries, such as Turkey, Korea, east European and Mediterranean countries, have reduced their comparative disadvantage in *middle-high tech* products, as reflected in diminishing net imports of these goods. Regarding high-technology goods, Germany remains a net importer.[13]

In Germany, firm entry has declined since the middle of the 1990s in both the new and the old states (relative to the labour force)[14]. The decline was particularly pronounced in the medium-high technology industries, in which the rate of new firm creation dropped by about a third between 1994 and 2001. By contrast, firm entry in the high-tech sector – where firm entries are most likely to make a substantial contribution to innovation and productivity growth – remained at about the same level as in 1993. Knowledge and technology intensive services recorded sizable increases in firm entry, though this did not fully compensate the losses in the medium-high tech field. As in many other European countries, employment growth rates of new firms appear much lower in Germany than in the United States. Moreover, entry rates in Germany appear to be relatively low, although cross country comparisons are difficult to make.[15]

Regulatory reform interacts with innovation

As outlined in the preceding section, key indicators point to Germany as a very innovative economy. Innovation activity appears largely concentrated on

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Box 5.4. The case of the pharmaceutical industry

Among the manufacturing industries in which Germany plays a leading role in world exports, the pharmaceuticals industry has fared least well relative to other main exporters (Table 5.1). The pharmaceutical industry has experienced substantial changes in innovation processes and, as a result, in marketing mechanisms. German firms adapted relatively slowly to these changes. For example, German pharmaceuticals were slower to divest non-pharmaceutical activities than their UK counterparts.

Table 5.1. **Market share in world exports of selected industries**
Top five exporters in 1991, per cent of world exports[1]

	1991	1995	2002
Pharmaceuticals			
Germany	**17.0**	**14.4**	**10.2**
United States	12.1	9.2	9.8
Switzerland	12.1	10.6	9.6
United Kingdom	11.3	10.6	9.1
France	10.2	9.6	9.1
Chemicals excluding pharmaceuticals			
Germany	**18.4**	**14.9**	**12.1**
United States	15.9	13.9	13.6
France	10.2	8.9	7.2
Netherlands	7.3	6.4	5.0
Japan	6.7	7.2	6.2
Machinery			
Germany	**20.2**	**17.2**	**16.1**
United States	16.1	15.1	16.3
Japan	14.0	15.1	10.4
Italy	8.7	8.3	7.9
United Kingdom	7.4	6.1	6.2
Vehicles			
Japan	18.5	16.5	13.4
Germany	**18.2**	**15.7**	**16.7**
United States	16.8	14.1	13.8
France	8.8	8.7	8.1
Canada	7.3	8.6	8.1

1. World exports measured in current US dollars.
Source: UN COMTRADE Database.

In the 1980s innovation strategies shifted to biotechnological techniques, replacing to some extent traditional chemical methods. These radical innovations in biotechnology have created strong incentives for new entry and much research is taking place in start-ups rather than established firms. In addition ongoing research in any one complex disease follows various distinct research trajectories. As a result of shortened product life cycles new distribution channels developed, requiring dramatically different competencies.*

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Box 5.4. **The case of the pharmaceutical industry** (*cont.*)

Biotechnology start-ups developed relatively slowly in Germany, compared with the UK whose pharmaceutical industry expanded rapidly. This difference is not attributable to price regulations as both countries have high generic drug penetration rates and firms are free to set prices of newly patented drugs. Relatively strong regulatory restrictions on genetic research in Germany are likely to explain part of the difference. Indeed, regulations were relaxed since 1993, while still being stricter than in the UK, and EU rules governing innovation in the pharmaceutical industry were harmonised in 1995. In recent years the number of German biotechnology firms has increased rapidly. Institutional factors such as stricter regulation in Germany might also have influenced different patterns of firm growth. While German pharmaceutical firms expanded mainly via acquisition of firms abroad, British firms mostly grew internally or through UK-based mergers.

* See Casper and Matraves (2003).

fields of traditional strength on export markets. By contrast, there is evidence that a smaller proportion of innovation activities are devoted to high-tech fields than in some other economies competing with Germany. Similarly, Germany has been less able than a number of other countries within the OECD to reap the benefits of ICT for raising the productivity performance in other industries (Box 5.5). Regulatory features of product, capital and labour markets affect an economy's capacity to innovate via various channels and may benefit different economic sectors to differing degrees. Interactions are often complex, and knowledge on their impact is deficient. Nevertheless, theoretical and empirical evidence allows a number of policy conclusions that are important to foster innovation. For example, empirical work suggests that administrative regulation can have significant adverse effects on the creation of enterprises, stemming not only from specific barriers to entry but also from administrative opacity more generally.[16] New firms make an important contribution to innovation because they can enter a market with the most productive combinations of inputs and, unlike established firms, do not incur adjustment costs when adopting new technologies and work practices.[17] Examples for such costs include making redundant or retraining employees, replacing machinery, or reorganising management procedures. Moreover, new firms create competitive pressures on existing firms, inducing them to innovate and improve performance, and lead to the replacement of obsolete firms. Accordingly, firm entry spurs innovation, contributing particularly strongly to productivity growth in

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"high-tech" sectors, notably in ICT-related industries.[18] Thus, regulatory reform fostering entry might be particularly effective in high-tech sectors. The example also shows that interactions between different regulations can play an important role. The contribution of enterprise start-ups to total factor productivity is likely to be larger the more stringent the employment protection regime is. This suggests that easing the regulatory environment for firm entry is likely to be of particular importance for the diffusion of innovation in countries where established firms face high costs of adjusting their production processes.

Protection of intellectual property rights and product market competition

Protecting intellectual property rights (IPRs) is important to raise incentives to innovate. While a key issue in the design of IPR protection is to find an efficient balance between raising the returns accruing to innovators by granting a certain degree of monopoly power and safeguarding the diffusion of innovation, evidence from surveys indicate that the degree of IPR protection within Germany is similar to the one in other high-income countries. Patent protection in Germany is among the strongest in the OECD, and the patent protection system has served as a model for other countries as well as for the European Patent Office.[19] The German system of protection for smaller inventions (*Gebrauchsmuster*), which do not qualify for patents, has also been adopted by many countries as it has been useful in encouraging incremental innovation and technological catch-up. In recent years further progress has been made in speeding up patent filing procedures in Germany, and the rights of producers to receive copyright fees have been widened.

To a large extent IPR protection worldwide – as opposed to protection by national law – is a relevant parameter determining the returns to innovation. IPR in Germany depends on policies of the European Union as well as the European Patent Convention, which established the European Patent Office.[20] The monetary cost of filing a patent at the European patent office is about five times higher than at the patent office of the US, notwithstanding significant reductions in application fees since 1996. High costs are in part attributable to legal complexity, as infringement cases need to be brought before the national courts of each country for which the European patent has been granted.[21] High costs are likely to weigh particularly heavily on small and medium-sized enterprises in Germany, as the proportion of small and medium sized enterprises which are engaged in innovation activity is large relative to other European countries.[22] The EU plans to introduce a "European Community Patent", linked with a European Community Patent Court. According to the European Commission, the European Community patent would be likely to reduce patenting costs substantially.

For a given level of IPR protection product market competition strengthens incentives to innovate, and therefore productivity growth. In a more competitive environment firms have to work harder to meet the performance of competing

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Box 5.5. The ICT sector in Germany

Rapid innovation has turned ICT into an important source of productivity improvements, through both the production and the use of ICT goods. Often the gains from the use of ICT appear to exceed those of ICT production. Specifically, ICT spurs economy-wide productivity growth via three channels:

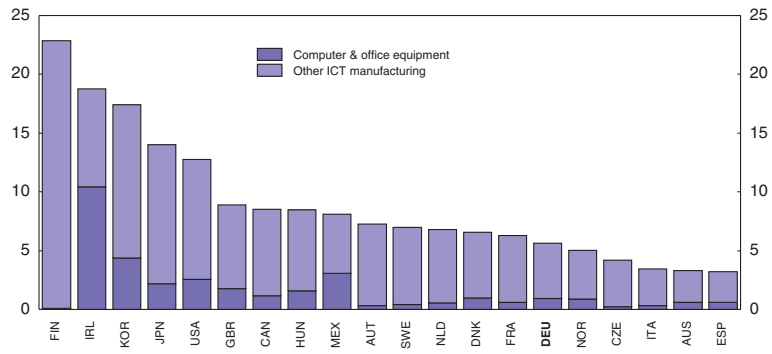
- Acceleration of productivity in the ICT producing sectors themselves and a growing size of the ICT producing sector.
- Increased use of ICT equipment in the production of other goods.
- Spill-over effects from the use of ICT, arising as a result of complementary innovations, *e.g.* in work practices or organization, as ICT equipment is introduced.[1]

Production of ICT goods and ICT has been less vivid in Germany than elsewhere…

Productivity growth in manufacturing of ICT goods has increased more strongly than in other industries in Germany, broadly mirroring similar developments in other countries, However, the share of ICT goods in manufacturing production in Germany is relatively small (Figure 5.4). This holds similarly for the share of ICT production in GDP. Moreover, recent growth of the sector does not suggest any catching up. Comparisons of the production of ICT services across

Figure 5.4. **Share of ICT in manufacturing value added, 2000**

Per cent



2. Or latest available year. Other ICT manufacturing includes communication equipment, insulated wire and cable and precision instruments.

Source: OECD, *STI Scoreboard* 2003.

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Box 5.5. **The ICT sector in Germany** (*cont.*)

countries are subject to considerable uncertainty due to differences in measurement.Available information suggests, however, that Germany occupies a middle rank.[2] The share of German innovation efforts devoted to ICT, as measured in R&D spending and the proportion of patents devoted to ICT is also relatively small.[3]

... which is also true for the application of ICT

Benefits from productivity improvements associated with ICT can be reaped by application of ICT goods without own production.[4] Nonetheless, the proportion of GDP spent on the acquisition of ICT equipment (hardware and software) in Germany of about 6½ per cent falls short of that in several other countries within the OECD, such as Finland (7 per cent) and the United States (above 8 per cent).[5] Relatively moderate spending on ICT is likely to account to some extent for the relatively small contribution of quality improvements in the capital stock (embodied technological change) to total factor productivity growth in Germany, as ICT is an important source of such quality improvements. Notably the US and Finland recorded substantially higher contributions of embodied technological change to total factor productivity than Germany (OECD, 2003a).

Moreover, although German industries with intensive use of ICT technologies recorded higher gains in labour productivity than other sectors, productivity growth in these industries did not accelerate in the second half of the 1990s, unlike in the ICT-use intensive industries in the US. This suggests that German firms have not yet reaped the productivity benefits arising from innovations that are complimentary to the introduction of ICT goods as much as US firms did, possibly because German firms have been slower in embracing changes in organization and work practices that would be required to fully take advantage of these benefits.[6]

... indicating scope for improvement in the policy framework

Some evidence suggests that the smaller degree of ICT capital deepening in Germany has been more significant than the smaller contribution of ICT production in explaining the relatively subdued overall contribution of ICT to productivity growth in Germany.[7] Since Germany also occupies a middle position concerning household access to computer and internet, it is likely that not all of the reasons for slow diffusion of ICT are specific to the business sector.[8] Moreover, the price differential of ICT investment goods *vis-à-vis* the US declined substantially in the late 1990s to about 7 per cent and is now among the lowest in the OECD.[9] Hence, a comprehensive picture of the forces preventing higher rates of innovation, investment and utilisation of ICT is difficult to obtain, but structural factors related to product and labour market regulations are likely to play a significant role.

– *Regulation of firm entry*: Firm entry is considerably more important in ICT manufacturing industries and services than in the business sector as a whole.[10] Empirical evidence also shows that firm entry spurs productivity growth strongly in the ICT producing sector. Moreover, new general-use technologies,

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Box 5.5. **The ICT sector in Germany** (*cont.*)

such as ICT, are more quickly diffused by new firms or firms with new management and organization, as these, unlike incumbent competitors, do not face the costs of changing existing organization and technology. In the second half of the 1990s Germany was among the OECD countries with the highest level of administrative burdens to start-ups.[11]

– *Regulation of telecommunications*: Broadband access to the internet is likely to provide scope for productivity improvements in e-commerce. Lack of competition between the various broadband technologies appears to be limiting the use of broadband, as the incumbent telecoms operator, *Deutsche Telekom*, adopted a pricing strategy for DSL services which inhibited market entry by competitors, although more recently resale contracts between *Deutsche Telekom* and market entrants have been increasing competition within the DSL sector.[12]

– *Product market regulation*: High intensity of competition in product markets forces firms to adopt new production technology more rapidly in order to increase productivity. Compared to the United States, German product markets are noticeably more regulated, reducing scope for experimenting with changes in business organization which would help generating productivity gains from ICT use.

– *Lack of* ICT-*specialists*. Human capital plays an important role in realizing productivity gains from ICT use. Life-long learning helps match the skills of workers to new technologies such as ICT. The proportion of highly skilled ICT workers in the workforce in Germany is below the EU-15 average, and so is its growth. One third of German enterprises state that lack of ICT skills is preventing ICT investment. Participation in life-long learning in Germany is low in international comparison.[13]

– *Labour market regulation*: Countries with labour market institutions that support labour mobility and firm creation may be better equipped to innovate in industries characterized by multiple and rapidly evolving technologies including most of the ICT industry.[14]

1. See OECD (2003f). Hempell (2002) and Van der Wiel (2001).
2. OECD (2003b). The comparison is based on ICT services include ICT equipment, telecommunications, computer and related services, as wholesaling and renting of ICT equipment are not available for Germany.
3. OECD (2003b).
4. Colecchia and Schreyer (2002) provide illustrative evidence that the levels of ICT us and production are not correlated across countries.
5. ZEW (2003c). Data from the European Information Technology Observatory which used nominal spending on ICT divided by nominal GDP. Spending in Germany was 0.2 per cent of GDP lower than the average of European countries consisting of the EU-15, Norway and Switzerland.
6. SVR (2003b).

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Box 5.5. **The ICT sector in Germany** (*cont.*)

7. See D. W. Jorgenson (2003), Figure 4. Jorgenson investigates only information technology rather than information and communication technology. In Jorgenson's study of the G7 countries, the contribution from IT capital deepening was smaller than in the US, the UK and Canada but larger than in Italy and France. Differences in the measurement of ICT prices affect the growth contribution of ICT investment on real economic growth (see *e.g.* Deutsche Bundesbank, 2000, 2001 for an estimate of the effect on measured real GDP growth in Germany). However, the studies quoted here make use of harmonized ICT price indices in order to eliminate price measurement effects.
8. OECD (2003b). Comparable data refer to 2000 and 2001.
9. OECD STI Department. Unpublished data.
10. OECD (2003a) and Brandt (2004).
11. Bassanini and Scarpetta (2002); Nicoletti *et al.* (1999).
12. Gordon and RWI (2002), ZEW (2003c), European Commission (2003).
13. See OECD (2003f), ZEW (2003c), Gordon and RWI (2002), European Commission (2003).
14. OECD (2003a).

firms, providing incentives to adjust work organisation and technologies to adopt best practice. Overall, empirical evidence suggests that regulation of product markets inhibiting competition reduces R&D spending and is negatively correlated with total factor productivity growth.[23] As reviewed in Chapter 4 above, there is considerable scope to improve competition in some sectors. This includes network industries such as the telecom sector, which is of particular importance for the ICT sector.

Reducing administrative over-head

Recent empirical work suggests that Germany occupies a middle rank among European countries (EU-15) with regard to the duration and cost of procedures to create a limited private company as well as with respect to the capital requirement for limited private companies.[24] Some OECD members outside the EU appear to have considerably lower start-up costs, notably New Zealand, Canada, Australia and the US.[25] One factor raising start-up costs for incorporated business in Germany appears to be multiple administrative contact points.[26] While start-up times and costs are relatively low for private partnerships,[27] the duration of business registration appears to be unnecessarily long, exceeding two months in about 70 per cent of registrations. There is also some indication that firm exit costs are on the high side: while reported duration of bankruptcy procedures is relatively low in comparison with other high-income OECD countries, associated costs

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appear to be above average.[28] Empirical evidence also suggests that the duration of creditors' access to bankrupt debtors deters firm creation.[29] The duration seems to be relatively long in Germany in comparison to many other European countries and the United States.[30] High exit costs have a negative impact on firm creation as firm closures are particularly likely among young firms.

Proliferating procedures contribute to administrative opacity, weighing particularly on small firms and therefore on firm entry and may favour established organised interests groups (see Chapter 4 above).[31] Excessive regulation is likely to contribute to perceptions among enterprises of overly time-consuming administrative procedures for granting permission to undertake investment projects, notwithstanding reductions in durations that were achieved in the mid 1990s.[32] Still, about 20 per cent of German enterprises state that insufficient flexibility of regulations or standards are hampering innovation, compared to less than 10 per cent for the average of countries participating in the European innovation survey. Similarly, the complexity of Germany's income tax system is likely to deter entry, as the costs of coping with complex tax rules are more difficult to shoulder for small firms. Indeed, surveyed small and medium-sized enterprises consider the simplification of tax legislation and the reduction of regulatory overheads one of the most important reform issues, with the smallest firms giving it the highest weight. Simplification of the tax code is considered more important than extending subsidies to the *Mittelstand*.[33]

Hence, more efforts to simplify the administrative burdens associated with the foundation and growth of companies could enhance innovation. To this end one-stop shops should be established by regional governments, and regulations should be scrutinised and streamlined. While the government has launched an *Initiative to Reduce Bureaucracy*, profound deregulation also requires action on the side of the states and communities because sub-central authorities are often responsible in areas where administrative burdens are high.[34] Moreover, steps to reduce administrative costs associated with existing regulation should be complemented by steps to avoid unnecessary costs imposed by new regulation (see Chapter 4 above). Substantial simplification of Germany's income tax code could probably go a long way to reduce entry barriers.

Developing risk capital markets

Profits generated by innovation in the business sector typically arise with a time lag. Therefore, as is the case with investment decisions, the availability of financing is crucial for innovation decisions. Moreover, innovation decisions are, by their nature, particularly risky and the expenses are often not associated with the purchase of capital goods that can be collateralized. Also, the distribution of information is often highly asymmetric – more so than is the case with investment – as an innovating firm will be better able to assess the potential prof-

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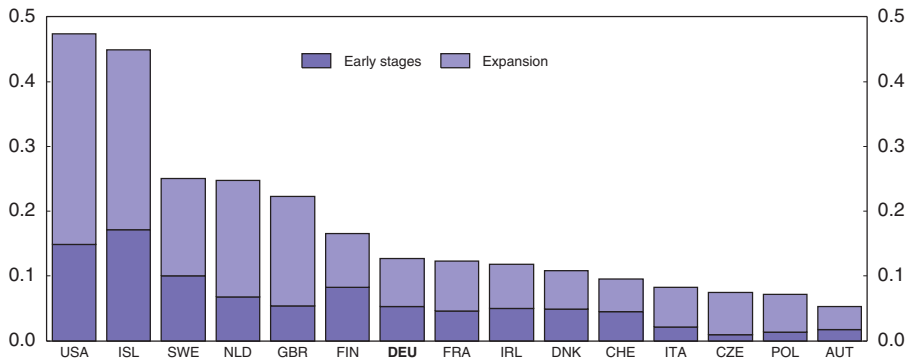
itability of an innovative project than a provider of external finance. In particular, much innovation in high-tech sectors, such as biotechnology, relies on entry of new firms.[35] The large proportion of intangible assets (such as human capital or a new product idea) involved in innovation in these industries aggravates the asymmetry of information between the firm and a potential provider of finance so that credit and conventional equity financing are typically not provided to these start-ups. Venture capitalists fill the void, typically providing management services as well for these companies to overcome the informational problems. The scope for cross-country VC capital flows is limited since management services are not easily exported because they require local knowledge and presence.[36] Hence, well functioning markets for risk and venture capital are indispensable for innovation projects. In the same vein, the taxation of enterprises affects innovation not only through the overall tax burden it places on enterprises but also through its effects on the structure of financing and on risk-taking.

Venture capital financing is relatively small in scale in Germany, amounting to about one fifth of the level (expressed as a percentage of GDP) in the United States on average between 1998 and 2001. The size of the VC market is also smaller than in many European countries, notwithstanding generous public sector support, relative to the size of the market (Figure 5.5). The closure of the *Neuer Markt* stock exchange segment for young technology firms in 2003 was a blow to the venture capital industry, as the *Neuer Markt* provided about 75 per cent of the initial public offerings for VC-backed firms between 1998 and 2000.[37] Both supply

Figure 5.5. **Managed venture capital investment, 1999-2002**[1]

Per cent GDP



1. Total investment flow over period. Venture capital investment in country i is investment managed by venture capital funds located in country i, it is not the amount invested in country i.

Source: OECD, based on data from EVCA (Europe) and NVCA (United States).

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and demand factors are likely to contribute to the small size. The preponderance of cumulative innovation in established firms is likely to reduce the demand for VC. However, net outflows of VC managed in Germany have been close to zero, suggesting that domestic supply factors are also at work.[38]

In particular, a liquid stock market is crucial for the supply of venture capital as it provides an efficient exit channel for venture capitalists having financed start-ups.[39] Hence, broadening the stock market would support the emergence of a more active VC market. However, initial public offerings on the stock market are small in Germany, and the number of joint stock companies has been falling steadily for several years. The small number of German firms listed on the stock market has been attributed *inter alia* to the traditional preponderance of universal banks. Universal banks may have relatively higher opportunity costs to engage in the issue business,[40] which may substitute their loan business. The fact that local and state governments act as guaranteeing authorities for public sector banks might also have played a role. Such guarantees lower the cost of capital, which is likely to induce banks to extend credit at less stringent conditions than would prevail if competition were undistorted. Government guarantees are being phased out over the next years.[41] Legislation came into force in February 2004 facilitating activities of investment funds. I*nter alia*, capital requirements for investment funds were reduced and hedge funds can now be issued. Supervision of investment fund providers was also simplified. These measures support the development of equity markets, although policies could be more pro-active. Future reforms could address legal requirements for setting up a joint stock company (A*ktiengesellschaft*) so as to foster the formation of small listed corporations.[42]

A number of tax rules generate disincentives for venture capitalists to provide management services to their client firms and reduce incentives to invest in start-ups rather than established firms. Similarly, some rules hamper investment in equity capital and the broadening of equity capital markets.

- Under certain conditions investors are not taxed on the gains from the sale of a company's equity in the venture capital firm's portfolio if the venture capital firm does not supply management services to the client firm. By contrast, if management services are supplied, the gains are typically taxed as business income.[43] In addition, the gains then become liable to the business tax which is likely to discourage foreign investors.[44]
- While capital gains on stock options in large established firms, used as an incentive device for management, are typically not taxed, performance incentives for the managers of venture capital firms (carried interest) are subject to income tax. New legislation, effective from July 2004 onwards, reduces this difference in tax treatment of performance incentives between managers of venture capital firms and managers of large established firms, subjecting half of the carried interest to taxation.

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- Business angels – wealthy individuals investing in start-up firms who concentrate on developing companies at the seed stage – are subjected to capital gains tax[45] whereas investors in private equity firms financing management buy-ins or buy-outs are typically tax exempt. This favours investing into restructuring of established firms over investment in start-ups.
- Limits on offsetting losses against gains from the sale of certain types of equity holdings used in start-up financing ("silent partnerships", *Stille Beteiligungen*) discourage providing capital for a start-up.
- Retained earnings of incorporated companies are taxed more favourably than distributed profits at the level of the individual investor. Since start-up firms cannot typically resort to retained earnings as a source of finance but depend on equity, this tax structure is also likely to divert flows from new to established incorporated firms.
- If the owner of a company changes, for example as a result of an initial public offering of a start-up, the company cannot, in general, offset previous losses against future profits for corporate income taxation purposes if a substantial amount of additional capital is injected into the newly acquired firm.[46] This provision discourages initial public offerings and innovation of start-ups expecting to raise external equity. The provision also discourages the supply of capital to firms after a trade sale or public offering.
- Inheritance tax rules favour non-equity ownership of enterprises as well as real estate over assets in equity. While this provision is meant to protect established small and medium sized enterprises, it might hamper the development of the equity market. The inheritance tax law is currently under review. A possible road of reform would be to harmonise taxation of the different types of assets while assigning the inheritance tax liability the status of subordinated debt. This would address the liquidity problems that can arise as a result of the inheritance tax liability.
- Calculations of the Council of Economic Advisers show that average effective tax rates are lower for private partnerships than for incorporated companies.[47] Empirical evidence indicates that owner-managed firms appear to have a lower propensity to innovate than firms with appointed managers, arguably because more widespread ownership spreads the risks stemming from innovative business strategies,[48] Although limited liability might also encourage enterprises to take risks.The reductions in marginal personal income tax rates in 2004 and 2005 will increase the tax advantage of unincorporated companies further, reinforcing the need to modify company taxation.

Moreover, since returns on equity (both retained earnings and externally raised equity) of corporations resident in Germany are taxed more heavily in Germany

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than in major European countries, German parent companies have incentives to provide equity financing to foreign affiliates (while foreign firms face tax-induced disincentives to provide equity to affiliates in Germany).[49] This in turn might imply that capitalisation in Germany becomes thinner, which might hamper risk financing. Hence, tax rules should be modified and specific provisions abolished such that they cease to penalise early stage financing of start-ups and the involvement of venture capital firms in the management of start-ups, as well as to hamper the development of equity markets more generally. In the same vein, a recent sharpening of the limits to carrying forward losses might also be reconsidered as they might reduce the propensity to take risks and hence to innovate.[50]

Adapting labour market institutions and work practices

Institutional features of the labour market can have a significant impact on the structure of innovation and adoption of new technologies. Worker competencies that are firm-specific appear to play an important role in "cumulative" innovation. By contrast, firms engaged in more "radical" innovation are more likely to rely on adjustments of their workforce so as to adjust skill compositions or change work practices. Accordingly, industries in which innovation is mainly cumulative are more likely to benefit from incentives for firm-provided training, and to suffer less from the adverse affects of institutions that hamper the re-allocation of labour, than firms that are engaged in radical innovation (Box 5.1 above). Indeed, countries with strict employment protection legislation combined with coordinated wage setting tend to concentrate their R&D spending on those industries where cumulative, as opposed to radical innovation predominates.[51] At the same time, the evidence is that companies are more likely to realise substantial productivity gains from adopting new technologies, such as ICT, if they undertake complementary innovations in production processes or work organization.[52] Hence, labour market legislation that hampers reorganisation of production processes and work practices is likely both to reduce the realised productivity gains associated with the adoption of new technologies and to slow their diffusion. Similar effects are to be expected from labour market policies that discourage entry or growth of new firms. Indeed, the strictness of EPL and the proportion of investment spending devoted to ICT are negatively correlated across OECD countries, suggesting that strict EPL, through increased adjustment cost, reduces the absorption of new technologies. This is particularly likely to hold if incentives for retraining of workers are insufficient to adjust to new skill requirements.

In Germany, several institutional features of the labour market tend to support the attachment of workers to firms and hence the accumulation of firm-specific human capital, notably the dual apprenticeship system, which is sustained by the absence of poaching of workers by other firms,[53] and relatively stringent EPL. These factors are likely to have contributed to the pattern of

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specialisation of German innovation in industries that are largely characterized by cumulative innovation. On the other hand, labour market institutions appear less adapted to support the reallocation of labour, which may have diverted activity away from areas with particularly rapid absorption of new technologies (Box 5.5 above).

Adapting labour market institutions and work practices to new demands may pay off in terms of productivity gains associated with swifter adoption of new technologies or more active innovation in certain areas. For example, exempting small firms from dismissal protection is likely to contribute to the generation of innovations by facilitating creation of new firms. Quick adaptation of vocational training curricula to new market needs increases the attractiveness of plant-based vocational training and is necessary to support rapid diffusion of innovation. Progress on these and related scores has recently been made, but there remains scope for further reform as discussed in Chapter 3 above.

Securing the supply of highly qualified workers

The supply of well trained and highly qualified labour is a key ingredient in the generation and diffusion of innovation. This includes the fact that the international location decision of enterprises is influenced by the availability of highly qualified employees and research personnel. Hence, securing a high performance of all segments of the education system is called for. Also, policies should foster the attractiveness of Germany for highly qualified labour from abroad.

Indeed, there is a secular trend toward employment with higher qualification, which Germany shares with other countries within the OECD. The share of those employed in German industry and the services sector with academic education increased by about a third since the beginning of the 1990s. In manufacturing, employment of persons with academic qualification increased substantially in absolute terms while overall employment fell significantly.[54] To a considerable extent this development appears to be related to the demand for personnel in R&D. The employment share of high-qualified personnel (as measured by academic and technical occupations) in German research-intensive industry and knowledge-intensive services exceeds the EU average significantly. Even so, the margin declined in the second half of the 1990s.[55] But there is evidence that even in periods of high general unemployment firms find it increasingly difficult to recruit highly qualified labour. For example, in a recent survey, sampling enterprises of all size classes in industry and among providers of technical services, some 42 and 64 per cent of the firms reported problems in filling vacancies for engineers in the recession and boom years, respectively, around the turn of the last decade. R&D turns out to be the main activity adversely affected, and 76 per cent of the firms concerned perceive a slowing of the innovation process as a consequence.[56] Most firms concerned (68 per cent) consider a lack of interest in

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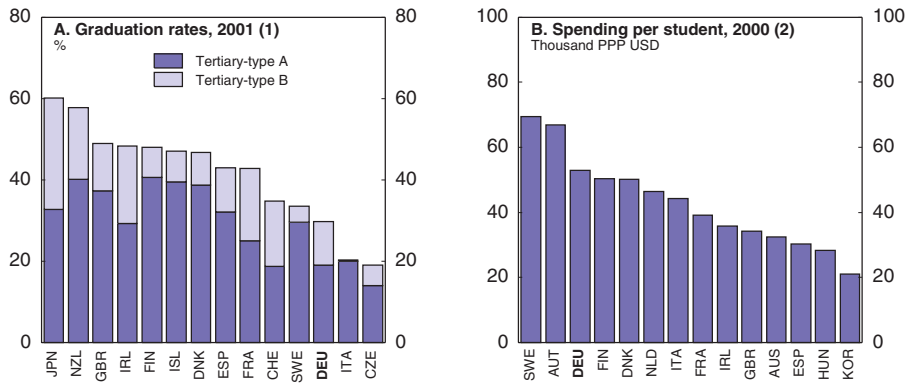
taking up engineering studies as the main force behind the supply constraint. Migration of employees abroad, while seen as an issue, is considered the least important factor (19 per cent).

Efficiency in the tertiary education system needs to be raised

Tertiary graduation rates in Germany are among the lowest within the OECD (Figure 5.6, panel A). The number of graduates declined since the middle of the 1990s, including those in science and engineering. After the middle of the 1990s, inflow into the university system picked up, from 28 per cent of the relevant age group in 1998 to 36 per cent in 2003. Still, inflow rates into tertiary studies are low by international comparison.[57] At the same time, average expenditures of tertiary education per graduate are among the highest within the OECD. Relatively long average study durations, especially at the universities,[58] are an important determinant behind this outcome (Figure 5.6, panel B). At graduation, German students have about the highest age within the OECD. Drop-out rates from tertiary studies of about 30 per cent, while about average among OECD countries, add further evidence that the efficiency of the system should improve. Indeed, long study periods tend to discourage investment in education unless they are matched by proportionate gains in productive capacity that are reflected in subsequent earnings. In general, the latter condition does not appear to hold, suggesting

Figure 5.6. **Graduation rates and spending in tertiary education**



1. Rates are estimated as [number of graduates]/[population of typical graduation age]. Tertiary-type A programmes provide qualifications for advanced research or higher skill professions. Tertiary-type B programmes focus on practical, technical or occupational skills.
2. Cumulative spending on educational institutions over study duration (average duration of tertiary studies).

Source: OECD, *Education at a Glance 2003.*

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that studies may become less focused when they are stretched. Indeed, in Germany private financial returns on tertiary education are low by international comparison, and this relates largely to the long duration of studies.[59] Moreover, longer working lives would raise the returns to education (see Chapter 3).

Over the last years, the importance of securing high standards of university education has increasingly been recognised in the policy debate, and induced reforms in some areas. In particular, to shorten study times, universities were given the option to offer bachelor and master degrees, and by 2010 these will become mandatory. A system of national quality assurance by accreditation has been introduced for the new bachelor and master degrees, in line with EU requirements. Also, the new carrier path of a "junior professorship" was introduced, and the requirement in several disciplines to produce a second thesis (H*abilitation*) as a prerequisite for a full professorship will be phased out by 2010. Some part of the pay for professors with tenure has been made performance dependent. These changes are perceived as insufficient by policy makers, and the government plans now to create a small number of highly funded "high performance universities". Indeed, more fundamental reform appears necessary to increase the capacity of the education system to produce high-quality output.

In this context, it is important to note that the performance of the tertiary education system is to a certain degree conditional on the performance of pre-tertiary education. Germany's low scoring in the PISA exercise suggests that improving the effectiveness of secondary education could make a substantial contribution to raising the efficiency of tertiary education.[60] Moreover, the organisation of students' access to universities is likely to be partly responsible for the low rates of students in tertiary education. The share of students being formally eligible for university access within the relevant age cohorts is low in Germany by international comparison. As a rule, eligibility to university access is confined to students with a higher non-vocational degree (A*bitur* or equivalent). Vocationally-oriented degrees in secondary education mostly do not qualify for university access, and where they do few students chose this track to access a university. Those who are eligible for tertiary education can in principle freely choose their tertiary institution. If the students' demand exceeds the number of places offered, by university and by study, students are allocated to universities by a federal administration, the selection criterion being the students' grades in secondary education and their waiting time. The government has drafted legislation to allow universities to choose up to 60 per cent of their students in courses in which entry is restricted. This would be a welcome step, as the system of university access is likely to be one factor preventing the potential among young people for tertiary education from being fully exhausted, and may produce an inefficient match between universities and students. Efforts should be made to insure that non-academic tracks of secondary education prepare better for direct entry to university education, so as to reduce the formal divide between vocational and non-

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vocational degrees. Moreover, the bureaucratic allocation of students to universities should be given up entirely to give universities further discretion to choose students.

The German tertiary education system relies heavily on government funding, which accounts for more than 90 per cent of total finance. This is somewhat below the European average, while in non-European member countries of the OECD funding from non-governmental sources covers between more than 30 per cent (Canada) and more than 70 per cent (Korea) of total outlays. The operating costs of universities are largely financed out of the L*änder* budgets while the federal government and the states cofinance investment. Most of university funding reflects the regional development of capacities as agreed between the federal government and the states, with the notable exception of special research projects, where public sector funding is based on evaluations of independent scientists. As debt of the federal and the state governments is mounting (see Chapter 2 above), finances of tertiary education institutions is increasingly coming under pressure, with some governments cutting back their funding. Simultaneously Germany is aiming to increase its cohort-specific graduation rate by 10 percentage points over the next years, which is further increasing the pressure on the general government budget. Moreover, additional funding will be required to implement the plans of the federal government to promote high-performance universities.

All of these pressures reinforce the need to improve the allocative efficiency of funding tertiary education. In recent years first steps have been made in this direction. The L*änder* are implementing models of performance-oriented allocation to varying degrees. In some federal states budgetary allocations are associated with the number of graduates rather than the number of students. This provides incentives to shorten study times and reduce the frequency of studies without a degree. However, the L*änder* are allowing at most 8 per cent of university funding to be determined in this way. While some L*änder* link substantially larger proportions of university funding to indicators, these are mostly input-oriented (such as the number of teaching staff) or demand-oriented (such as the number of students), resulting in weaker performance incentives.[61] Much more is thus required. Reforms should quickly move on implementing a higher degree of competition among universities, with the allocation of public sector funding more strongly linked to performance indicators. Several countries within the OECD have moved in this direction, Austria being a recent example. Moreover, more use should be made of non-government funding. Indeed, within the OECD the level of funds spent for tertiary education tends to be higher the more countries rely on other sources of financing apart from government spending, notably student fees and grants.[62] In Germany tuition fees are virtually absent and, apart from special cases, prohibited by federal legislation. Putting in place a framework that allowed universities to levy tuition fees, complete with an income contingent loan-employment scheme, would seem to be a useful step. This would enable students

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to act as purchasers of the universities' services at the institution of their choice, providing a powerful incentive for universities to improve their services. Fees would also prevent students from considering tertiary education as a costless good, contributing to shorter and more efficient studies from the demand side as well. Income contingent credit schemes were recently introduced in the United Kingdom and are applied for more than a decade in Australia and New Zealand.[63] In Australia, fees had little impact on the socio-economic mix of students, while in New Zealand the share of ethnic minorities more than doubled since the beginning of the 1990s and participation in tertiary education almost doubled since the introduction of student fees in 1992. This suggests that, so long as tuition fees were strongly additional to present funds, and not a substitute, they would be accepted.[64]

To make competition between universities viable, they should be given more autonomy with respect to the use of funds and personnel management to improve the quality of their services. Regulatory reform in this field would also improve incentives for non-governmental funding. At present, personnel are subject to tight employment regulations that restrain the scope of universities to attract highly qualified personnel and pose disincentives for employees to engage in high-quality teaching and research. Professors (including the recently introduced junior professor) have the status of tenured civil servants, salaried by the Land. Life time tenure is granted by the state government, with virtually no possibility of the universities to condition tenure on performance. While recent legislation allows for performance related components of pay for professors, this is not true for junior professorships and other research or teaching personnel. In the same vein, universities have little scope to attract highly qualified personnel via pay differentiation.[65] Also, professors are subject to certain teaching obligations that are fixed by state regulation. This makes contractual arrangements between the universities and its academic personnel about the relative weight of teaching and research more cumbersome.

... and the qualification level of net immigration needs to be improved

There is some evidence that Germany, like other European countries, is experiencing some outflow of highly qualified labour, in particular to the United States.[66] The number of German science and engineering employees working in the US increased by about 10 per cent (25 000) between 1995 and 1999 and appears large in comparison to the other larger non-English speaking EU countries. Regarding inflow from abroad, in recent years a steep increase was recorded in the number of foreigners studying information technology at German universities.[67] While the influx signals that German tertiary education in this field is internationally attractive, it is also likely to reflect that studying in Germany is effectively subsidised by the absence of student fees. More generally, over the last three years a

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limited number of work permits were granted to ICT professionals. Overall, however, migration to Germany appears significantly biased toward low qualifications. Since the hiring freeze of manufacturing "guest workers" in 1973, immigration is driven by family reunification of former guest workers who settled in Germany, asylum seekers, refugees from civil wars and ethnic Germans from eastern Europe. On average, immigrants have a significantly lower qualification than the domestic population.[68]

Most of the regulatory features concerning product, labour and capital markets discussed above can also influence the location decision of high-qualified labour and of enterprises. Progress on these scores therefore also improves Germany's position in the international competition for knowledge and capital. Moreover, the size of wage *premia* for skills are a potentially important determinant of a location's attractiveness for the high-qualified. If wage *premia* for qualification are relatively small, incentives to emigrate increase with the level of qualification. The distribution of gross wages in Germany appears to be more compressed than in some other countries, notably the United States, limiting the extension at the upper tail of the distribution. Empirical evidence suggests that compression derives to a certain degree from relatively high skill levels among German workers in the lower part of the wage distribution. However, there are other factors that compress the German wage distribution from below while hampering it to spread out to higher pay.

Germany's relatively high taxation of high incomes makes it costly for enterprises to provide high take-home pay in favour of the high-qualified. As outlined in Chapter 2 above, according to a study comparing six EU countries, Switzerland and the US Germany belonged to the countries with the highest average effective taxation of wages for high-qualified personnel before the income tax reductions in 2004 and 2005 (see Figure 2.2 above).[69] This position is set to improve, however, once the reductions have been fully phased in. Moreover, the rigid pay schemes for public sector employees in research institutions and universities restrain the scope for competitive remuneration in these large segments of the German research sector. In line with these findings, empirical research suggests that the pre-tax wage *premia* associated with an extra year spent in education is significantly lower in Germany than in the United States.[70]

In sum, these findings reinforce the need for fiscal reform that reduces the taxation of labour, including the high-qualified, and reform in the social transfer system and labour relations reducing effective minimum wages at the low end of the pay scale (see Chapters 2 and 3 above). Reforming the funding of universities is also necessary, as outlined above. New immigration legislation that aims at introducing qualification standards for immigrants seeking dependent and self-employment in Germany has been decided. Legislation along this line that raises the qualification profile of immigrants would probably yield significant benefits.

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Government support can be effective but requires a favourable regulatory environment

Externalities from innovation can be substantial, justifying some government support for innovation activities.[71] Government intervention is most necessary where markets completely fail to provide research which is beneficial to society. This is most apparent in basic research, which as a result of its public goods characteristics, is not provided through the market, and in Germany is conducted in public sector research institutions (see Box 5.2 above). In the business sector, positive externalities come about, *inter alia*, because innovators cannot appropriate the full return of their effort, given that protection of intellectual property rights is limited. All OECD countries, including Germany, provide subsidies to enterprises for innovation in terms of transfers or tax advantages for R&D spending. In Germany government support for innovation in the business sector for civilian purposes is below the levels in Japan, the UK and France but above those in Finland and the US.[72] Innovation is subsidised entirely through transfers. The following issues warrant particular attention:

- As is the case with other subsidies, support for innovations need to be carefully designed to limit distortion and assure effectiveness. While evaluations of some German programmes indicate that support has raised R&D spending of participating firms over and above the received amount of public support,[73] not all programmes have been subjected to evaluation. Hence, evaluation needs to be stepped up.
- The subsidy programmes appear to be complex in comparison to other countries.[74] Relatively high administrative costs for both the programmes' administrators as well as the enterprises are likely to hamper participation of small and medium sized enterprises and favour "insiders" – firms that are already familiar with the programmes. The latter may contribute to the finding that government-supported business R&D spending is more concentrated on large firms than total business R&D spending,[75] notwithstanding recent increases in the number of enterprises picking up government support. Programmes should therefore be simplified. In the same vein, the general nature of externalities suggests that criteria for participation in R&D support programmes should be widened. However, safeguards – thorough *ex ante* assessment and *ex post* evaluation – need to be in place to limit rent-seeking.

Special support conditions for the new Länder should be reduced

Considerable public sector resources were devoted over the last decade or so to support R&D in the new states. With enterprises in the east hardly meeting the same set of eligibility conditions for financial aid that applied to enterprises in the west, the policy response was to design various special subsidy

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programmes that broadened the coverage of aid in the east. This policy has contributed to widening the share of research intensive sectors in the economy. However, some indicators suggest that scope to increase the efficiency of policies affecting innovation might be significant. While the share of companies with continuous R&D activity is higher in the east than in the west the opposite is true for the contribution of research-intensive sectors to value added. Also, the return from innovation relative to its costs is substantially lower than in the west. This is mainly true for process innovation while the difference is only modest for product innovation.[76]

Various structural factors are likely to account for the relatively weak return to R&D in the new states. Much is attributable to the fact that in comparison to the old states the size distribution of firms is heavily skewed toward smaller enterprises.[77] This reduces the share of innovation associated with large firms whose innovative potential is normally high, notably in terms of internal financing capacity, potential spin-offs of new innovative firms and demand for intermediate inputs from other business. Regarding smaller enterprises, however, firm entry has been declining for several years and enterprise growth is relatively weak, both weighing on innovation. In the same vein, endowment with equity capital – as a share of the firms' balance sheets – is significantly smaller in eastern German enterprises than in western ones, reducing the capacity to engage in risky innovation activities. Relatively high wage settlements in eastern Germany can reinforce this weakness by putting profit margins under pressure. Indeed, empirical evidence for the new states, as well as for Germany overall, suggests that firms respond to wage increases that exceed underlying labour productivity growth by scaling back investment.[78]

Also, to some extent, subsidisation is likely to have supported R&D activity in areas with low returns. Indeed, in eastern Germany some 90 per cent of manufacturing enterprises with continuous R&D activity receive financial aid from the public sector, as opposed to thirty per cent in the west (Figure 5.7). While surveys indicate that only one third of supported firms would be active in R&D without support, generous special aid raises incentives for rent seeking, associated with the risk that an artificially high share of innovation activities might not stand the market test in later years.

Hence, especially for the new states, reforming economic framework conditions in capital, product and labour markets and in the education system is necessary to raise the effectiveness of government support for innovation. Without such reform, support programmes are bound to remain of limited effectiveness, implying undue fiscal burdens for the general government. Moreover, there is a risk that continuing generous special support programmes for enterprises in the new states perpetuates inefficient economic structures.[79] Therefore, special support conditions should be reduced.

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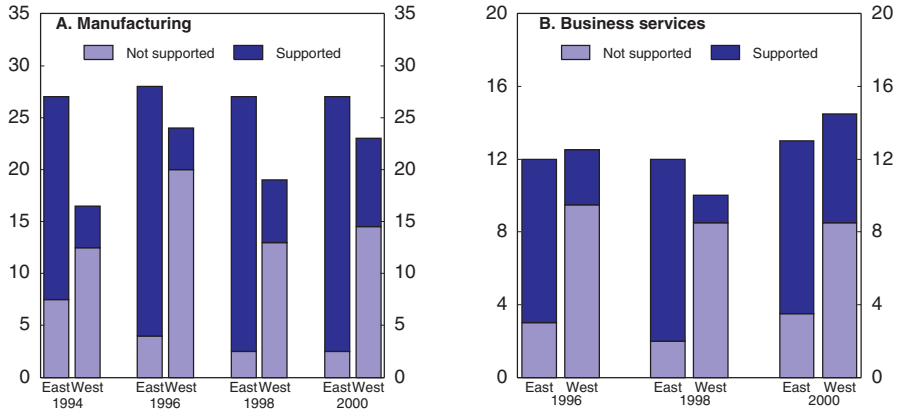
Figure 5.7. **Share of enterprises with continuous R&D**
Per cent



Source: Zentrum fŸr EuropŠische Wirtschaftsforschung (ZEW).

Summary

Germany belongs to the countries with strong innovation activity in the OECD, although innovation effort may have weakened somewhat in recent years. While productivity growth in Germany has slowed, there is scope to improve economic framework conditions in a number of areas so as to enhance the economy's capacity to innovate and reap the full benefits of innovation for raising productivity and employment. Reducing administrative opacity would support the creation of new enterprises, which are particularly important in developing and adopting new technologies. Simplification of Germany's complex income tax code is important in this respect. Also, various features of the tax system hamper the development of risk capital markets, calling for reform. A positive impact on innovation activities, in particular in fields that are associated with rapid technological change, can be expected from policy action that supports the capacity of the labour market to reallocate labour. In the same vein, provisions need to be made to secure the availability of a well trained and highly qualified workforce in the future. This reinforces the need to improve the efficiency and the access to the tertiary education system and to adopt measures that raise the qualification profile of immigrants. Finally, government aid programmes to enterprises in favour of innovation should be more subjected to evaluation. At the same time, support programmes should be simplified, while special subsidies for the new states should be reduced.

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Notes

1. A one per cent increase in a country's R&D spending was estimated to generate an increase in total factor productivity of 0.1 and 0.3 per cent, see Guellec and van Pottelsberge (2001), Nadiri (1993) and OECD (2003a).
2. See Sheehan and Wyckoff (2003) for analysis on implications of boosting government R&D spending for the demand for high-qualified personnel.
3. Eurostat (2004). Participating countries include the EU15 countries, Iceland and Norway.
4. See Belitz (2003), OECD (2003b).
5. The use of patents to protect innovations appears to be particularly widespread in the sectors where Germany performs strongly. This might explain to some extent why German patenting figures suggest a better performance, relative to the OECD, than its R&D spending.
6. See Frietsch and Breitschopf (2003) and Legler (2003).
7. Some OECD countries, including the US and the UK, also improved tax advantages of R&D spending substantially in the 1990s. These tax advantages are not included in public-sector financed R&D figures.
8. According to the overall proportion of innovating firms in the EC *Innovation Surveys* of 1996 and 1992 Germany's position moved from second to third. Survey data are not consistent across years, so that only rank comparisons are used (European Commission, 2001).
9. *Bundesministerium für Bildung und Forschung* (2002).
10. See also Legler (2003).
11. See Frietsch and Breitschopf (2003), p. 45. The authors also show that the lack of specialisation of German patenting in fast-growing sectors depressed patent growth in Germany. See also: OECD (2003b).
12. A relatively small proportion of R&D spending is devoted to high-tech industries (OECD, 2003b).
13. OECD (2003b), p. 147.
14. ZEW (2003b). Declining East German rates reflect the end of the post-unification entry boom.
15. OECD (2003a). The sample is based on social security data for western German between 1989 and 1994. EUROSTAT has published firm entry data based on trade registration and deregistration which show relatively high firm creation rates in Germany between 1996 and 2000 if compared with other European countries. However, according to Eurostat, these German data are not comparable to those of other countries. For

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example, the German figures are likely to be inflated by registration of inactive firms, before 1999. See European Commission (2002b).

16. The absolute cost of administrative regulation to enterprises is often invariant with respect to firm size so that the implied burden weighs more heavily for entering firms, which tend to be relatively small.
17. E.*g.*, Mortensen and Pissarides (1994).
18. Scarpetta *et al.* (2002). In addition, firm entry rates are above-average in ICT-related industries and services as well as pharmaceuticals OECD (2003a), Chapter 4, and Brandt (2004).
19. See Park and Wagh (2002).
20. Member states of the European Patent Convention include the EU member countries as well as some European states that are not members of the EU.
21. Martinez and Guellec (2004).
22. E*uropean* I*nnovation* S*urvey*, European Commission (2002).
23. Bassanini and Ernst (2002a).
24. See European Commission (2002).
25. World Bank (2003).
26. European Commission (2002).
27. European Commission (2002).
28. World Bank (2003).
29. Brandt (2004).
30. According to UNICE (2000).
31. OECD (2003c).
32. See, *e.g.* BDI (2003), IHK Hamburg (2002). IHK Hamburg reports on surveys of enterprises which indicate that commercial building permit applications took about four months to be completed in major German cities. 20 per cent of this time period lapsed between application and the onset of the administrative procedure, suggesting scope for reducing opacity in the application process.
33. Creditereform (2002) and *esbf* survey, quoted in Hommel and Schneider (2003).
34. For example, most of the measures suggested in a pilot project on reducing bureaucracy in the state of Bremen, which aims at identifying and testing scope for reducing administrative costs for Germany as a whole, require implementation by subcentral authorities. Freie Hansestadt Bremen, Handelskammer Bremen (2003).
35. See *e.g.* ZEW (2003a).
36. This refers to the flows of capital from venture capital firms to start-ups. Venture capital firms themselves may still raise funds from foreign investors. Venture capital supplied by domestic venture capital firms largely determines how much venture capital is available to domestic firms, as reflected in the close correlation between the amount of VC raised and the amount of VC invested across countries. See also OECD (2003b).
37. Franzke *et al.* 2003.
38. Pension funds make a relatively small contribution to venture capital financing in Germany. In the UK or the US, by contrast, pension funds contribute a large proportion of VC funds, reflecting the more developed pension funds industry in these two countries in general. Private pension assets have reached 60 per cent of GDP in the UK and

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US, but only 17 per cent in Germany. Moreover, in Germany at least a third of private pension wealth is kept as retained earnings by enterprises within occupational pension schemes (see Yermo, 2003 and FAZ, 2003). In the US, permission for pension funds to invest substantial amounts into risky funds, including venture capital, contributed to the expansion of the venture capital market. More than half of the funds at the disposal of German venture capitalists is provided by banks, considerably more than in the UK and the US. As a result of bank ownership of venture capital firms, venture capitalists in Germany may have less expertise which is specific to the sector in which their client firms operate (see Schertler and Stolpe, 2003).

39. Divestment through the stock market provides an opportunity for the venture capitalist to signal its success and for the entrepreneur to regain control of the company. See Black and Gilson (1998).

40. According to Monopolkommission (1998).

41. See OECD (2001a).

42. See Franzke *et al.* (2003).

43. Venture capital firms which do not provide management services above the monitoring activities of a shareholder are classified as private wealth managers. Investors in these venture capital firms are subject to capital gains tax on the sale of shares of the venture capital firm's client companies. The tax rules exempt capital gains from taxation if the investor's share of the client company, through the venture capital fund, is below 1 per cent and is kept for at least one year. Investors in a venture capital firm providing management services, by contrast, do not qualify for the tax exemption.

44. While domestic investors can offset the business tax against income tax, foreign investors cannot do so.

45. This is because their shareholdings tend to exceed the 1 per cent threshold. See endnote 43.

46. Losses cannot be offset against future profits if the acquired firm's assets are increased by more than 100 per cent, unless the capital injection is necessary to ensure the survival of the firm.

47. SVR (2003a). Effective tax rates are calculated averaging over financing methods, using observed financing patterns as weights and include the tax burden of the local enterprise tax.

48. Czarnitzki and Kraft (2003).

49. See SVR (2003). The comparator countries include France, the UK, Ireland, Italy, the Netherlands, Sweden and Spain.

50. Specifically, losses from earlier years can be fully deducted from profits if the latter do not exceed € 1 million. Above this threshold losses can be deducted only up to 60 per cent of profits. As a result, for any given expected return, riskier projects are taxed more heavily. The adverse effect on risk taking associated with this measure will be larger for non-incorporated companies which are subject to the progressive marginal tax rates of income taxation, as higher profits resulting from successful innovations will tend to be taxed under higher marginal tax rates.

51. See Bassanini and Ernst (2002a).

52. OECD (2003f).

53. See *e.g.* Casper and Matraves (2003).

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54. For example, employment of high-skill workers has helped firms reap productivity gains from ICT use (OECD 2003f).
55. High-qualified personnel is measured by occupational ISCO-88 groups two and three (academic and technical or equivalent occupations). Within the EU the share of high-qualified personnel is higher only in Sweden, the Netherlands and Finland. See: special evaluation of employment by qualification, based on the EU Labour Force Survey, *Bundesministerium für Bildung und Forschung* (2001).
56. *Zentrum für Europäische Wirtschaftsforschung* (ZEW) and VDI *Nachrichten* (2004), *Fachkräftemangel bei Ingenieuren – aktuelle Situation and Perspektiven*.
57. See OECD (2003d).
58. Courses are shorter at the more occupationally oriented *Fachhochschulen* than at the academically oriented *Universitäten*.
59. See Blöndal *et al.* (2002). Incentives to invest in higher education are also likely to be strengthened if firm entry rates are high. New firms with the potential to expand are often set up by high-skilled workers. Opportunities to set up a business therefore increases highly skilled workers' job opportunities and reduces the job market risks highly-skilled workers are exposed to. The favourable effect of firm entry on the supply of skilled labour is backed up by cross country empirical evidence (Dullek *et al.*, 2003).
60. See OECD (2001c). Also: OECD (2002).
61. Leszczensky and Orr (2004).
62. See OECD (2004).
63. See OECD (2004).
64. See Lundsgaard and Turner (2004) on the introduction of study fees in the UK. According to a recent survey by the Forsa institute, commissioned by Stifterverband für die Deutsche Wissenschaft. According to the poll, 59 per cent of the students are in favour of paying a fee of € 500 per semester, provided both the funds are earmarked to their university so as to improve the quality of the university and there are loans available that only need to be redeemed after the end of studies if income exceeds a certain threshold. By contrast, 94 per cent of the students reject study fees if these are not earmarked, acting just as an additional source of revenues for the purpose of relieving the governments' budgets. In the population at large 67 per cent are in favour of fees earmarked to the university and 72 per cent are against fees if they are not earmarked. In the same survey three years earlier a smaller share of 47 per cent of the students responded in favour of fees (quoted in *Handelsblatt*, 11th December 2003).
65. For example, the fact that university professors of German citizenship are civil servants, who are not liable to pay social charges, can pose a barrier to attract foreigners, who are subject to such charges.
66. See European Commission (2003).
67. See Engeln *et al.* (2003).
68. See for example: Constant and Massey (2002) Constant and Shachmurove (2003). At its peak in 2001 immigration to Germany totaled 879 200 persons and emigration out of Germany 606 500 persons. For international comparisons of educational attainments of foreign and national populations see OECD (2003e).
69. Elschner *et al.* (2003).
70. See Psacharopoulos and Patrinos (2002).
71. See Griliches (2001) and Jones (1998).

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72. Rammer *et al.* (2004).
73. Czarnitzki *et al.* (2002), Czarnitzki and Fier (2003).
74. Rammer *et al.* (2004).
75. While enterprises with fewer than 250 employers received 7.3 per cent of government-financed R&D, they conducted 9.3 per cent of all business R&D. OECD (2003b).
76. See Deutsches Institut für Wirtschaftsforschung, (2003).
77. See Rammer and Czarnitzki (2003).
78. For an investigation, based on firm-level data, of the impact of wage setting on productivity and investment in eastern Germany see: Lehmann (2003). For an empirical investigation of the impact of wage setting on investment for Germany overall, based on aggregate data, see Weyerstrass and Klaus (2003).
79. High rates of subsidisation in the 1990s, notably for capital intensive production and construction, have biased the structure in the eastern economy. See OECD (2001a). For more detail see also Wurzel (2001).

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Annex. **Progress on structural reform – a condensed review**

Key challenges	Action taken	Assessment	Follow-up recommendation
Panel A: Labour markets			
Remove obstacles to supply labour			
Reduce incentives for early retirement	Phased increase in early retirement age on account of unemployment. Shortened duration of unemployment benefit entitlement for older workers. Phasing-out of search requirement exemptions for older workers.	Significant progress in reducing disincentives for labour supply of older people. Some tax and benefit-induced incentives for effective early retirement remain.	Consider stricter application of job search requirements regardless of age. Announce increase in statutory retirement age and reduce disincentives to work longer. Remove incentives for effective early retirement.
Remove disincentives for women to work.	Investment in fulltime schooling is being co-financed by the federal government.	Child benefits in present form are associated with disincentives for female labour supply. Spouses of earners face higher effective taxation of work than single earners.	Reallocate resources for child care support to the users of child care facilities. Reduce effective taxation of second earners. Consider introducing charges for health co-insurance for non-working spouses.
Increase the effectiveness of labour market policies	Shortened duration of unemployment benefits. Unified benefit systems and tightened eligibility. Reduced withdrawal rates of social assistance, widened subsidized payroll taxes for low paid jobs and subsidies for small self-employment. Increased resources for placement services. Abolished re-entitlement to unemployment benefit after ALMP participation.	Reforms in unemployment related transfers increase incentives to take up employment. Preferential benefit eligibility conditions for older unemployed remain. The option for local governments to provide services to the unemployed may compromise some of the efficiency gains of benefit reform. Subsidized contribution rate scheme for low paid jobs and support for small self employment may be little effective in their present form.	Assess ending preferential eligibility conditions for unemployment benefits of older unemployed. Unify responsibility for placement services in one agency. Develop performance incentives for local labour offices. Consider targeting support for low-paid jobs to benefit recipients. Evaluate all active labour market measures and discontinue immediately programmes that are not effective. Further strengthen placement activities.

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Annex. **Progress on structural reform – a condensed review** (*cont.*)

Key challenges	Action taken	Assessment	Follow-up recommendation
Remove obstacles to demand labour			
Widen the scope for wage determination at the company level so as to better align wage contracts with labour market conditions.	Social partners make use of firm level opening clauses for collective agreements.	Opening clauses often require agreement of the social partners which may result in insufficient wage differentiation.	Consider legislative action to allow more firm level agreements if use of opening clauses turns out insufficient. Further confine the administrative extension of collective agreements.
Reduce costs associated with employment protection legislation (EPL)	Limited range of social criteria to be considered in dismissals, streamlined legal procedures. Widened exemption of small firms from dismissal law.	Measures likely to improve labour market flexibility. Legal costs may remain high; risk of labour market segmentation between well protected permanent and unstable employment.	Evaluate the impact of recent reforms of EPL and consider further easing accordingly, including further raising the firm size threshold of EPL.
Exploit temporary work agencies' (TWAs) employment potential	Lifted restrictions on employment contracts of TWAs workers; introduction of collective bargaining for TWA workers.	Temporary contract legislation limits reemployment of workers by TWAs. Collective bargaining may reduce low-skill employment in TWAs.	Ease re-employment restrictions on TWAs; evaluate impact of collective bargaining on low-skill employment and reconsider the rule mandating equal working conditions of TWA workers with those of regular workers if necessary.
Reduce tax wedge on labour	Lowered income tax rates. Introduced cost containment measures in the health care system to create room for rate reductions. The pension replacement rate is being curbed.	Pension and health care reform are steps in the right direction. Scope for more efficiency improvements remains in the health care system, and over the next decades substantial increases in contribution rates for pensions are still expected. Challenges of ageing are pressing in the social security systems.	Widen the scope for selective contracting and managed care in the health care system. Announce increase in the statutory retirement age and reduce disincentives to work longer. Raise efficiency in the provision of long-term care.

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Annex. **Progress on structural reform – a condensed review** (*cont.*)

Key challenges	Action taken	Assessment	Follow-up recommendation
Panel B: Product market regulation and innovation			
Improve competition in the product markets			
Make public procurement more competitive	The government is planning to simplify complex procurement rules.	Involvement of business associations in rule-setting may hamper competition. A relatively small proportion of procurement is published at EU level.	Discontinue participation of business associations in the setting of procurement rules. Increase international publication of procurement at EU level.
Open the handicrafts sector to more competition	Qualification-related entry requirements in crafts largely dropped or relaxed.	Measures will increase competition, but 90 percent of enterprises remain subject to entry requirements	Further deregulate handicrafts entry.
Deregulate professional services	The government is planning to phase out the binding fee schedule for engineers and architects.	Professional services are more stringently regulated in Germany than in many other OECD countries.	Eliminate legally binding fee schedules for architects and engineers and prohibit recommended price schedules. Consider further deregulation of professional services and explore scope for market determined prices.
Phase out state aid	Cuts across a wide range of subsidies are part of measures to consolidate the government budget.	Cuts are move in the right direction. State aid in Germany remains higher than in several European high-income countries.	Phase out state aid except where it addresses market failures.
Deregulate the retail sector	Restrictions on the discounting of prices have been lifted.	General prohibition of below-cost pricing limits competition.	Abolish prohibition of below-cost pricing when predatory market power is absent.

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Annex. **Progress on structural reform – a condensed review** (*cont.*)

Key challenges	Action taken	Assessment	Follow-up recommendation
Improve regulation of network industries	Introduced regulator for the gas and electricity sectors. Amended telecoms regulation legislation.	Non-discriminatory access at non-excessive prices to networks needs to improve. The network industry regulator may not be sufficiently perceived as independent from the government.	Fully privatise the Telecoms and the Post Office. Facilitate the construction of competing networks in telecoms. Ensure ex-ante price regulation is applied when significant market power is present. Further unbundle energy generation, distribution and transmission in the electricity and gas sectors. Consider ownership unbundling when privatising Deutsche Bahn.
Improving the framework conditions for innovation			
Remove obstacles to firm creation.	Eased administrative burdens on firms and employment protection legislation for start-ups. Raised threshold for application of dismissal legislation.	Handicrafts entry remains excessively regulated. The complexity of the tax code is particularly costly for new and small firms. Setting up a limited liability company is more costly than in other high-income countries. Tax rules favour provision of finance to established rather than new firms.	Further deregulate handicrafts entry. Cut tax expenditures and reduce the complexity of the tax system. Strengthen regulatory impact analysis, reduce costs related to registering companies and to EPL further. Modify the tax code so as to remove disincentives to provide capital to start-ups.
Increase the efficiency of tertiary education.	Introduced option for bachelor and master degrees and first elements of performance-oriented remuneration and university funding introduced. Planned competition-based selection of a small number of additionally-funded high performance universities.	Performance incentives need to improve in the tertiary education sector as a whole.	Improve access to university entry. Give universities more discretion to choose students. Link the allocation of public sector funding more strongly to performance and give universities more autonomy.

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Annex. **Progress on structural reform – a condensed review** (*cont.*)

Key challenges	Action taken	Assessment	Follow-up recommendation
Facilitate immigration of qualified workers.	New immigration legislation to introduce qualification standards for immigrants has been adopted. Reduced top marginal income tax rates. Pay differentiation for full professors allowed.	Scope for competitive remuneration remains restrained. Tax wedge still relatively high. Immigration biased towards low-skill workers.	Further facilitate immigration of qualified workers. Increase efficiency of tertiary education. Reduce tax wedge on labour.
Evaluate government-funded innovation programmes and eliminate ineffective subsidies.	More frequent evaluation of subsidy programmes and public sector research institutes.	Not all government subsidy programmes have yet been evaluated.	Establish a timetable of evaluations of public sector research institutes by independent commissions. Evaluate all subsidy programmes and discontinue ineffective programmes. Reduce special support conditions for the new *Länder*.

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Glossary of acronyms

ALMP	Active Labour Market Policy Measure
APW	Average production worker
BDI	*Bundesverband der Deutschen Industrie*
BMBF	Federal Ministry for Education and Research (*Bundesministerium für Bildung and Forschung*)
BMF	Federal Finance Ministry (*Bundesministerium der Finanzen*)
BMGS	Federal Ministry for Health and Social Security (*Bundesministerium für Gesundheit und Soziale Sicherung*)
CEE	Central European Communities
DRG	Diagnosis Related Groups (Payments for Hospitals)
DTAG	*Deutsche Telekom* AG
EPL	Employment Protection Legislation
FEA	Federal Employment Agency (*Bundesagentur für Arbeit*)
HGF	Helmholtz Association of German Research Centres
Ich AG	Me Inc. (subsidy scheme for the unemployed to take up self-employment)
ICT	Information and Communication Technology
IHK	*Industrie- und Handelskammer*
IPRs	Intellectual Property Rights
ISCO	International Standard Classification of Occupations
PES	Public Employment Service
R&D	Research and Development
SME	Small and Medium sized Enterprise
TE	Tax expenditure
TFP	Total factor productivity
TWA	Temporary Work Agency
UNICE	Union of Industrial and Employers' Confederations in Europe
VC	Venture capital
VHA	Veterans Health Administration
ZEW	*Zentrum für Europäische Wirtschaftsforschung*

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OECD PUBLICATIONS, 2, rue André-Pascal, 75775 PARIS CEDEX 16
PRINTED IN FRANCE
(10 2004 12 1 P) ISBN 92-64-10840-8 – No. 53645 2004
ISSN 0376-6438